The Untaxables? Effective Capital Gains Taxes with Diversified Portfolios

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Abstract

Capital gains are central to the taxation of the very rich, yet realization taxes allow investors to defer payment. Despite its policy relevance, a measure of long-run effective taxation capturing optimal deferral has been missing because of technical challenges. I close this gap with a model that represents portfolios by distributions of embedded gains and losses. It describes a dynamic trading problem in which today's sales change tomorrow's tax opportunities. I establish when harvesting losses and selling the least appreciated positions is optimal over any horizon. Aggregate downside risk can reverse this rule. I express the consumption benefits of tax-optimal trading as effective tax rates and equivalent returns. The calibration matches historical U.S. market conditions and the OECD average capital gains tax rate. A 25% tax on realized gains and dividends sustains the same consumption perpetuity as an accrual tax of about 14%. The corresponding return premium is 74 basis points: at 25%, an accrual-taxed portfolio needs this additional expected annual return to match consumption under realization taxes. A cohort-model extension with carry-over basis at death predicts that revenue under realization taxes is 49% lower than under accrual taxes. Continued deferral thus limits the revenue and progressivity of realization-based capital gains taxes.

JEL Classification: H21, H24, D14, G11

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I Introduction

Capital gains are central to the income and taxation of the very rich (Piketty et al., 2023, Saez and Zucman, 2021, Scheuer and Slemrod, 2020, 2021). In 2021, net realized capital gains accounted for 58% of income among the richest U.S. taxpayers.[1] Their taxation therefore matters for both public revenue and the distribution of tax burdens. However, the official ("statutory") tax rate on investment profits overstates what these investors effectively pay. Because capital gains are commonly taxed when assets are sold ("realized"), investors can postpone taxes by retaining appreciated asset positions ("deferral") and reduce current taxes by realizing losses ("loss-harvesting"). For wealthy households with broad portfolios, and for a wider population of households using recent fintech innovations[2], these deferral and loss-harvesting incentives are a recurring feature of taxation. One central question is therefore how large an advantage realization taxation gives investors relative to taxes levied as income accrues, such as labor income taxes.

This paper develops a framework for optimal trading in large taxable portfolios, where today's sales change tomorrow's tax opportunities. To make realization and accrual taxes comparable, I derive accrual-equivalent tax rates from the outcomes of optimal trading. I find that taxing upon realization rather than accrual has large quantitative effects on investor consumption and government revenue. At the baseline, optimal trading reduces the investor's effective tax rate and the government's revenue flow to little more than half of what the statutory rate suggests.[3] Investors achieve these savings by continually deferring the sale of the most appreciated remaining positions, sustaining a large stock of untaxed gains and a permanent shortfall in tax revenue.

I find that, at a statutory rate of 25%, optimal trading perpetually sustains the same constant real consumption as an accrual tax of about 14%. I call this consumption-equivalent accrual rate the *effective tax rate*. The baseline combines a 25% realization tax, corresponding to the 2025 OECD average capital gains tax,[4] with market conditions calibrated to historical U.S. asset returns, inflation rates, and corporate dividend payout practices. Relative to the OECD average labor tax wedge of 35%,[5] the effective tax advantage is therefore about 21 percentage points, more than twice the 10-point gap suggested by statutory rates. This additional advantage weakens tax progressivity because capital gains are concentrated among the very rich. Matching the labor tax wedge would require a realization tax of about 57% on gains and dividends. At baseline market conditions, the effective rate is roughly linear in the statutory

[1] Author's calculation from Internal Revenue Service (2021, Table 1.4, tax year 2021). Net realized capital gains comprise taxable net gains and capital-gain distributions, less deductible net capital losses.

[2] Robo-advisors are one example. Reported assets in 2026 include Wealthfront's $93.2 billion in platform assets, Betterment's over $65 billion under management, and Wealthsimple's C$124.8 billion under administration (Nasr, 2026).

[3] The revenue comparison assumes carry-over basis at death: heirs retain the existing tax basis, so the result requires no forgiveness of unrealized gains.

[4] Author's calculation for 28 OECD countries with realization taxes, using statutory rates on long-held listed shares for non-substantial shareholdings. Source: Mengden (2025, Table 2). The mean is 24.75%.

[5] Single, childless worker earning the national average wage in 2025; income tax and employee and employer social contributions, net of benefits, as a share of total labor costs (OECD, 2026, Table 1.2).

rate: a one-percentage-point statutory increase raises the effective rate by approximately 0.60 percentage points.

The same consumption advantage can also be expressed in expected-return units, connecting the effective tax rate to return-alpha measures in finance. Here, the *return alpha* is the additional annual expected return an accrual-taxed portfolio would need, at the same statutory rate, to sustain the consumption achievable through optimal realization trading. At the baseline, this alpha is 73.8 basis points (bp): 44.4 bp come from single-asset deferral and 29.4 from selecting across multiple assets.

Because the effective tax rate and return alpha measure consumption equivalence rather than government collections, I also derive a revenue-equivalent accrual rate in a cohort version of the model with accidental bequests. This rate reproduces the revenue from realization taxation at every date along the economy's balanced-growth path, holding the aggregate wealth path fixed across tax systems. At the baseline statutory rate of 25%, the revenue-equivalent rate is 13% under carry-over basis at death, close to the consumption-equivalent rate of 14%. Realization taxation therefore collects 49% less revenue than accrual taxation at the same statutory rate. This shortfall persists even though heirs retain the existing tax basis and no unrealized gains are forgiven at death.

My quantitative analysis contributes to the literature in two ways. First, it provides unified measures of the realization advantage, connecting effective tax rates in public finance with return alphas in finance. Second, it quantifies this advantage and its sensitivity to statutory tax rates and market conditions, and assesses its implications for government revenue.

Quantifying these effects requires a tractable model of optimal trading. My first theoretical contribution is therefore to identify the relevant characteristics of a large taxable portfolio and define a class of recursive problems under realization taxation. An intuitive economic insight applies: within a portfolio, the identity of an individual asset becomes negligible once idiosyncratic risk is diversified. This allows me to represent the portfolio by the distribution of market value across tax lots—the state representation of the Tax Portfolio Model. A tax lot groups assets whose sale realizes the same gain or loss per dollar of value sold; these are precisely the assets that have accrued the same cumulative gross return since purchase. My second theoretical contribution is to characterize trading rules that determine which assets should be sold today once future trading opportunities are taken into account.

The two theoretical contributions overcome two obstacles that have long constrained the literature on portfolio choice under realization taxation. The first is computational. Realization taxation requires tracking both market value and tax basis for every asset, while choosing consumption, liquidation, and reinvestment over many periods. The resulting state and choice spaces quickly become too large to handle with conventional numerical methods. The second is theoretical. With finitely many assets, trades motivated by risk and taxes compete: a tax-favorable sale can leave the wrong risk exposure, while a risk-motivated rebalance can trigger taxes. A simple rule for which asset to sell first may therefore not exist. These difficulties help explain why analytical characterizations often focus on a single asset

(Constantinides, 1984), multi-asset studies rely on back-testing and trading heuristics (Fischer and Gallmeyer, 2016, Haugh et al., 2016), and life-cycle models generally retain only one or two risky assets (Dammon et al., 2001a, 2004).

My approach is to let the number of assets grow without bound in a portfolio with many small positions. As asset-level risk diversifies, choosing which securities to sell within a tax lot no longer affects the portfolio's risk or expected return. Security-specific risk diversifies away in this limit, but common aggregate shocks remain. Individual assets can therefore lose value even when the diversified portfolio as a whole earns a nonnegative return. The portfolio can then be represented as a distribution of market value across tax lots. Section II establishes this state-reduction result formally, but the intuition is simple: consider two assets with the same expected return. If the portfolio spreads risk across many small positions, selling either asset has a negligible effect on the risk and expected return of the retained portfolio. Their tax consequences, however, depend on their tax lots. Per dollar sold, the more appreciated asset realizes the larger gain and triggers the larger tax payment. Hence, in the large-portfolio limit, if two liquidation rules each sell the same amount from every tax lot, choosing different securities within a lot does not affect consumption or utility. By contrast, shifting sales across tax lots changes realized gains, current taxes, and which embedded losses remain available to shelter future gains. The distribution of asset value across tax lots is therefore the relevant state of a large taxable portfolio. Intuitively, the investor chooses how much to sell from baskets of securities grouped by their accrued returns.

Representing the portfolio this way makes trading operations intuitive. Let an investor hold a portfolio $w(\cdot)$, a distribution of market value over cumulative gross return bins $r > 0$. Because newly purchased assets have gross return one, investment adds a point mass at tax lot $r = 1$. Selling assets subtracts value from the tax lots sold, and period returns transport the end-of-period distribution into the next period. After-tax liquidation value equals the total value sold across tax lots minus taxes, which are calculated lot by lot and summed. This structure accommodates the three main tax systems analyzed in this paper, which differ in how they treat realized losses. Full use of losses (FUL) pays an immediate rebate on realized losses. No use of losses (NUL) taxes realized gains and ignores realized losses. Limited use of losses (LUL) offsets realized gains with realized losses and carries any unused loss forward as a tax credit. Section III develops this structure formally.

Using these ingredients, I define two classes of recursive problems, over both finite and infinite horizons. The first class requires finding the largest constant real withdrawal the portfolio can sustain. Its infinite-horizon version—the perpetuity problem—provides the basis for the taxation measures in this paper. By abstracting from life-cycle mechanisms that make consumption timing specific to each investor, it provides a common benchmark for comparing tax systems. The second class consists of classical time-separable discounted-utility problems, allowing consumption to vary over time. This formulation extends life-cycle models in the spirit of Dammon et al. (2001a) to a diversified continuum of assets. Both classes use the same portfolio state and trading structure. Section IV establishes that these

problems are well defined, that backward iteration converges, and that optimal policies are attained for a broad class of tax systems.

To solve either recursive problem, and hence to construct effective taxation and return alpha measures, I establish what optimal trading actually looks like. The analysis generalizes a rule familiar to investors: selling loss positions and reinvesting the proceeds, then meeting cash needs by selling the least appreciated remaining positions. This prescription goes back to Constantinides (1983, 1984), appears in standard investments curricula (Bodie et al., 2013), and is automated by robo-advisors. Formally, the loss-harvesting cutoff liquidation (LHCL) strategy class harvests a lower interval of losses and funds consumption by selling remaining positions in ascending order of cumulative return. Liquidation stops at a cutoff tax lot once the required cash target is reached. I characterize when this strategy yields optimal trades in Section IV.

To prove optimality, I use arguments that compare trades that fund the same current consumption and ask which trade leaves the investor with a continuation portfolio better suited to fund future consumption. For this comparison under full use of losses taxation, I introduce the right-domination order: one portfolio right-dominates another if it offers at least as much after-tax liquidation value and contains at least as much market value in the right tail, i.e., above every cumulative-return cutoff. The first condition ensures that the dominant portfolio can generate weakly more liquidity in the short run; the latter preserves wealth in the more appreciated tax lots. Keeping these positions invested postpones the largest embedded tax bills. Moreover, additional positive returns translate more directly into after-tax returns on more appreciated positions: a given percentage increase in market value produces a larger percentage increase in after-tax liquidation value.

The recursive argument shows that the continuation value function is increasing in this order: a right-dominant continuation portfolio is worth at least as much to the investor as any dominated one. Thus, among trades that fund the same current consumption, the investor prefers a trade whose continuation portfolio right-dominates that of the alternative.

LHCL strategies produce the required portfolio ordering in two steps. Harvesting losses for reinvestment brings their tax benefits forward and shifts market value from low-return tax lots ($r < 1$) into newly purchased positions ($r = 1$). Raising additional cash from the least appreciated remaining positions preserves market value in the right tail. To guarantee that right dominance survives subsequent returns, the diversified portfolio must earn a nonnegative net period return in every aggregate state. Under this return-support condition, each feasible trade can be matched by an LHCL trade that funds the same consumption and guarantees a right-dominant continuation portfolio. Thus, trades from the LHCL class are optimal under full use of losses and the support condition.

If the return-support condition fails, two portfolios that were ordered immediately after trading need not remain so after returns are drawn. The greater sensitivity of already appreciated positions to additional returns works in reverse when those returns are negative: a given percentage decline in market value produces a larger percentage decline in after-tax

liquidation value for more appreciated positions. This can make selling appreciated positions and retaining losses optimal.

The tax treatment of losses also matters. When realizing losses generates neither refunds nor tax credits (as under NUL), the investor may prefer to keep them invested as a shield against future appreciation. In both cases, LHCL strategies are no longer guaranteed to be optimal. Under full use of losses, reinvestment costs can make harvesting small losses unattractive and shrink the harvest region, but leave LHCL optimality unaffected. Finally, my results extend to limited use of losses with perfect carryforward and frictionless reinvestment, under the same return-support assumptions and a dominance order that also accounts for unused loss credits.

Once the optimal policy is known, effective taxation becomes computable. Section V does this in the setting without aggregate risk, where LHCL strategies are optimal, and studies how the effective rate changes under alternative policy configurations and market conditions. The baseline allows losses to offset gains, with perfect carryforward and frictionless reinvestment.[6] The value of realization discretion then depends on the opportunities to fund consumption while preserving appreciated positions. Greater return dispersion creates more losses and small gains from which the investor can fund consumption, lowering effective taxation. Dividends force taxable payouts, while inflation creates taxable nominal gains without increasing expected real returns. Allowing investors to borrow leaves the effective tax rate unchanged if interest is not deductible, while deductible interest acts like a loss credit and can lower it sharply.

The effective tax rate captures the total value of realization discretion but does not separate its sources. Section V.B uses the return alpha to decompose this advantage into the contributions of single-asset deferral and tax-lot selection. A single diversified fund offers deferral and risk diversification, while holding many assets also allows the investor to select which gains and losses to realize. Expressing both contributions in common annual expected-return units provides a benchmark for interpreting empirical estimates of tax alpha under explicit tax and market assumptions.

Section VI then aggregates the same optimal trading policies in a cohort economy with accidental bequests to study government revenue. Holding the aggregate wealth path fixed across tax systems, I derive the accrual tax rate that reproduces realization-tax revenue along the balanced-growth path. I compare three treatments of gains at death: forgiveness through a step-up in basis, carry-over of basis to heirs, and taxation of remaining gains. These provisions substantially affect collections: at the baseline, step-up yields about 29% less revenue and constructive realization about 34% more revenue than carry-over. Nevertheless, taxing upon realization rather than accrual carries a substantial revenue cost under all three treatments. Even if each individual asset is eventually sold, new unrealized gains replace those realized, sustaining an aggregate stock of untaxed gains and a permanent revenue shortfall.

[6]The numerical benchmark uses a 5% expected real return, 30% return dispersion, 2% inflation, a 30% dividend payout rate, and a 25% statutory tax rate. The investor can carry unused loss credits forward without expiry and reinvest without cost.

In sum, this paper isolates the investor-side trading margin and provides a flexible framework for analyzing the benefits investors derive from the discretion built into realization-based capital gains taxation. To focus on this margin, I hold the return process fixed and study how investors choose among tax lots over time.[7] The large-portfolio result assumes that assets share a common expected return and that security-specific risk diversifies across many balanced positions (Online Appendix Section III.A.2 extends the model to heterogeneous, nondiversifiable factor exposures when the distribution of factor loadings is independent of current tax lots). The framework therefore applies most directly to investors with broad taxable portfolios and access to tax-aware trading. Belda (2026) studies the complementary asset-pricing margin in an equilibrium with a single risky asset. The framework also provides a foundation for studying life-cycle consumption and portfolio choice with many risky assets. In follow-up work, Bennhoff and Scheuer (2026) use it to analyze a quantitative life-cycle model with many assets and aggregate risk, comparing alternative treatments of capital gains at death.

Related Literature

This paper studies investors whose taxable portfolios are broad enough to make tax-lot selection a recurring economic choice. A recent literature documents that such investors matter. Capital gains are central to the income and taxation of the very rich (Piketty et al., 2023, Saez and Zucman, 2021, Scheuer and Slemrod, 2020, 2021). Bach et al. (2020) and Fagereng et al. (2020) document persistent heterogeneity in returns to wealth. Balloch and Richers (2023) study the portfolios of wealthy households, while Gabaix et al. (2023) document their security-level holdings, flows, and rebalancing. Msall and Næss (2026) show that gains held until death have direct fiscal consequences through their treatment at death. How investors should trade such portfolios is the question taken up by the portfolio-choice literature on realization taxation.

The modern portfolio-choice literature on realization taxation begins with Constantinides (1983, 1984). Constantinides (1983) derives the canonical logic of realizing losses and deferring gains in an environment in which investors can offset appreciated positions without realizing them.[8] Constantinides (1984) studies differential taxation of short- and long-term gains and the option to reset an asset's holding period. Dybvig and Koo (1996) formulate a dynamic exact-basis portfolio problem and solve numerical examples with small portfolios. DeMiguel and Uppal (2005) solve terminal-wealth problems with exact basis for short-sale-constrained investors holding a small number of risky assets. Other contributions develop numerical methods for richer portfolios and tax environments. Haugh et al. (2016) analyze dynamic tax-

[7] Embedded gains are primarily investor-specific state variables because they depend on individual purchase histories. They therefore need not aggregate into an economy-wide state. Their first-order effects on equilibrium asset prices are not mechanically determined by the portfolio problem studied here.

[8] Constantinides' many-asset characterization permits "shorting against the box": short selling an appreciated asset for liquidity while retaining the long position and its deferred gain. I exclude this transaction because it separates liquidity from tax realization. Tax codes may also neutralize it; in the United States, for example, IRC §1259 generally treats it as a constructive sale.

aware policies under limited loss offsets, Fischer and Gallmeyer (2016) study heuristic trading rules with transaction costs, and Moehle et al. (2021) incorporate exact tax lots and wash-sale restrictions into a convex-optimization approach. These papers substantially broaden the class of tax-aware portfolio problems that can be solved numerically.

A larger life-cycle literature jointly studies consumption, investment, and tax-basis dynamics. Dammon et al. (2001a,b) analyze portfolios with one or two risky assets, and Dammon et al. (2004) add tax-deferred accounts and asset-location choices. Fischer et al. (2013) find small welfare gains from tax-optimal trading over simpler policies in a life-cycle model with one risky asset and labor income. My quantitative results, however, show that optimal realization choices can substantially reduce effective taxation and raise sustainable portfolio consumption, highlighting the importance of these incentives for the richest households, whose resources depend heavily on financial wealth. This branch connects to the earlier asset-location analyses of Auerbach and King (1983), Black (1980), Tepper (1981), and to the later work of Garlappi and Huang (2006), Huang (2008). Later work isolates particular features of the tax environment: short-selling costs, wash-sale constraints, limits on loss rebates, loss carryovers, tax-deferred accounts, transaction costs, and year-end taxation.[9] Exact-basis models must track asset- and lot-specific state variables, so the state expands rapidly with the number of assets and purchase histories. This helps explain why explicit life-cycle models generally remain limited to one or two risky assets. This paper replaces asset-specific basis histories with a distribution of value across tax lots and uses that distribution to characterize long-horizon trading. Bennhoff and Scheuer (2026) build on the Tax Portfolio Model to study a quantitative life-cycle problem with many assets and aggregate risk and compare alternative treatments of capital gains at death.

The state representation also connects to theories of large portfolios and economies represented by distributions. Chamberlain and Rothschild (1983), Ross (1976) establish approximate factor structures under which security-specific risk can be diversified. Al-Najjar (1998) develops factor pricing in a large asset economy, while Khan and Sun (1997, 2003) extend arbitrage pricing to economies with an asset continuum. In these theories, systematic exposures remain relevant for pricing. This paper applies their large-portfolio logic under common expected returns and diversifiable residual risk.

A separate public-finance literature studies realization responses, effective capital-gains taxation, and government revenue. Poterba (2002) surveys how taxation affects household portfolio behavior, Alan et al. (2010) document portfolio responses, and Agersnap and Zidar (2021) estimate realization elasticities and revenue-maximizing tax rates. Ehling et al. (2026) directly connect optimized portfolio choice to aggregate realizations and revenue by simulating investors under historical returns and capital-gains tax rates. The empirical and reduced-form branches of this literature generally study how total realizations respond

[9] Gallmeyer et al. (2006) study two risky assets and costly short selling; Jensen and Marekwica (2011) analyze wash-sale constraints; Marekwica (2012) limits rebates on realized losses; Ehling et al. (2018) introduce carryover losses; Fischer and Gallmeyer (2017) combine limited loss use with taxable and tax-deferred accounts; and Dai et al. (2026) study transaction costs and year-end taxation.

without deriving which assets investors optimally sell. This paper supplies that position-selection mechanism. For the revenue comparison, the cohort version of my model draws on Yaari's (1965) treatment of uncertain lifetimes and Blanchard's (1985) cohort structure with a constant mortality rate. I use this demographic structure to aggregate the tax payments generated by the model's optimal perpetuity policies in an economy with accidental bequests.

An effective-tax literature translates realization taxation into an accrual-equivalent rate. Glenday and Davies (1990), Poterba (1989), Protopapadakis (1983) infer such rates from accrued and realized gains or by equating terminal wealth, present values, or internal rates of return; Glenday and Davies (1990) also distinguish investor-side equivalence from revenue equivalence. The effective tax rate and closed-cohort rate extend these two criteria to an optimally managed many-asset perpetuity. A finance literature expresses the benefits of tax-aware trading in annual-return units: Chaudhuri et al. (2020) estimate a historical tax-loss-harvesting alpha similar in magnitude to my baseline total alpha, while Mainardi (2026) measures tax alpha from differences in household tax expenses. The return alphas developed here likewise translate realization benefits into expected-return units, but decompose the total alpha into single-asset deferral and an additive contribution from tax-lot selection. The total alpha is dual to the effective tax rate, so the two measures connect the effective-rate language of public finance to the alpha language of finance within the same optimal-trading problem.

II Tax Motives Dominate in Large Portfolios

This section formalizes and proves the claim that in large portfolios studied in this paper, liquidation decisions are governed at first order by tax considerations rather than by security-specific risk management. I show that, as the number of assets grows, the distribution of portfolio value across *tax lots*—positions grouped by the same gross cumulative return—becomes the only relevant state variable for the limiting portfolio problem.

Consider a two-period investor who chooses which assets to liquidate from a portfolio of n securities. Denote by $\boldsymbol{\alpha}^n = (\alpha_1^n, \ldots, \alpha_n^n) \in [0,1]^n$ the vector of liquidation shares, where α_i^n is the fraction of asset i's current market value liquidated at date 1. The investor has expected utility

$$U^n(\boldsymbol{\alpha}^n) := \mathbb{E}\Big[u\big(c_1^n(\boldsymbol{\alpha}^n), c_2^n(\boldsymbol{\alpha}^n; Z, \boldsymbol{R}^n)\big)\Big], \tag{1}$$

Utility is defined over date-1 and date-2 consumption, c_1^n and c_2^n. Both depend on the date-1 liquidation choice $\boldsymbol{\alpha}^n$, while c_2^n also depends on the aggregate state Z and the vector of asset-level returns $\boldsymbol{R}^n$. The function u is continuous, increasing, concave, and continuously differentiable. At date 1, the investor is endowed with n securities with current market values $\omega_i^n \geq 0$, collected in $\boldsymbol{\omega}^n$, which amount to total market value $\sum_{i=1}^n \omega_i^n = M$. Each security i also has an exogenously given cumulative gross return $r_i > 0$ since purchase, so its cost basis is ω_i^n / r_i. Assets with the same accrued return carry the same embedded gain per unit of market value and are therefore in the same tax lot.

The liquidated market value, realized gain, and date-1 consumption are

$$\mathrm{M}_1^n(\boldsymbol{\alpha}^n) := \sum_{i=1}^n \alpha_i^n \omega_i^n, \qquad \mathrm{CG}_1^n(\boldsymbol{\alpha}^n) := \sum_{i=1}^n \alpha_i^n \omega_i^n \Big(1 - \frac{1}{r_i}\Big), \qquad c_1^n = \mathrm{M}_1^n - \mathrm{T}_1(\mathrm{CG}_1^n),$$

whereby date-1 capital gains taxes are determined by the continuous map T_1. This formulation takes realized gains as the primitive tax base. The map T_1 summarizes date-1 taxation, while the date-2 map T_2 will allow taxes to depend both on current realized gains and on gains realized at date 1.

The unsold shares are liquidated at date 2. Between dates 1 and 2, each asset earns a positive gross return R_i^n. Asset returns have a common finite mean $\mathbb{E}[R_i^n] = 1 + \mu > 1$ for every $i \leq n$, but their marginal distributions and variances may otherwise differ. Asset-specific variances and cross-asset dependence are summarized by the positive semidefinite covariance matrix Σ^n. In addition, an independent aggregate factor $Z \sim f_Z$ with $\mathbb{E}[Z] = 1$ and $\mathbb{E}[Z^2] < \infty$ is common to all assets and cannot be diversified away. Date-2 liquidated market value and realized gain are therefore

$$\mathrm{M}_2^n(\boldsymbol{\alpha}^n; Z, \boldsymbol{R}^n) := \sum_{i=1}^n (1 - \alpha_i^n)\omega_i^n Z R_i^n, \qquad \mathrm{CG}_2^n(\boldsymbol{\alpha}^n; Z, \boldsymbol{R}^n) := \mathrm{M}_2^n - \sum_{i=1}^n \frac{(1-\alpha_i^n)\omega_i^n}{r_i}.$$

Given continuous date-2 taxes $\mathrm{T}_2(\mathrm{CG}_2, \mathrm{CG}_1)$, future consumption is

$$c_2^n = \mathrm{M}_2^n - \mathrm{T}_2(\mathrm{CG}_2^n, \mathrm{CG}_1^n).$$

Random variates and taxes are assumed well-behaved, such that date-2 tax payments are bounded *a.s.* and that the tax levied on a realization never exceeds the asset's market value.

Now consider a sequence of such utility-maximization problems indexed by n, in which the investor gains access to an ever larger asset universe while the total market value of her endowment remains fixed at M. For simplicity, assume that the set of embedded cumulative returns that appears along the sequence, $\mathcal{R} := \{r_i : i \in \mathbb{N}\}$, is finite (Section III removes this requirement). Also assume that, for every $r \in \mathcal{R}$, the total market value invested in lots indexed by r converges to some number $w(r) \geq 0$.[10] For each r, define the function $w : \mathcal{R} \to \mathbb{R}_+$ via

$$w(r) := \lim_{n\to\infty} \sum_{i \in I^n(r)} \omega_i^n, \qquad I^n(r) := \{i \leq n : r_i = r\}.$$

Thus, w is the limiting distribution of value across tax lots, and $w(r)$ is the value of all lots whose embedded cumulative return equals r. By construction, the total market value is $M = \sum_{r\in\mathcal{R}} \sum_{i\in I^n(r)} \omega_i^n$, and therefore, in the limit, $M = \sum_{r\in\mathcal{R}} w(r)$.

Two regularity conditions ensure that the expanding asset universe indeed diversifies idiosyncratic risk. First, *balancedness* requires that no single position dominates: call the triangular array $\{\boldsymbol{\omega}^n\}_{n\in\mathbb{N}}$ balanced if $\max_{i\leq n} \omega_i^n / M \to 0$. As more securities become available, the portfolio broadens with them rather than remaining concentrated in a few positions. Second, *diversifiability* requires that the new assets represent genuinely new securities: call the covariance sequence $\{\Sigma^n\}_{n\in\mathbb{N}}$ diversifiable if the corresponding sequence of largest eigen-

[10] This is a well-posedness condition for the limiting portfolio problem: although each individual position becomes negligible, new positions are added as n grows, so one requires that the lot-level sums settle down.

values is uniformly bounded. This rules out adding assets whose variance explodes with n, or assets that become nearly perfectly correlated with the existing ones, as would happen if asset n were a mere linear combination of existing ones.[11]

The next theorem formalizes this reduction. Two investors whose portfolios share the same asymptotic tax-lot distribution $w(r)$ achieve asymptotically equivalent consumption and utility if their liquidations generate the same aggregate liquidated market value $\mathrm{M}_1 := \lim \mathrm{M}_1^n$ and realized gain $\mathrm{CG}_1 := \lim \mathrm{CG}_1^n$. Which combinations of M_1 and CG_1 can be achieved depends on how much value sits in the lots at each r, making the function w the relevant portfolio state. However, neither the security-level composition of the portfolio nor the identity of the liquidated assets nor the covariance structure among returns matters in the limit. Aggregate risk remains present, since "perfect diversification" here refers only to asset-level residual risk.

Theorem 1 (Asymptotic Sufficiency of the Tax-Lot Distribution)**.** *Consider two balanced and diversifiable sequential portfolio problems with endowments and covariance matrices $\{(\boldsymbol{\omega}^n, \Sigma^n)\}$ and $\{(\bar{\boldsymbol{\omega}}^n, \bar{\Sigma}^n)\}$. Let their return vectors $\{\boldsymbol{R}^n\}$ and $\{\bar{\boldsymbol{R}}^n\}$ satisfy $\mathbb{E}[\boldsymbol{R}^n] = \mathbb{E}[\bar{\boldsymbol{R}}^n] = (1+\mu)\mathbf{1}_n$ for every n and some common $\mu > 0$; component distributions and covariance structures may otherwise differ across assets and between the two portfolio problems. Suppose $\{\boldsymbol{\omega}^n\}$ and $\{\bar{\boldsymbol{\omega}}^n\}$ have the same limiting tax-lot distribution w and let $\{\boldsymbol{\alpha}^n\}$ and $\{\bar{\boldsymbol{\alpha}}^n\}$ be any two sequences of liquidation shares such that, for every $r \in \mathcal{R}$, the lot-level liquidation limits exist:*

$$w_\alpha(r) := \lim_{n\to\infty} \sum_{i \in I^n(r)} \alpha_i^n \omega_i^n, \qquad \bar{w}_{\bar{\alpha}}(r) := \lim_{n\to\infty} \sum_{i \in I^n(r)} \bar{\alpha}_i^n \bar{\omega}_i^n.$$

Let (c_1^n, c_2^n, U^n) and $(\bar{c}_1^n, \bar{c}_2^n, \bar{U}^n)$ denote consumption and utility under $(\{\boldsymbol{\alpha}^n\}, \boldsymbol{\omega}^n, \Sigma^n)$ and $(\{\bar{\boldsymbol{\alpha}}^n\}, \bar{\boldsymbol{\omega}}^n, \bar{\Sigma}^n)$, respectively. Then, i) implies ii):

i) The limiting liquidated market value and realized gain coincide:

$$\sum_{r\in\mathcal{R}} w_\alpha(r) = \mathrm{M}_1 = \sum_{r\in\mathcal{R}} \bar{w}_{\bar{\alpha}}(r), \qquad \sum_{r\in\mathcal{R}} w_\alpha(r)\big(1-\tfrac{1}{r}\big) = \mathrm{CG}_1 = \sum_{r\in\mathcal{R}} \bar{w}_{\bar{\alpha}}(r)\big(1-\tfrac{1}{r}\big).$$

ii) The two sequences are asymptotically equivalent in consumption and utility:

$$\lim_{n\to\infty} |c_1^n - \bar{c}_1^n| = 0, \quad \operatorname*{plim}_{n\to\infty} |c_2^n - \bar{c}_2^n| \to 0, \quad \lim_{n\to\infty} \left|U^n - \bar{U}^n\right| = 0.$$

All proofs are provided in the Online Appendix.

A transparent way to implement the theorem's matching conditions on $\boldsymbol{\alpha}^n$ and $\bar{\boldsymbol{\alpha}}^n$ is to choose the same limiting liquidation $\tilde{w}(r) \in [0, w(r)]$ from each tax lot. Because both portfolios converge to the same distribution w, any such lot-level liquidation feasible for one investor is feasible for the other. It leaves both with the same post-liquidation distribution $w - \tilde{w}$, and therefore with the same current proceeds, realized gains, and retained tax basis. Conditional on the aggregate state, security-specific return differences diversify away, so the portfolios have the same limiting date-2 market value; if all remaining assets are then liquidated, their taxable gains and consumption coincide as well. This construction also provides the formal

[11] One could weaken the technical condition by allowing the largest eigenvalue to grow slowly with n; the boundedness assumption keeps the statement transparent. Balancedness and diversifiability are analogous to the conditions ensuring well-diversified portfolios in Ross (1976) and the approximate factor structure of Chamberlain and Rothschild (1983).

foundation for the *Tax Portfolio Model* developed in the next section. Under a common return density, identical post-liquidation distributions evolve into identical next-period distributions. Matching subsequent liquidations lot by lot can therefore match realized gains and portfolio states at every later date. The distribution $w(\cdot)$ and the liquidation function $\tilde{w}(\cdot) \leq w(\cdot)$ are thus the natural state and control objects of the Tax Portfolio Model.

Theorem 1 is an asymptotic equivalence and is silent on how fast security-level selection becomes irrelevant. The next corollary supplies the rate: reshuffling *which* securities are sold within tax lots moves consumption and expected utility at rate $n^{-1/2}$, while moving liquidation *across* tax lots changes tax payments by $O(1)$. This is the quantitative content of the claim that tax motives dominate in large portfolios.

Corollary 1 ($\sqrt{n}$-decay of security-selection effects)**.** *In the setting of Theorem 1, fix one endowment sequence $\{(\boldsymbol{\omega}^n, \Sigma^n)\}$, strengthen balancedness to $\max_{i \leq n} \omega_i^n \leq CM/n$ for some $C \geq 1$, and bound the largest covariance eigenvalues by $\bar{\lambda} := \sup_n \lambda_{\max}(\Sigma^n) < \infty$. Let taxes be linear, $\mathrm{T}_t(\mathrm{CG}) = \tau\,\mathrm{CG}$, and let $\{\boldsymbol{\alpha}^n\}$ and $\{\bar{\boldsymbol{\alpha}}^n\}$ be two liquidation sequences with identical lot-level liquidations at every n:*

$$\sum_{i \in I^n(r)} \alpha_i^n \omega_i^n = \sum_{i \in I^n(r)} \bar{\alpha}_i^n \omega_i^n \qquad \text{for all } r \in \mathcal{R}.$$

Define the relative date-2 consumption gap by $g_2^n := (\bar{c}_2^n - c_2^n)/M$. Then $c_1^n = \bar{c}_1^n$, $\mathbb{E}[g_2^n \mid Z] = 0$, and

$$\mathrm{sd}(g_2^n) \;\leq\; (1-\tau)\sqrt{\bar{\lambda}\, C\, \mathbb{E}[Z^2]}\; \frac{1}{\sqrt{n}}.$$

If, in addition, marginal utility in the second argument is bounded by L, then $|U^n - \bar{U}^n| \leq LM\mathbb{E}|g_2^n| \leq LM\,\mathrm{sd}(g_2^n) = O(n^{-1/2})$.

The bound implies small residual security-selection risk even in moderately sized portfolios. Consider an equally weighted portfolio of 100 assets with independent Gaussian residual returns of 30% volatility, no aggregate shock, and a 25% tax rate. For two trading rules implementing the same tax-lot liquidations, the standard deviation of their consumption difference is at most 2.25% of initial portfolio wealth. To express the remaining security-selection risk in consumption units, take proportional liquidation within each tax lot as the benchmark: sell the same fraction of every security within a given lot. Under these assumptions, this minimizes consumption variance among trading rules implementing the same tax-lot liquidations. With CARA utility over date-2 consumption and absolute risk aversion $a = 5/M$, the certainty-equivalent consumption loss from using another rule is at most 0.13% of initial portfolio wealth for 100 assets and 0.03% for 500 assets.[12]

In sum, large portfolios permit us to ignore the identities of the assets, as long as we keep track of their distribution across tax lots. The preceding bound suggests that this approximation can also be useful for moderately sized portfolios. Under the common-expected-return assumption, historical embedded returns contain no information about

[12] Let $\bar{c}_2^n$ denote consumption under proportional liquidation. Liquidation differences sum to zero within each tax lot, so independent residual returns with identical variances imply $\mathrm{Cov}(g_2^n, \bar{c}_2^n) = 0$. Joint Gaussianity then makes the consumption gap independent of benchmark consumption. Under $u_2(c) = -e^{-ac}$, the certainty-equivalent loss, expressed as a share of initial wealth, is therefore $\frac{aM}{2}\,\mathrm{var}(g_2^n) \leq 2.5\,\frac{0.225^2}{n}$. This gives 0.13% at $n = 100$ and 0.03% at $n = 500$. The choice $aM = 5$ corresponds to relative risk aversion of 5 at consumption M.

expected future performance, while differences in marginal return distributions and variances diversify away (provided the covariance sequence remains diversifiable). Asset identity can matter again when expected returns or nondiversifiable factor exposures are heterogeneous. Online Appendix Section III.A.2 considers an extension with heterogeneous factor exposures.

Section III now uses the insights of Theorem 1 to build a model of tradable, taxable, and fully diversified portfolios.

III The Tax Portfolio Model

In this section, I introduce the core objects of the *Tax Portfolio Model* (TPM): portfolios, liquidation choices, tax operators, and the portfolio law of motion. I then show how aggregate risk fits into the framework. For the remainder of the paper, unless stated otherwise, idiosyncratic period returns have a common and continuous marginal density f on support $D \subseteq \mathbb{R}_{>0}$.[13]

III.A Portfolios, Trading, and Taxation

Assets, Returns and Portfolios. Let ω_i denote the current market value of asset $i \in [0,1]$ and $\breve{\omega}_i$ its market value at purchase. The cumulative gross return of asset i is $r(i) := \omega_i/\breve{\omega}_i \in (0,\infty)$. The common marginal return $R \sim f$ satisfies $\mathbb{E}[R] - 1 = \mu \geq 0$, while dependence across assets is mostly unrestricted; see Section II. Theorem 1 allows me to suppress asset identities and describe a portfolio by its distribution of market value across tax lots. I represent this distribution by a nonnegative, Lebesgue-integrable function $w : (0,\infty) \to \mathbb{R}_{\geq 0}$, and denote the set of portfolios by $\mathcal{W}$. Whenever I take a portfolio as the endowment for an optimization problem, I assume that its market value, liquidation value, and tax liability, defined below, are finite.

In lot-level examples, $\boldsymbol{\delta}_r$ denotes a Dirac portfolio that places one unit of market value in the tax lot defined by accrued return r. Note that the tax lot represented by $\boldsymbol{\delta}_r$ still aggregates infinitely many underlying assets with the same accrued return.

The quantity $w(r)\ \mathrm{d}r$ represents the market value held in tax lots with cumulative gross returns between r and $r + \mathrm{d}r$. Integrating over these lots, $\int_0^{\bar{r}} w(s)\ \mathrm{d}s$, gives the value of all assets with cumulative gross returns at most $\bar{r}$. Setting $\bar{r} = \infty$ yields the market value of the entire portfolio. Dividing $w(r)$ by the cumulative gross return r converts market value into its corresponding cost basis. Integrating $w(r)/r$ over tax lots gives the total cost basis, and subtracting this from market value gives the embedded net capital gain. These three portfolio statistics are

$$\mathrm{M}(w) := \int_0^\infty w(r)\ \mathrm{d}r, \qquad \mathrm{B}(w) := \int_0^\infty \frac{w(r)}{r}\ \mathrm{d}r, \qquad \mathrm{CG}(w) := \mathrm{M}(w) - \mathrm{B}(w).$$

For example, the portfolio $w = 2\boldsymbol{\delta}_{0.8} + \boldsymbol{\delta}_1 + 3\boldsymbol{\delta}_{1.5}$ has market value 6, basis $2/0.8 + 1 + 3/1.5 = 5.5$, and an embedded net gain of 0.5. Thus, the coefficients on the Dirac lots are dollar market

[13] In two-period problems, it is enough that investors lack asset-level information that would let them select securities with systematically different marginal return distributions.

values, while their subscripts carry the tax-basis information.

Asset Sales and Purchases. Investors generate liquidity by choosing which parts of their portfolio to sell. A feasible liquidation portfolio $\widetilde{w}$ cannot contain negative sales or exceed the holdings in any tax lot. Thus,

$$\widetilde{w} \in [0, w], \qquad [0, w] := \{v \in \mathcal{W} : 0 \leq v(r) \leq w(r) \text{ for every } r > 0\}.$$

This set of feasible liquidations is the order interval between the zero portfolio and w. The unsold portion $w - \widetilde{w} \in \mathcal{W}$ is the post-liquidation portfolio.

Returning to this example, selling $\widetilde{w} = \boldsymbol{\delta}_{0.8} + \boldsymbol{\delta}_{1.5}$ from w raises $\mathrm{M}(\widetilde{w}) = 2$ dollars before tax and realizes the net gain $\mathrm{CG}(\widetilde{w}) = 2 - (1/0.8 + 1/1.5) = 1/12$. Panel (a) of Figure 1 shows this liquidation: gold denotes the portions sold, while burgundy denotes the post-liquidation portfolio $w - \widetilde{w} = \boldsymbol{\delta}_{0.8} + \boldsymbol{\delta}_1 + 2\boldsymbol{\delta}_{1.5}$.

To compute the liquidity generated by a sale, one must know the taxes it triggers. Start with the simple FUL-case of a constant statutory rate $\tau \in (0, 1)$. Selling market value $\widetilde{w}(r)$ from the lots at r triggers tax

$$\tau\big(\widetilde{w}(r) - \widetilde{w}(r)/r\big)\,\mathrm{d}r = t_{\mathrm{FUL}}(r)\widetilde{w}(r)\,\mathrm{d}r,$$

where $t_{\mathrm{FUL}}(r) = \tau\frac{r-1}{r}$ is the tax share per unit sold from a lot at r. Integrating over lots yields total tax liability, $\mathrm{T}_{\mathrm{FUL}}(\widetilde{w}) = \int t_{\mathrm{FUL}}(r)\widetilde{w}(r)\,\mathrm{d}r = \tau\,\mathrm{CG}(\widetilde{w})$. More generally, one can take either a tax share function $t : r \mapsto t(r) < 1$ satisfying the no-tax-at-zero-gain condition $t(1) = 0$, or the tax operator T itself, as primitive. Definition 1 introduces this object together with liquidation value.

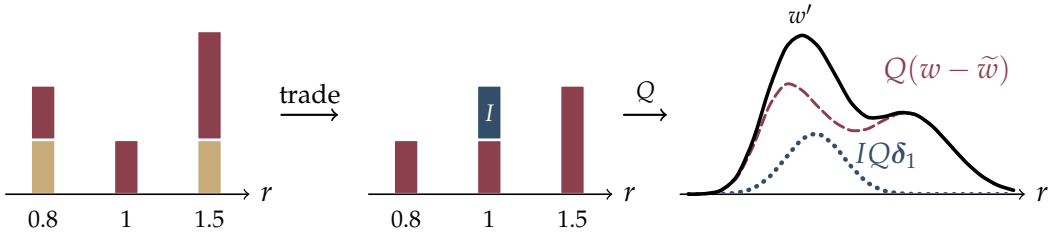
Definition 1 (Tax Liability and Liquidation Value)**.** *A tax operator* T *is a linear functional which maps a portfolio* w *to its 'tax liability'* $\mathrm{T}(w)$*. The functional* $\mathrm{K} := \mathrm{M} - \mathrm{T}$ *yields the 'liquidation value'* $\mathrm{K}(w)$*. Additionally, if* T *is defined through a tax share function,* $t(r)$ *with* $t(1) = 0$*, then* $k(r) = 1 - t(r)$ *is the 'payoff function' and*

$$\mathrm{T}(w) := \int_0^\infty w(r)t(r)\,\mathrm{d}r, \qquad \mathrm{K}(w) = \int_0^\infty w(r)k(r)\,\mathrm{d}r. \tag{2}$$

The integral representation in (2) covers the canonical tax systems that operate through lot-specific tax shares such as FUL and NUL (no use of losses). Later, when I turn to LUL, the relevant tax object can no longer be written in this simple way, and T will be taken as primitive, with a second argument for potential carry-forwards.

Generally, the investor's problem is to choose a liquidation portfolio $\widetilde{w}$ that delivers the required current liquidity $\mathrm{K}(\widetilde{w})$ while leaving an optimal post-liquidation portfolio $w - \widetilde{w}$. The current sale determines taxes and cash today through $\mathrm{K}(\widetilde{w})$, while the residual market value $\mathrm{M}(w - \widetilde{w})$ determines how much principal is carried into the next period. Only in some cases does solving this problem amount to maximizing $\mathrm{M}(w - \widetilde{w})$ subject to a liquidity target.

In addition to selling assets, an investor can purchase them. Investment adds value to the lot at $r = 1$, where acquisition price and tax basis coincide. Formally, investing $I > 0$ adds a Dirac mass $(\delta_I I)\boldsymbol{\delta}_1$ to w, where $\delta_I \in [0, 1]$ is an investment friction. The law of motion then



(a) choose a sale (b) retain and invest at $r = 1$ (c) derived next-period state

Fig. 1. TPM mechanics for $w = 2\delta_{0.8} + \delta_1 + 3\delta_{1.5}$. Panel (a) selects the liquidation portfolio $\tilde{w} = \delta_{0.8} + \delta_{1.5}$, panel (b) adds investment $I = 1$ at $r = 1$, and panel (c) applies the portfolio law of motion. Burgundy denotes retained value, gold sold value, and navy new investment.

diffuses the point mass into the next-period portfolio. Liquidation plus (re-)investment is the only way to shift value across lots.[14] In the running example, frictionless investment of $I = 1$ adds δ_1 to the post-liquidation portfolio. Panel (b) of Figure 1 shows the resulting post-trade portfolio, with the new investment in navy at $r = 1$.

Given a portfolio w and exogenous liquid resources $y \geq 0$, a feasible control consists of a liquidation portfolio $\tilde{w} \in [0, w]$ and a nonnegative investment I that does not exceed available liquidity. The feasible-control correspondence is thus

$$\Gamma(w) := \{(\tilde{w}, I) : \tilde{w} \in [0, w], \quad 0 \leq I \leq y + \mathrm{K}(\tilde{w})\}.$$

A trading-strategy class specifies which feasible controls are allowed at each portfolio: it is a correspondence Γ^S satisfying $\Gamma^S(w) \subseteq \Gamma(w)$ for every $w \in \mathcal{W}$.

A natural generalization of the canonical trading strategy in Constantinides (1983) is to harvest losses up to a threshold and, if additional liquidity is required, sell the least appreciated remaining tax lots. I call these loss-harvesting cutoff-liquidation (LHCL) strategies. I additionally define strict LHCL (sLHCL) strategies as those with no "round trips": there are no simultaneous investments and realizations of positive gains.

Definition 2 (LHCL and sLHCL Strategies). *Fix a portfolio w and a harvest interval $(0, h]$, $h \geq 0$. For a cutoff $a \in [h, \infty]$, current consumption under liquidation $w\mathbf{1}_{(0,a]}$ is $y + \mathrm{K}(w\mathbf{1}_{(0,a]}) - I$. The LHCL correspondence indexed by h is*

$$\Gamma_h^L(w) := \{(w\mathbf{1}_{(0,a]}, I) \in \Gamma(w) : a \in [h, \infty]\}.$$

The sLHCL strategy class imposes the no-roundtrip restriction

$$\Gamma_h^{sL}(w) := \{(w\mathbf{1}_{(0,a]}, I) \in \Gamma_h^L(w) : a > h \Rightarrow I = 0\}.$$

The Portfolio Law of Motion. A law of motion maps today's liquidation and investment choices into tomorrow's portfolio, w'. Fix a post-liquidation portfolio $w^p \in \mathcal{W}$. Value currently in tax lot s is scaled by the realized period return R and therefore appears tomorrow in tax lot sR. Averaging over the common marginal return distribution yields the next-period tax-lot

[14] Note that, in a finite-asset description, repeated purchases of the same security would also raise lot-identification rules such as FIFO or LIFO. These constraints do not bind in the TPM because latent asset richness lets sold and retained positions be treated as marginal slices of a large asset pool.

distribution. Before adding new investment or aggregate risk, the value of all assets with total return at most r in the next period is

$$\int_0^\infty \int_0^{r/R} w^p(s) R \, \mathrm{d}s \, f(R) \, \mathrm{d}R = \int_0^r (Q w^p)(t) \, \mathrm{d}t,$$

where Q is the transition operator. Differentiating gives the pointwise law of motion

$$Q w^p(r) = \int_0^\infty w^p\left(\frac{r}{R}\right) f(R) \, \mathrm{d}R.$$

Applied to new investment, Q gives $I\delta_I \chi(r)$, where $\chi(r) := Q\boldsymbol{\delta}_1(r) = rf(r)$. Definition 3 collects Q, introduces the additional aggregate-risk operator A_z, and combines liquidation, investment, and aggregate risk in the transition maps G_z. Aggregate risk causes all assets to appreciate by the same factor and thus shifts mass uniformly between tax lots.

Definition 3 (Portfolio Law of Motion). *For any portfolio $w \in \mathcal{W}$, define*

$$Qw(r) := \int_0^\infty w(r/R) f(R) \; \mathrm{d}R.$$

For aggregate risk Z with finite support $\mathcal{Z} \subset (0, \infty)$, write $p_z := \mathbb{P}(Z = z)$ and define $A_z : \mathcal{W} \to \mathcal{W}$ by $A_z w(r) := w(r/z)$. Without aggregate risk, take $\mathcal{Z} = \{1\}$ and drop z-subscripts. Given $\widetilde{w} \in [0, w]$ and $I \geq 0$, the next-period portfolio conditional on $z \in \mathcal{Z}$ is

$$G_z(w, \widetilde{w}, I) := A_z\big(Q(w - \widetilde{w}) + I\delta_I \chi\big); \qquad \chi(r) := rf(r) = Q\boldsymbol{\delta}_1(r). \tag{3}$$

Panel (c) of Figure 1 completes the example. Without aggregate risk and with frictionless investment, the next-period state is $w' = Q(w - \widetilde{w}) + IQ\boldsymbol{\delta}_1$. The dashed burgundy curve is the transitioned post-liquidation portfolio $Q(w - \widetilde{w})$, the dotted navy curve is the transitioned investment $IQ\boldsymbol{\delta}_1$, and the solid black curve is their sum w'.

Thus, the TPM reduces security-level dynamics to linear operators on tax-lot distributions: diffuse the post-liquidation portfolio across lots according to f, add the contribution of new purchases, and scale lot returns by the aggregate realization. This yields closed-form laws of motion for market value and liquidation value under FUL taxation, shown in Lemma 1.

Lemma 1. *Consider $\mathrm{T}_{\mathrm{FUL}}$ with statutory rate $\tau \in [0, 1]$. Given any portfolio w, liquidation portfolio $\widetilde{w} \in [0, w]$ and investment $I \geq 0$, set $w^p = w - \widetilde{w}$. Conditional on aggregate realization $z \in \mathcal{Z}$, write $w'_z := G_z(w, \widetilde{w}, I)$. The laws of motion for market value, basis, capital gains, and liquidation value are*

$$\mathrm{M}(w'_z) = z\mathbb{E}[R](\mathrm{M}(w^p) + \delta_I I), \qquad \mathrm{B}(w'_z) = \mathrm{B}(w^p) + \delta_I I,$$

$$\mathrm{CG}(w'_z) = \mathrm{M}(w'_z) - \mathrm{B}(w'_z), \quad \textit{and} \tag{4}$$

$$\mathrm{K}_{\mathrm{FUL}}(w'_z) = \mathrm{M}(w'_z) - \tau \, \mathrm{CG}(w'_z). \tag{5}$$

Intuitively, post-liquidation market value grows at rate $\mathbb{E}[R]$ and is then scaled by the realized aggregate shock z, while the cost basis changes only through sales and new purchases. Capital gains are the difference between the two. The identities in (4) do not depend on the tax system; (5) additionally uses the linearity of $\mathrm{T}_{\mathrm{FUL}}$.

Three Tax Operators. I now define three tax systems within this framework: i) full-use-of-losses (FUL) taxation $\mathrm{T}_{\mathrm{FUL}}$, ii) taxation without use of losses (NUL) $\mathrm{T}_{\mathrm{NUL}}$, and iii) limited-use-of-losses (LUL) taxation $\mathrm{T}_{\mathrm{LUL}}$. i) is the simplest of all realization-based tax systems,

where every capital gain $w(r) - w(r)/r$ is taxed at a constant rate, τ. This case gives rise to a tax operator $\mathrm{T}_{\mathrm{FUL}}$ with $t_{\mathrm{FUL}}(r) = \tau\frac{r-1}{r}$ (recall Definition 1 and the preceding example). The associated payoff share is $k_{\mathrm{FUL}}(r) := 1 - \tau + \tau/r$, so $\mathrm{K}_{\mathrm{FUL}}(w) = \int_0^\infty w(r)k_{\mathrm{FUL}}(r)\,\mathrm{d}r$. Importantly, a negative capital gain leads to a negative tax. This fact is sometimes recast as a feature, with authors claiming that the negative capital gains tax reflects a tax offset 'elsewhere'. However, the TPM is meant to model all assets subject to capital gains taxation jointly. Hence, unless capital losses can be offset against other income classes at the same rate, $\mathrm{T}_{\mathrm{FUL}}$ misstates refundability. Two points mitigate this concern. First, when portfolios are fully appreciated, which may hold for long-term investors, loss refunds are irrelevant. Second, one can bound the other extreme with operator ii) $\mathrm{T}_{\mathrm{NUL}}$: *no use of losses*. The tax function for this operator is then $t_{\mathrm{nc}}(r) = \tau\frac{r-1}{r}\mathbf{1}_{\{r\geq 1\}}$. Optimal liquidation rules and investor behavior can differ markedly across $\mathrm{T}_{\mathrm{FUL}}$ and $\mathrm{T}_{\mathrm{NUL}}$.

Case iii) is a closer representation of the U.S. and most OECD systems which feature limited use of losses: realized losses can offset realized gains, but do not generate cash refunds. Additionally, if *carry-forward* is allowed, unused losses from past periods can be used to offset gains in the current period. This is modeled by introducing a state variable, $F \geq 0$. If initially at zero, liquidating $\widetilde{w}$ carrying a net loss increases the carry-forward to $F = -\mathrm{CG}(\widetilde{w})$. Next period, liquidating some $\widehat{w}$ with $\mathrm{CG}(\widehat{w}) \geq F > 0$, uses up all carry-forward losses and triggers a tax payment of $\tau(\mathrm{CG}(\widehat{w}) - \delta_F F)$. Here, I introduce $\delta_F \in [0,1]$ as a reduced-form carry-forward parameter. The benchmark $\delta_F = 1$ corresponds to perfect carry-forward. Values $\delta_F < 1$ capture any force that lowers the future value of current unused losses, including explicit time limits, offset restrictions, or even a small probability of regulatory change. In this sense, $\delta_F = 1$ is a knife-edge case, while $\delta_F = 0$ represents regimes in which unused losses expire completely.[15]

Accordingly, I define the *limited-use-of-losses tax operator* featuring a constant statutory tax rate τ and carry-forward tax credits as $\mathrm{T}_{\mathrm{LUL}}(w,F) := \left[\tau(\mathrm{CG}(w) - \delta_F F)\right]_+$ (where $[\,\cdot\,]_+$ is the maximum of its argument and zero). Its liquidation functional is $\mathrm{K}_{\mathrm{LUL}}(w,F) := \mathrm{M}(w) - \mathrm{T}_{\mathrm{LUL}}(w,F)$. The aggregate transition extends G_z by adding the future carry-forward state, $F_{\mathrm{LUL}}(\widetilde{w},F) := \left[F - \mathrm{CG}(\widetilde{w})\right]_+$. Given a state (w,F) and control $(\widetilde{w},I)$, define the law of motion and feasible correspondence by:

$$G_{\mathrm{LUL},z}(w,F,\widetilde{w},I) := \big(G_z(w,\widetilde{w},I), F_{\mathrm{LUL}}(\widetilde{w},F)\big),$$
$$\Gamma(w,F) := \{(\widetilde{w},I) : \widetilde{w} \in [0,w],\ 0 \leq I \leq y + \mathrm{K}_{\mathrm{LUL}}(\widetilde{w},F)\}.$$

Note that, if the tax operator is clear from context, I drop the subscripts on the laws of motion and correspondences to avoid clutter. With the main definitions now in place, I

[15] Perfect carry-forward systems—where capital losses can generally be carried forward indefinitely and offset against future capital gains without major structural restrictions—include Australia, Canada, Ireland, Israel, Norway, and the United Kingdom. Imperfect carry-forward systems include Germany, where loss utilization is effectively constrained by the "minimum taxation" rule limiting the share of income that can be offset in very high-income years, and the United States, where capital losses are separated into short- and long-term buckets before cross-netting and where excess losses are subject to specific offset ordering rules. Time-limited carry-forward regimes, where the usefulness of losses depends on the investor's realization horizon, include Japan (3 years), Spain (4 years), Finland, Greece, Poland, Portugal, and Czechia (typically 5 years), and France and Mexico (10 years).

illustrate their economic content with the examples in the following section.

III.B Examples and Failure of Canonical Trading Rules

The familiar loss-harvesting prescription is to sell depreciated tax lots, reinvest the proceeds, and raise any additional cash from the least appreciated remaining positions. Definition 2 makes this prescription precise through the LHCL strategy classes Γ_h^L and their no-roundtrip sLHCL subclasses Γ_h^{sL}. Under symmetric loss offsets, frictionless investment, and sufficiently favorable aggregate returns, the recursive results in Section IV show that the Bellman problems studied there can be restricted to these strategy sets without loss. The following numerical examples show the economics behind each qualification. Throughout the examples, set the asset-level return to $R = 1$ and let all return variation enter through the aggregate gross return Z, which is deterministic unless stated otherwise.

1. No current loss offset. Let the statutory rate be $\tau = 1/2$, let $Z = 2$, and start with $w = \boldsymbol{\delta}_{1/2} + \boldsymbol{\delta}_{3/4}$. Under $\mathrm{T}_{\mathrm{NUL}}$, either tax lot raises one dollar today because a realized loss receives no refund. A lower-cutoff rule sells $\boldsymbol{\delta}_{1/2}$. The retained $\boldsymbol{\delta}_{3/4}$ becomes a two-dollar position at $r = 3/2$ and has continuation liquidation value

$$2k_{\mathrm{NUL}}(3/2) = 2\left(\frac{1}{2} + \frac{1}{3}\right) = \frac{5}{3}.$$

Selling $\boldsymbol{\delta}_{3/4}$ instead preserves $\boldsymbol{\delta}_{1/2}$, which becomes a two-dollar position exactly at basis, $r = 1$. It therefore liquidates for $2 > 5/3$ tomorrow. The investor optimally sells the *less* depreciated position and preserves the deeper loss as a future tax shield. The reversal comes entirely from the no-offset kink of k_{NUL} at $r = 1$.

2. Reinvestment friction. Under symmetric loss offsets, reinvestment friction shrinks the harvest region. Set $\tau = 1/2$, let the deterministic aggregate gross return be $Z = z > 1$, and consider one dollar in tax lot r, represented by $\boldsymbol{\delta}_r$. Let $L(r)$ and $H(r)$ denote next-period liquidation value when the position is left in place or harvested and reinvested. Then

$$L(r) = zk_{\mathrm{FUL}}(rz) = \frac{zr+1}{2r}, \qquad H(r) = \delta_I k_{\mathrm{FUL}}(r)\left[zk_{\mathrm{FUL}}(z)\right] = \frac{\delta_I(r+1)(z+1)}{4r},$$
$$H(r) - L(r) = \frac{\delta_I(r+1)(z+1) - 2(zr+1)}{4r}.$$

If $\delta_I > 2/(z+1)$, harvesting is weakly optimal exactly when

$$r \le r_H(\delta_I, z) := \frac{\delta_I(z+1) - 2}{2z - \delta_I(z+1)}.$$

Lowering δ_I shrinks the cutoff because only sufficiently deep losses generate a tax subsidy large enough to cover the value destroyed in reinvestment. If $\delta_I \le 2/(z+1)$, the harvest region is empty. Frictionless investment, $\delta_I = 1$, gives $r_H = 1$ for all z. Otherwise, whether it is ex post optimal to harvest a given tax lot at r depends on the aggregate realization Z. Thus, only lots $r \le \min_{z \in \mathrm{supp}(Z)} r_H(\delta_I, z)$ are guaranteed to be optimal to harvest. Section IV.C

formalizes this logic and computes the corresponding harvest region under linear taxation for general aggregate-return support and reinvestment friction.

3. Leaky carryforward. Carryforward leakage can reverse the lower-cutoff prescription even when reinvestment is frictionless. As in the NUL case in Example 1, realizing a loss can weaken its value as a future tax shield. Here, the loss becomes a carryforward credit that may lose value before it can offset taxable gains. The investor may therefore prefer to realize the smaller loss and preserve the deeper loss.

4. Aggregate downside and risk aversion. Consider $\mathrm{T}_{\mathrm{FUL}}$ with $\tau = 1/2$ and $w = 3\boldsymbol{\delta}_1 + 4\boldsymbol{\delta}_2$. Selling either $3\boldsymbol{\delta}_1$ or $4\boldsymbol{\delta}_2$ raises exactly three dollars today because $\mathrm{K}_{\mathrm{FUL}}(4\boldsymbol{\delta}_2) = 3$. If the aggregate realization is $Z = 1/2$, selling $3\boldsymbol{\delta}_1$ and retaining $4\boldsymbol{\delta}_2$ leaves continuation liquidation value $4\,\mathrm{K}_{\mathrm{FUL}}(A_{1/2}\boldsymbol{\delta}_2) = 2$. Selling $4\boldsymbol{\delta}_2$ instead and retaining $3\boldsymbol{\delta}_1$ leaves $3\,\mathrm{K}_{\mathrm{FUL}}(A_{1/2}\boldsymbol{\delta}_1) = 9/4$. Hence the better trade sells the right tail. Intuitively, selling $4\boldsymbol{\delta}_2$ uses four dollars of market value to raise three, leaving only three rather than four dollars exposed to the aggregate shock.

The same force matters under risk aversion. Let Z be positive and nondegenerate with $\mathbb{E}[Z] = 1$, and let date-2 utility be strictly increasing and concave. Selling $3\boldsymbol{\delta}_1$ and retaining $4\boldsymbol{\delta}_2$ gives $c_2^{\text{sell}\,\delta_1} = 2Z + 1$, whereas selling $4\boldsymbol{\delta}_2$ and retaining $3\boldsymbol{\delta}_1$ gives $c_2^{\text{sell}\,\delta_2} = 3Z/2 + 3/2$. Both expressions have expected value three, but the former is a mean-preserving spread of the latter. A risk-averse investor therefore sells $4\boldsymbol{\delta}_2$. This example can be easily generalized to arbitrary portfolios $w \in \mathcal{W}$.

5. One-step versus recursive trading. Finally, under $\mathrm{T}_{\mathrm{NUL}}$, a liquidation ranking that maximizes next-period value need not maximize the value left several dates later. The no-offset kink at $r = 1$ means that a ranking can change as retained tax lots compound relative to cost basis. Set $\tau = 1/2$, keep $R = 1$, and let the deterministic aggregate gross return be $Z = 3$ in each period. Start from $w = \boldsymbol{\delta}_{1/2} + (7/5)\boldsymbol{\delta}_{7/3}$ and require one unit of consumption at each of the first two dates. The position $(7/5)\boldsymbol{\delta}_{7/3}$ has current liquidation value one because $(7/5)k_{\mathrm{NUL}}(7/3) = 1$. If retained for one period, it would be worth $12/5$, below the loss tax lot's $5/2$. A rule that sells to maximize date-2 liquidation value therefore sells the appreciated position first. At date two it sells $6/5$ units from the compounded loss tax lot to finance the second unit. Liquidating what remains at date three gives

$$\left(3 - \frac{6}{5}\right) 3k_{\mathrm{NUL}}(9/2) = \frac{33}{10}.$$

The alternative sells $\boldsymbol{\delta}_{1/2}$ first, keeps the larger appreciated position, and at date two sells $7/4$ units from it. Its date-three value is

$$\left(\frac{21}{5} - \frac{7}{4}\right) 3k_{\mathrm{NUL}}(21) = \frac{77}{20}.$$

Thus the strategy that maximizes date-2 liquidation value leaves less market value at date two and only 3.30 at date three, whereas reversing its first trade leaves 3.85. This failure is

driven by T_{NUL}: the no-offset kink prevents the one-period liquidation ranking from being preserved recursively. Section IV instead asks when a liquidation ranking survives recursive continuation. Under T_{FUL}—and under perfect-carryforward T_{LUL} in the corresponding frictionless benchmark—the recursive results restore LHCL optimality when the aggregate-return support condition holds.

Together, these examples delimit the canonical trading rule. Absence of current-loss offsets can make a deeper loss worth preserving; reinvestment costs shrink the harvest region; carryforward leakage changes which loss should be realized; aggregate downside and risk aversion can favor selling appreciated tax lots; and, without restrictions on T, a longer horizon can reverse a one-step ranking. Section IV identifies conditions under which the lower-cutoff ranking survives these recursive comparisons.

IV The Recursive Framework

This section formulates two classes of recursive portfolio problems. The *perpetuity problem* determines the largest constant real consumption level that a portfolio can sustain indefinitely under a given tax system and market conditions. These consumption levels are used in the quantitative applications. The *discounted-utility problem* allows consumption to vary over time and accommodates life-cycle preferences.

Scope of the trading results. I characterize the structure of optimal trading under conditions on taxation, returns, and investment frictions. Under these conditions, tax-arbitrage arguments replace any feasible trade with one that preserves current consumption and weakly improves the continuation portfolio in every aggregate state. The resulting strategy classes are independent of preferences. Intuitively, the investor lets her preferences determine her consumption choice and delegates the structure of liquidation and investment to the optimal trading strategy. When the conditions fail, this separation can break down. Trading then involves trade-offs whose resolution depends on preferences and return probabilities, so tax-arbitrage arguments alone cannot generally characterize optimal trading.

Sections IV.A and IV.B formulate these problems and establish convergence and policy attainment. Sections IV.C and IV.D then characterize optimal trading under full and limited use of losses. Appendices IV.A–IV.D mirror these four subsections and collect their assumptions, proofs, and extensions. Readers mainly interested in the quantitative applications may proceed to Sections V and VI.

IV.A Two Recursive Problems

Both problems use the same feasible trades and portfolio transition from Section III. The formulation below applies to tax-share systems: each dollar sold from tax lot r pays a fixed after-tax amount $k(r)$, and liquidation proceeds add across lots. Assumption 2 in

Appendix IV.A.1 gives the formal conditions. Limited use of losses with carryforward falls outside this class because proceeds depend on net realized gains and unused losses. Section IV.D extends the recursions by adding unused losses to the state.

Given exogenous income $y \geq 0$, a feasible control $(\widetilde{w}, I) \in \Gamma(w)$ yields current consumption $c(\widetilde{w}, I) := y + \mathrm{K}(\widetilde{w}) - I$. Continuation states are given via the maps $G_z(w, \widetilde{w}, I)$. The discounted-utility ($\beta < 1$) Bellman operator is

$$(\mathbb{T}V)(w) := \sup_{(\widetilde{w},I)\in\Gamma(w)} \Big\{ u(c(\widetilde{w}, I)) + \beta \sum_z p_z V\big(G_z(w, \widetilde{w}, I)\big) \Big\}. \tag{6}$$

This operator chooses trades to maximize current consumption utility plus discounted expected continuation utility, and can be used for finite- and infinite-horizon consumption and saving choices involving portfolios of many assets.

Without aggregate risk, a perpetuity is a constant real consumption payout that can be sustained indefinitely from a given portfolio. Let $\mathcal{C}(w)$ denote the set of such payouts feasible from w:

$$\mathcal{C}(w) := \left\{ c \geq 0 : \begin{array}{l} \exists (w_t, \widetilde{w}_t, I_t)_{t\geq 0} \text{ with } w_0 = w, \\ (\widetilde{w}_t, I_t) \in \Gamma(w_t), \quad w_{t+1} = G(w_t, \widetilde{w}_t, I_t), \\ c \leq c(\widetilde{w}_t, I_t) \text{ for every } t \geq 0 \end{array} \right\}. \tag{7}$$

The object of interest is $\sup \mathcal{C}(w)$, the portfolio's optimal perpetuity value.

I focus on this value to isolate the pure advantage of tax-optimal trading from life-cycle consumption choices. An infinite horizon avoids choosing a terminal liquidation date or specifying what happens to wealth at that date. The comparison also requires neither a subjective discount factor nor assumptions about utility curvature, retirement, or desired bequests.

Linear accrual taxation provides a simple benchmark. With frictionless investment and a constant real return $\mu > 0$, a unit investment sustains a maximal payoff of exactly $\mu(1-\tau)$ each period. Under linear realization taxation at the same positive rate τ, tax-aware trading increases the sustainable payoff.

How can one characterize $\sup \mathcal{C}(w)$? A recursive formulation helps. A constant payout must be feasible today and remain sustainable from the continuation portfolio. Given a candidate continuation value function V, the smaller of current consumption and continuation value measures the payout supported by both. This motivates the sup-min operator

$$(\mathbb{P}V)(w) := \sup_{(\widetilde{w},I)\in\Gamma(w)} \min\Big\{ c(\widetilde{w}, I),\ \sum_z p_z V\big(G_z(w, \widetilde{w}, I)\big) \Big\}. \tag{8}$$

The operator uses expected continuation value when aggregate risk is present. Its interpretation suggests looking for a fixed point $V = \mathbb{P}V$ that, without aggregate risk, assigns to each portfolio its optimal perpetuity value, $V(w) = \sup \mathcal{C}(w)$. This interpretation needs to be proved.

One way to construct a candidate for this value function is to begin with a finite-horizon consumption-smoothing problem: what is the largest constant payout sustainable over a given number of periods? With only the current period remaining, the largest payout is

income plus liquidation value. Each application of $\mathbb{P}$ adds one future period, giving the canonical sequence

$$V_0^{\mathbb{P}}(w) := \sup_{(\widetilde{w},I)\in\Gamma(w)} c(\widetilde{w},I) = y + \mathrm{K}(w), \qquad V_{n+1}^{\mathbb{P}} := \mathbb{P}V_n^{\mathbb{P}}. \tag{9}$$

Extending the horizon means the portfolio must sustain the payout for longer, so the maximal constant payout $V_n^{\mathbb{P}}(w)$ weakly decreases in n. The following corollary identifies the finite-horizon values and connects their limit to the feasible perpetuity set.

Corollary 2 (Perpetuity interpretation without aggregate risk)**.** *Suppose Assumptions 1 and 2 hold and there is no aggregate risk. For the canonical sequence in (9):*

i) $V_n^{\mathbb{P}}(w)$ is the largest constant payout sustainable in the current period and for n future periods from w.

ii) The sequence approximates the optimal perpetuity value:

$$\lim_{n\to\infty} V_n^{\mathbb{P}}(w) = \sup \mathcal{C}(w).$$

The proof is in Appendix IV.A.2.

Corollary 2 establishes that extending the finite payout horizon approaches the supremum of the feasible perpetuity set. The remaining question is whether this limit also solves the Bellman equation $V = \mathbb{P}V$. In the next subsection, I establish convergence to Bellman fixed points for both the sup-min operator in (8) and the discounted-utility operator in (6). This, in turn, allows me to prove the existence of optimal trades at any horizon and analyze their structure.

IV.B Convergence and Optimal Policies

I now show that backward iteration converges to a stationary value function and that optimal trades exist at finite horizons and at the stationary limit under the conditions below.

The regularity conditions are collected in Appendix IV.A.1 for portfolios and liquidation, and Appendix IV.B.1 for utility. They control portfolio growth and allow bounded, logarithmic, or sublinear utility. Appendix IV.B.1 also defines the weighted value-function spaces $\mathcal{B}_{W_p}$, where $p \geq 0$ is a parameter of the norm. For discounted utility, what matters for my results is only that some admissible choice yields a space on which the Bellman operator is a contraction. By choosing the space appropriately, its modulus can be made arbitrarily close to $\beta\underline{\lambda}_I$, defined there.

Backward iteration. The discounted-utility operator $\mathbb{T}$ has a unique fixed point, and backward iteration converges to it. The sup-min operator can have several fixed points (for example, $\mathbb{P}0 = 0$), but the canonical sequence in (9) converges to the one relevant to the perpetuity problem.

Proposition 1 (Backward iteration)**.** *Suppose Assumptions 1 and 2 hold.*

i) Under Assumption 3, with $\beta\underline{\lambda}_I < 1$ in case (iii), there is an admissible $p \geq 0$ such that $\mathbb{T}$ is a contraction on $\mathcal{B}_{W_p}(\mathcal{W})$. It has a unique fixed point $V^{\mathbb{T}}$ in this space, and its iterates converge to $V^{\mathbb{T}}$ in the corresponding norm.

ii) The canonical sup-min sequence in (9) decreases pointwise to $V^{\mathbb{P}}$, the greatest fixed point of $\mathbb{P}$:[16]

$$V^{\mathbb{P}}(w) = \lim_{n\to\infty} V_n^{\mathbb{P}}(w), \qquad V^{\mathbb{P}} = \mathbb{P}V^{\mathbb{P}}.$$

Part (i) establishes that the Tax Portfolio Model supports classical time-separable recursive models of consumption and saving: backward iteration converges to a unique stationary value function. Together with Corollary 2, part (ii) establishes that, without aggregate risk, the optimal perpetuity value $\sup \mathcal{C}(w)$ solves the Bellman equation in (8). Corollaries IV.B.1 and IV.B.2 give the corresponding conditions for CARA, CRRA, and log utility.

Both Bellman operators also preserve monotonicity and concavity, and the discounted-utility fixed point inherits these properties under the convergence conditions above; Appendix IV.B.4 states and proves this additional result.

Policy attainment. To characterize optimal trading strategies, I first need to establish that optimal trades exist. Theorem 2 shows that each Bellman supremum is achieved by a feasible trade, both at finite horizons and at the stationary limit.

Theorem 2 (Policy attainment)**.** *Suppose Assumptions 1 and 2 hold, together with Assumption 3 for the discounted-utility problem. Let $\mathbb{H} \in \{\mathbb{T}, \mathbb{P}\}$. For $\mathbb{T}$, initialize iteration with any weak-$*$ continuous $V_0^{\mathbb{T}}$ in the relevant space $\mathcal{B}_{W_p}$. For $\mathbb{P}$, use the canonical sequence in (9). Then:*

i) Every finite-horizon value $V_n^{\mathbb{H}}$ is weak-$*$ continuous on each norm-bounded state set.

ii) Every finite-horizon Bellman supremum, including the one defining $V_0^{\mathbb{P}}$, is attained. The corresponding argmax correspondences are nonempty-valued, weak-$*$ compact-valued, and upper hemicontinuous on norm-bounded state sets.

iii) The stationary values from Proposition 1 also have nonempty, weak-$*$ compact argmax sets at every state. On norm-bounded state sets, $V^{\mathbb{T}}$ is weak-$*$ continuous and $V^{\mathbb{P}}$ is weak-$*$ upper semicontinuous.

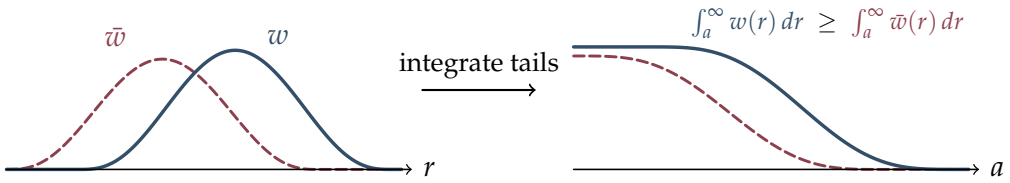
Appendix IV.B provides the proofs and details on the continuity notions used here.

Theorem 2 establishes that optimal trades exist for tax operators with a tax-share representation. I now specialize to $\mathrm{T}_{\mathrm{FUL}}$ to characterize these trades under linear taxation with full use of losses. Section IV.D revisits these trading results under limited use of losses with carryforward.

IV.C Recursive Optimal Trading under Linear Taxation

Under $\mathrm{T}_{\mathrm{FUL}}$, I show that both recursive problems in (6) and (8) admit optimal LHCL trades at finite horizons and at their stationary limits. These trades harvest a preference-independent

[16] Every fixed point $U = \mathbb{P}U$ satisfies $U \leq V_0^{\mathbb{P}}$, since current consumption bounds the sup-min objective. Monotonicity of $\mathbb{P}$ then gives $U \leq V_n^{\mathbb{P}}$ for every n, and hence $U \leq V^{\mathbb{P}}$.



(a) tax-lot distributions (b) tail test for every a

Fig. 2. Right dominance. Solid navy denotes w and dashed burgundy $\bar{w}$. Integrating each distribution above a threshold a gives panel (b). The navy curve lies above the burgundy curve at every threshold, so w passes the tail test. It right-dominates only if it also satisfies $\mathrm{K}(w) \geq \mathrm{K}(\bar{w})$. Curves are schematic.

core loss region and arrange liquidation as a cutoff. Restricting trading to this class leaves the full value functions unchanged. However, these results rely on a sharp condition: nonnegative net period returns in every aggregate state, $z\mathbb{E}[R] - 1 \geq 0$. If, in addition, investment is frictionless, the core harvest region contains *all* losses, and an optimal control can be chosen as a *strict* LHCL trade, which avoids round-trips (see Definition 2).

Dominance on portfolios. My argument compares continuation portfolios using a partial order which I call *right dominance*, introduced in Definition 4. This order identifies portfolio improvements that weakly increase the values of both recursive problems. Throughout this subsection, I write $\mathrm{K} = \mathrm{K}_{\mathrm{FUL}}$, where $k_{\mathrm{FUL}}(r) = 1 - \tau + \tau/r$ and $\tau \in [0, 1)$.

Definition 4 (Right dominance). *A portfolio $w \in \mathcal{W}$ right-dominates $\bar{w} \in \mathcal{W}$, written $w \succeq_R \bar{w}$, if*

$$\int_a^\infty w(r)\,dr \geq \int_a^\infty \bar{w}(r)\,dr \qquad \text{for every } a > 0, \tag{10}$$

$$\mathrm{K}(w) \geq \mathrm{K}(\bar{w}). \tag{11}$$

A value function V is right-increasing if $w \succeq_R \bar{w}$ implies $V(w) \geq V(\bar{w})$.

Figure 2 illustrates the upper-tail requirement in Eq. (10). To see why the liquidation-value requirement places an additional restriction on preservation of right dominance under returns, consider two portfolios with the same current liquidation value. For $0 < \tau < 1$, let $w_\infty = \boldsymbol{\delta}_\infty/(1-\tau)$ denote the limiting portfolio concentrated in assets that have appreciated from zero, and let $w_r = \boldsymbol{\delta}_r/k_{\mathrm{FUL}}(r)$ denote a portfolio concentrated in a finite tax lot. Both liquidate for one unit today, but w_∞ right-dominates w_r, i.e., $w_\infty \succeq_R w_r$. A common gross return $g > 0$ gives

$$\mathrm{K}(A_g w_\infty) = g \gtrless 1 + (1-\tau)\,\mathrm{M}(w_r)(g-1) = \mathrm{K}(A_g w_r) \quad \Longleftrightarrow \quad g \gtrless 1.$$

Intuitively, assets in the limiting tax lot remain there after any return, and yield $(1-\tau)$ times their market value after taxes. Appreciation and depreciation therefore translate directly into proportional changes in liquidation value. For a finite tax lot, appreciation lowers the after-tax share and depreciation raises it, dampening the change in liquidation value. Consequently, w_∞ continues to right-dominate w_r exactly when the net return is nonnegative, $g - 1 \geq 0$ (see also the aggregate-downside example in Section III.B).

With idiosyncratic and aggregate returns, g in the liquidation-value comparison is replaced

by $z\mathbb{E}[R]$, and the return condition follows. Assume $w \succeq_R \bar{w}$. Then, the liquidation-value inequality $\mathrm{K}(A_z Q w) \geq \mathrm{K}(A_z Q \bar{w})$ is preserved for every such pair in every aggregate state if and only if $z\mathbb{E}[R] - 1 \geq 0$ in every state. Additionally, the upper-tail comparison (10) is preserved because returns move tax lots by positive multiplicative factors. Therefore, under the support condition, the law of motion preserves the right-dominance order.[17]

Proposition 3 establishes this fact along with the observation that the finite-horizon and stationary value functions are right-increasing. A right-dominating portfolio therefore has weakly higher value at each horizon, and its dominance survives the intervening returns.

Proposition 3 (Right dominance under returns and backward iteration). *Suppose Assumptions 1 and 2 hold,* $\mathrm{K} = \mathrm{K}_{\mathrm{FUL}}$, *and* $z\mathbb{E}[R] \geq 1$ *for every* $z \in \mathcal{Z}$, *together with Assumption 3 for the discounted-utility problem. Initialize the discounted-utility iteration with* $V_0^{\mathbb{T}} = 0$, *and use the canonical sup-min sequence in* (9).

i) If $w \succeq_R \bar{w}$, *then* $A_z Q w \succeq_R A_z Q \bar{w}$ *for every* z. *Adding a common investment portfolio preserves this comparison:* $A_z(Q w + \delta_I I\chi) \succeq_R A_z(Q\bar{w} + \delta_I I\chi)$, $I \geq 0$.

ii) For every $n \geq 0$, $V_n^{\mathbb{T}}$ *and* $V_n^{\mathbb{P}}$ *are right-increasing.*

iii) The canonical limit $V^{\mathbb{P}}$ *is right-increasing. Under the convergence conditions of Proposition 1(i),* $V^{\mathbb{T}}$ *is also right-increasing.*

The proof uses induction, starting with the observation that the initial value functions $V_0^{\mathbb{T}} = 0$ and $V_0^{\mathbb{P}} = y + \mathrm{K}$ are right-increasing. Appendix IV.C.1 proves that the Bellman operators preserve this property; the convergence results of Proposition 1 then carry it to the limits.

For a given current consumption level, a right-dominating end-of-period portfolio is therefore preferable. More generally, it suffices to find trades whose next-period portfolios right-dominate those of a candidate trade in every aggregate state. I now turn to constructing such trades.

Dominance-inducing trades. Intuitively, loss harvesting and cutoff liquidation induce right dominance. Harvesting low-r tax lots and reinvesting the proceeds shifts holdings toward higher tax lots. Additional liquidation is best taken from the left: among all ways to raise a given amount of after-tax cash, selling a lower interval $(0, a]$ preserves the most market value in every right tail of the residual portfolio.[18]

Suppose for the moment that the *core harvest region* $\mathcal{H}^L = (0, r_H]$ is known. For every tax lot in this region, harvesting and reinvesting its proceeds right-dominates retaining it in every aggregate state. I first use this property to construct an LHCL replacement for a given trade, then determine the region itself.

[17] The tax operator matters even when net returns are positive. Under $\mathrm{T}_{\mathrm{NUL}}$, let $\tau = 1/2$, let $R = 2$ deterministically, and write $K_{\mathrm{NUL}}(w) = \int k_{\mathrm{NUL}}(r) w(r)\, dr$. Then $\bar{w} = (4/3)\boldsymbol{\delta}_2$ right-dominates $w = \boldsymbol{\delta}_{1/2}$, with $K_{\mathrm{NUL}}(\bar{w}) = K_{\mathrm{NUL}}(w) = 1$. After returns, $Q\bar{w} = (8/3)\boldsymbol{\delta}_4$ and $Qw = 2\boldsymbol{\delta}_1$, but $K_{\mathrm{NUL}}(Q\bar{w}) = 5/3 < 2 = K_{\mathrm{NUL}}(Qw)$. The no-offset kink reverses the liquidation-value comparison while the upper-tail ranking survives.

[18] Because the tax paid per dollar of after-tax liquidity, $1/k_{\mathrm{FUL}}(r) - 1$, increases with r, this sale also maximizes the total market value retained.

Fix now a candidate control $(\widetilde{w}^{(0)}, I^{(0)}) \in \Gamma(w)$, generating net withdrawal $q = \mathrm{K}(\widetilde{w}^{(0)}) - I^{(0)}$ and current consumption $y + q$. Its end-of-period portfolio is $w - \widetilde{w}^{(0)} + \delta_I I^{(0)} \boldsymbol{\delta}_1$, and its next-period state is $G_z(w, \widetilde{w}^{(0)}, I^{(0)})$. The following construction holds q fixed and improves this next-period state in the right-dominance order for every z.[19]

First, harvest any unsold tax lots in the core and add their after-tax proceeds to investment, leaving net withdrawal unchanged. This gives a trade $(\widetilde{w}^{(1)}, I^{(1)}) \in \Gamma^H(w)$. Here, $\Gamma^H(w)$ is the set of all feasible trades that liquidate the entire core while leaving sales outside it unrestricted.[20]

Second, replace the remaining liquidation by extending a lower cutoff, preserving total liquidation proceeds and the investment choice from the first step. The resulting trade belongs to $\Gamma^L_{r_H}(w)$—the LHCL class from Definition 2 consisting of feasible cutoff trades $(w\mathbf{1}_{(0,a]}, I)$ with $a \geq r_H$. Each step therefore weakly improves the next-period portfolio in every aggregate state while preserving current consumption. Proposition 4 records these two improvements.

Proposition 4 (Core harvesting and cutoff replacement). *Suppose* $\mathrm{K} = \mathrm{K}_{\mathrm{FUL}}$ *and* $z\mathbb{E}[R] \geq 1$ *for every* $z \in \mathcal{Z}$.

i) For every $w \in \mathcal{W}$ *and every feasible trade* $(\widetilde{w}^{(0)}, I^{(0)}) \in \Gamma(w)$, *there is a trade* $(\widetilde{w}^{(1)}, I^{(1)}) \in \Gamma^H(w)$ *with the same current consumption and* $G_z(w, \widetilde{w}^{(1)}, I^{(1)}) \succeq_R G_z(w, \widetilde{w}^{(0)}, I^{(0)})$ *for every* z.

ii) For every $(\widetilde{w}^{(1)}, I^{(1)}) \in \Gamma^H(w)$, *there is a cutoff* $a \geq r_H$ *such that* $(w\mathbf{1}_{(0,a]}, I^{(1)}) \in \Gamma^L_{r_H}(w)$ *has the same liquidation proceeds and current consumption, and* $G_z(w, w\mathbf{1}_{(0,a]}, I^{(1)}) \succeq_R G_z(w, \widetilde{w}^{(1)}, I^{(1)})$ *for every* z.

The construction requires a harvest region whose lots can be sold and reinvested without reducing the next-period portfolio in the right-dominance order. To determine this core harvest region, I examine how the loss-harvesting step induces right dominance. Compare harvesting with retaining one unit of market value in tax lot $r \leq 1$, represented by $\boldsymbol{\delta}_r$. Harvesting and reinvesting its after-tax proceeds contributes $A_z(\delta_I k_{\mathrm{FUL}}(r)\chi)$ to the next-period portfolio, whereas retaining it contributes $A_z Q \boldsymbol{\delta}_r$. The statewise harvest region is therefore

$$\begin{aligned}\mathcal{H}^L_z &:= \{r \in (0,1] : A_z(\delta_I k_{\mathrm{FUL}}(r)\chi) \succeq_R A_z Q \boldsymbol{\delta}_r\} \\ &= \left\{ r \in (0,1] : \begin{array}{l} \delta_I k_{\mathrm{FUL}}(r) \geq 1, \\ \delta_I k_{\mathrm{FUL}}(r)\{z(1-\tau)\mathbb{E}[R] + \tau\} \geq z(1-\tau)\mathbb{E}[R] + \tau/r \end{array} \right\} \\ &= (0, r^H_z].\end{aligned} \tag{12}$$

The first inequality requires reinvestment to buy at least as much market value as the retained lot. The second requires at least as much next-period liquidation value. Under $z\mathbb{E}[R] - 1 \geq 0$, these conditions define a lower interval, with $r^H_z = 0$ when the region is empty.

[19] With reinvestment costs, the harvesting replacement need not right-dominate the candidate's end-of-period portfolio. Harvesting a unit in tax lot r and reinvesting its proceeds gives end-of-period liquidation value $\delta_I k_{\mathrm{FUL}}(r)$, compared with $k_{\mathrm{FUL}}(r)$ if retained. The core conditions in (12) establish dominance after returns, which is the comparison required by the Bellman objectives.

[20] Precisely, $\Gamma^H(w) := \{(\widetilde{w}, I) \in \Gamma(w) : w\mathbf{1}_{(0,r_H]} \leq \widetilde{w}\}$. The LHCL correspondence $\Gamma^L_h(w)$ is defined in Definition 2. Here, r_H is the guaranteed harvest cutoff, whereas a is the actual sale cutoff. At the core endpoint, $\Gamma^L_{r_H}(w) \subseteq \Gamma^H(w) \subseteq \Gamma(w)$.

The core harvest region is their intersection, $\mathcal{H}^L := \bigcap_{z\in\mathcal{Z}} \mathcal{H}^L_z = (0, r_H]$, where $r_H := \min_{z\in\mathcal{Z}} r^H_z$. Harvesting and reinvesting any tax lot in this core right-dominates retaining it under every aggregate realization. Losses outside the core may still be worth harvesting. With costless reinvestment, the core contains all losses: if $\delta_I = 1$, the maintained return condition implies $\mathcal{H}^L_z = (0, 1]$ for every z, and hence $r_H = 1$.

Higher aggregate returns and lower reinvestment costs weakly expand the harvest region; the effect of the tax rate is ambiguous. Conditional on the tail test $\delta_I k_{\mathrm{FUL}}(r) \geq 1$, stronger aggregate growth benefits the reinvested portfolio at least as much as the retained position, so a higher z weakly expands $\mathcal{H}^L_z$. A higher δ_I returns more of the sale proceeds to the portfolio, so a smaller tax refund suffices to cover reinvestment costs. A higher tax rate increases the refund from realizing a loss but also changes the taxation of future returns on the reinvested proceeds; its net effect on the harvest region can therefore go either way. Any change that expands every statewise interval also expands their intersection, the core $\mathcal{H}^L$.

Recursive LHCL optimality. With the core harvest region determined, Proposition 4 supplies an LHCL replacement for every feasible trade. Proposition 3 establishes that the relevant value functions are right-increasing, so these replacements cannot lower either Bellman objective. Since Theorem 2 establishes attainment, applying the replacements to a maximizer produces an optimal LHCL trade.

To state the implications for backward iteration and stationary values formally, let $\mathbb{T}^L$ and $\mathbb{P}^L$ denote the restricted discounted-utility and sup-min operators with the feasible set $\Gamma(w)$ replaced by $\Gamma^L_{r_H}(w)$. Initialize both discounted-utility iterations at zero and both sup-min iterations at $y + \mathrm{K}$. The latter is also the maximal current payout in the restricted problem, since full liquidation with zero investment is an LHCL trade.

Theorem 3 (Recursive LHCL optimality)**.** *Under the assumptions of Proposition 3, the following statements hold. For the discounted-utility stationary conclusions, suppose additionally that the convergence conditions of Proposition 1(i) hold.*

i) For every $n \geq 0$, full and LHCL-restricted backward iteration give the same values: $V^{\mathbb{T}}_n = V^{\mathbb{T},L}_n$ and $V^{\mathbb{P}}_n = V^{\mathbb{P},L}_n$.

ii) The discounted-utility sequences converge in the relevant norm to the common limit $V^{\mathbb{T}} = V^{\mathbb{T},L}$. The canonical sup-min sequences converge pointwise to the common limit $V^{\mathbb{P}} = V^{\mathbb{P},L}$. Each limit is a fixed point of both its full and restricted Bellman operators.

iii) At every state, each finite-horizon and stationary Bellman maximum admits a maximizer in $\Gamma^L_{r_H}(w)$. If the full maximizer is unique, it belongs to this class.

Theorem 3 allows the search for an optimal trade to be restricted to a sale cutoff $a \geq r_H$ and an investment choice I. This restriction preserves the full value functions at every finite horizon and at the stationary limit. Appendix IV.C gives the proof.

However, the theorem leaves two aspects of the optimal trade open: whether to combine investment with sales of appreciated tax lots, and how far to extend liquidation beyond the

guaranteed harvest core. Beyond r_H, probabilities, preferences, and frictions determine the optimal cutoff. In particular, harvesting and reinvesting losses outside a core with $r_H < 1$ shifts holdings toward higher tax lots, but investment costs can reduce liquidation value. Such harvesting need not induce right dominance in every aggregate state.

Without reinvestment costs, a third replacement step rules out simultaneous investment and sales of appreciated positions. Retaining appreciated positions right-dominates selling them and reinvesting their proceeds. A cutoff trade that combines sales of appreciated positions with positive investment can therefore be replaced by one with a lower sale cutoff and less investment, preserving current consumption and weakly improving the next-period portfolio in every aggregate state. Since the core contains all losses when $\delta_I = 1$, the resulting trade belongs to the sLHCL class $\Gamma_1^{sL}(w)$ from Definition 2, with harvest cutoff $h = 1$:

Corollary 4 (Frictionless sLHCL optimality). *Under the assumptions of Theorem 3, if investment is frictionless, $\delta_I = 1$, then $r_z^H = 1$ for every z. All conclusions of Theorem 3 hold with Γ_1^{sL} in place of $\Gamma_{r_H}^L$.*

Appendix IV.C.5 gives sufficient conditions for extending the portfolio comparisons to more general tax-share kernels. I next examine how these trading restrictions extend when unused losses must be carried forward.

IV.D Limited Use of Losses and Carryforward

With perfect loss carryforward and frictionless investment, an sLHCL trade also attains the discounted-utility Bellman maximum under LUL when $z\mathbb{E}[R] \geq 1$ for every aggregate realization. The Bellman problems extend to the state (w, F), where F records unused losses. Without aggregate risk, the sup-min recursion again gives the largest constant consumption sustainable forever from this state, including when carryforwards leak or investment is costly.

Carryforward taxation also changes the portfolio order required to prove the trading results: current resources depend nonlinearly on realized gains, while unused losses retain future tax value. Under perfect carryforward, write $\Psi(w, F) := \mathrm{K}_{\mathrm{FUL}}(w) + \tau F$ for the joint linear-tax value of the portfolio and unused losses. Say that $(\bar{w}, \bar{F})$ *carryforward right-dominates* (w, F), written $(\bar{w}, \bar{F}) \succeq_{\mathrm{CFRD}} (w, F)$, if $\int_a^\infty [\bar{w}(r) - w(r)]\, \mathrm{d}r \geq 0$ for every $a \geq 0$ and $\Psi(\bar{w}, \bar{F}) \geq \Psi(w, F)$. The tail comparison preserves the option value of high-r assets; the Ψ comparison preserves the joint after-tax value of the portfolio and unused losses. At $a = 0$, the tail test implies $\mathrm{M}(\bar{w}) \geq \mathrm{M}(w)$. Together, the two comparisons ensure that the dominant state can raise at least as much cash, because

$$\mathrm{K}_{\mathrm{LUL}}(w, F) = \min\{\mathrm{M}(w), \Psi(w, F)\}. \tag{13}$$

Write $\Gamma_{\mathrm{CF}}^{sL}(w, F)$ for the feasible cutoff controls that harvest every loss and rule out simultaneous reinvestment when liquidation extends into gains.

Proposition 5 (Dynamic sLHCL optimality with perfect carryforward)**.** *Suppose $\delta_F = \delta_I = 1$, Assumptions 1 and 3 hold, and $z\mathbb{E}[R] \geq 1$ for every aggregate realization z. Suppose also that the perfect-carryforward Bellman maxima are attained and that discounted value iteration is convergent under the product-state counterpart of Proposition 1. Starting from $V^{\mathbb{T}}_{\mathrm{LUL},0} \equiv 0$, for every finite horizon n and every state (w, F):*

i) $V^{\mathbb{T}}_{\mathrm{LUL},n}$ is increasing in carryforward right dominance;

ii) the full Bellman maximum admits a control in $\Gamma^{sL}_{\mathrm{CF}}(w, F)$, and full and restricted value iteration coincide;

iii) the limiting full and $\Gamma^{sL}_{\mathrm{CF}}$-restricted fixed points coincide, and the stationary full problem admits an sLHCL maximizer. If the full maximizer is unique, it belongs to $\Gamma^{sL}_{\mathrm{CF}}(w, F)$.

This proposition extends the discounted-utility conclusions of Proposition 3 and Corollary 4 to perfect carryforward: value functions respect the corresponding portfolio order, and the sLHCL restriction preserves finite-horizon and stationary values. The argument differs from the FUL proof at two steps. Equation (13) replaces the linear current-resource comparison. Matching the current net withdrawal then leaves the same end-of-period Ψ value, while a lower-cutoff replacement improves the upper tails. Both comparisons survive the law of motion: for any two end-of-period states, with Δ denoting the difference between them,

$$\Delta\Psi' = \Delta\Psi + (1-\tau)\big(z\mathbb{E}[R] - 1\big)\Delta \mathrm{M}. \tag{14}$$

The tail order implies $\Delta \mathrm{M} \geq 0$, so the same support condition as under FUL, $z\mathbb{E}[R] \geq 1$, preserves the comparison. Bellman induction then gives the proposition. Appendix IV.D provides the Bellman foundations, formal cutoff correspondence, and proof details.

Finally, a central difference from $\mathrm{T}_{\mathrm{FUL}}$ is that investment frictions can invalidate the sLHCL optimality result rather than merely shrink the core harvest region. For every $\delta_I < 1$, retaining a loss can yield more next-period consumption than harvesting and reinvesting it, even with perfect carryforward and favorable returns.[21] Under $\mathrm{T}_{\mathrm{FUL}}$, the immediate cash refund finances additional reinvestment that can compensate for the friction. Under $\mathrm{T}_{\mathrm{LUL}}$, a loss with no current gains to offset creates only a carryforward. It supplies no additional cash to reinvest, so harvesting and reinvesting reduce the market value earning subsequent returns.

V Investor-side tax measures

This section measures the investor's tax burden under optimal realization trading. I use the solution to the perpetuity problem in Section IV.A to derive an accrual-equivalent tax

[21] Let $\tau = 1/2$, $\delta_F = 1$, and $\delta_I \in [0, 1)$. Start from $w = \boldsymbol{\delta}_{1/2}$ with no unused losses or income, and suppose all assets earn a gross return of 2. Retaining the position gives $2\boldsymbol{\delta}_1$ next period, which liquidates for 2. Under $\mathrm{T}_{\mathrm{LUL}}$, harvesting raises 1 and creates a loss carryforward of 1. Reinvesting all proceeds gives $2\delta_I\boldsymbol{\delta}_2$ next period. Its gain of δ_I is fully offset by the carryforward, so it liquidates for $2\delta_I < 2$. Both choices deliver zero current consumption. Under $\mathrm{T}_{\mathrm{FUL}}$, harvesting also generates a cash refund of $1/2$. Reinvesting the resulting $3/2$ gives $3\delta_I\boldsymbol{\delta}_2$, with next-period liquidation value $9\delta_I/4$. Thus the lot belongs to the FUL core harvest region whenever $\delta_I \geq 8/9$, although harvesting under LUL yields less next-period consumption for every $\delta_I < 1$.

rate and a return alpha. The former connects to the public-finance literature on effective taxation (Poterba, 1989, Protopapadakis, 1983); the latter to the finance literature on returns from tax-aware trading (Chaudhuri et al., 2020). Both measures hold maximal sustainable consumption fixed across tax systems. Importantly, consumption equivalence need not imply revenue equivalence, as Glenday and Davies (1990) emphasize. Section VI therefore measures revenue separately.

V.A Effective tax rates

I express realization taxation as an equivalent tax upon accrual. Wages are taxed as they are earned, so an accrual-equivalent rate provides a useful benchmark for comparing capital and labor taxation. Earlier work also uses accrual equivalence (Glenday and Davies, 1990, Poterba, 1989, Protopapadakis, 1983). Here I apply this criterion to the maximal real perpetuity from an optimally managed portfolio.

Two features distinguish this benchmark. First, optimal trading continues over an infinite horizon, without a terminal liquidation date or a horizon set by the available data. Second, the comparison requires neither discounting nor a life-cycle specification: no retirement age, utility curvature, or target bequest is needed. Tax rules and market conditions enter as explicit parameters, so counterfactuals need not be tied to a particular historical episode.

Let $\mu := \mathbb{E}[R] - 1 > 0$ denote the expected real total return, including dividends. The accrual benchmark taxes this return each period. At an accrual tax rate s, a unit investment thus sustains $\mu(1-s)$ while preserving the principal. The effective rate is the value of s that matches maximal perpetual consumption under realization taxation.

Definition 5 (Market gains effective tax rate)**.** *Let $\bar{c}_M$ be maximal constant real consumption from a unit initial investment under optimal realization trading. The* market gains effective tax rate *(MGET) is the accrual tax rate on real total returns, including dividends, that sustains the same withdrawal:*

$$\mu(1-\tau^*) = \bar{c}_M, \qquad \tau^* = 1 - \frac{\bar{c}_M}{\mu}. \tag{15}$$

Scaling holdings, sales, reinvestment, and loss carryforwards by the same factor scales market values, realized gains, taxes, and feasible withdrawals proportionally (Sections III.A and IV.A–IV.D). Under accrual taxation, the same scaling follows from (15). Thus, without loss of generality, I normalize initial wealth to one.

The maximal perpetuity follows from the recursive problem in Section IV.A with no aggregate risk, frictionless reinvestment, and perfect LUL carryforward, extended by inflation and dividend payouts. Let $\pi > -1$ denote the inflation rate. With inflation, constant real consumption c requires nominal withdrawals $c(1+\pi)^t$ at date t. Dividends, in turn, convert part of the nominal excess return into a current taxable payout. Let $\varphi \in [0,1]$ denote this fraction. Here the initial investment occurs at date zero and the first trading date is $t = 1$; Section IV.A instead indexes the first trading date by $t = 0$.

Intuitively, higher inflation and dividend payouts raise effective taxes, other things equal. Inflation increases nominal taxable gains at unchanged real returns (Feldstein and Slemrod, 1978). Dividends reduce the scope for deferral: payouts can exceed consumption needs and trigger tax even when the investor would prefer to keep the return invested. Consistent with this incentive, Poterba (2004) finds evidence from U.S. data for 1935–2002 that a higher tax burden on dividends reduces long-run dividend payouts.

To incorporate both features, first convert R to the nominal gross return $(1+\pi)R$, then deduct dividend payouts. The return retained in each asset is

$$\widetilde{R} = 1 + (1-\varphi)\big[(1+\pi)R - 1\big]. \tag{16}$$

For the nominal portfolio w after returns and dividend payouts but before trading, the dividend flow is

$$d(w) = \left(\frac{(1+\pi)(1+\mu)}{\mathbb{E}[\widetilde{R}]} - 1\right) \mathrm{M}(w). \tag{17}$$

The ratio in (17) compares the gross nominal return including dividends with the gross return retained in the assets. Subtracting one gives dividends per dollar of portfolio value after payout, before personal taxes. Thus dividends provide cash without an asset sale but face the same tax rate as realized gains. Under LUL, losses offset both dividend payouts and realized gains. The next proposition extends the perpetuity set in equation (7) of Section IV.A to include these two features.

Proposition 6 (Perpetuities with dividends and inflation). *Fix a tax operator* $\mathrm{T} \in \{\mathrm{T}_{\mathrm{FUL}}, \mathrm{T}_{\mathrm{LUL}}\}$, *a statutory tax rate* $\tau \in [0,1)$, *and a real period-return distribution* f *with* $\mu > 0$. *Assume the regularity conditions of Theorem 2, with the extension in Section IV.D under LUL, for the primitives expressed in real units. Let* Q *be the law of motion induced by* $\widetilde{R}$ *and set* $\chi = Q\boldsymbol{\delta}_1$. *For a nominal loss carryforward* F, *write* $K(\widetilde{w}, F; w)$ *for cash available after tax from sales and dividends:*

$$K(\widetilde{w}, F; w) = \mathrm{M}(\widetilde{w}) + d(w) - \begin{cases} \tau\big[\mathrm{CG}(\widetilde{w}) + d(w)\big], & \mathrm{T} = \mathrm{T}_{\mathrm{FUL}}, \\ \tau\big[\mathrm{CG}(\widetilde{w}) + d(w) - F\big]_+, & \mathrm{T} = \mathrm{T}_{\mathrm{LUL}}. \end{cases} \tag{18}$$

Under LUL, the next loss carryforward is $F' = \big[F - \mathrm{CG}(\widetilde{w}) - d(w)\big]_+$; *under FUL, set* $F = F' = 0$. *Then:*

(i) *The set of sustainable real withdrawals from a unit initial investment is*

$$\mathcal{C}(\mathrm{T}; f, \tau, \pi, \varphi) = \left\{ c \geq 0 : \begin{array}{l} \exists\, (w_t, \widetilde{w}_t, I_t, F_t)_{t\geq 1} \text{ with } w_1 = \chi,\ F_1 = 0, \\ 0 \leq \widetilde{w}_t \leq w_t, \quad I_t \geq 0, \\ K(\widetilde{w}_t, F_t; w_t) \geq c(1+\pi)^t + I_t, \\ w_{t+1} = Q(w_t - \widetilde{w}_t) + I_t\chi, \\ \text{with the loss-balance rule above, for all } t \geq 1 \end{array} \right\}. \tag{19}$$

(ii) *The maximal real perpetuity and its effective rate satisfy*

$$\bar{c}_M = \max \mathcal{C}(\mathrm{T}; f, \tau, \pi, \varphi), \qquad \tau^* = 1 - \frac{\bar{c}_M}{\mu}. \tag{20}$$

Proposition 6 identifies the maximal perpetuity as the maximum of (19); the effective rate then follows from Definition 5. Setting $\varphi = \pi = 0$ recovers the earlier problem.

Finally, the special case in which f has no cross-sectional dispersion is worth highlighting. The investor then funds a perpetuity by selling fractions of a single diversified fund with a deterministic return. Deferral remains possible, but there is no choice among individual gains and losses. The quantitative results include this benchmark.

Statutory-rate pass-through. Holding market conditions fixed, $\partial\tau^*/\partial\tau$ measures how much a statutory-rate increase raises the effective rate. Pass-through incorporates the investor's deferral response, a margin that has been harder to quantify than the well-studied response of total investment (Alan et al., 2010, Poterba, 2002). An accrual tax at $\tau^*(\tau)$ sustains the same maximal perpetual consumption as realization taxation at τ. The investor is therefore indifferent between the two economies under this criterion. Revenue in the consumption-equivalent economy is $R^*(\tau) = X(\tau)\mu\tau^*(\tau)$, where $X(\tau)$ is long-run investment. Its revenue semi-elasticity adds the proportional investment response and pass-through divided by the effective rate.[22] Thus an empirical investment response and the model's pass-through estimate can jointly determine this benchmark's revenue response.

Computation. To approximate the maximal perpetuity, I search over constant real withdrawals c and simulate the optimal trading rule for $T = 200$ periods. After T periods, I check whether residual wealth has remained nondecreasing or above the screening bound. If so, I raise c; otherwise, I lower it. Bisection therefore narrows the interval between accepted and rejected withdrawals. The perpetuity operator converges under backward iteration (Section IV.B), motivating a finite-T approximation (Online Appendix Section V.A.2 gives the algorithm). For the selected withdrawal, the same trading rule generates the portfolio holdings, tax payments, accrued returns, and loss carryforwards used later in Section VI.

Calibration. Table 1 reports the baseline parameters and sources. Since market conditions can change over long horizons, I choose historically informed benchmarks rather than match a particular sample's moments. Extensive comparative statics show how the results vary across environments. In particular, the effective rate changes little with expected returns over the ranges considered.

Quantitative results. At the baseline, LUL sustains annual real consumption of about 4.30% of initial investment, compared with 3.75% under accrual taxation at the statutory rate. The effective rate is 13.9%: optimal trading reduces the effective burden by about 44%. Changing to FUL, by comparison, lowers the rate only slightly, to 13.7%. Delaying the use of losses therefore costs little in this calibration.

[22] Holding μ fixed and assuming $X(\tau) > 0$ and $\tau^*(\tau) > 0$, $\partial_\tau \log R^* = (\partial_\tau X)/X + (\partial_\tau \tau^*)/\tau^*$.

Table 1: Baseline calibration

Parameter	Value	Source or assumption
Expected real total return, μ	5%	Historical U.S. equity-return evidence; Mehra and Prescott (2003, Table 1).
Nominal return dispersion, σ	30%	U.S. stock-return volatility, 1962–1997; Campbell et al. (2001, Section I.B and Table III).
Inflation, π	2%	Inflation targets of the Federal Reserve and ECB (European Central Bank, 2025, Federal Reserve, 2025).
Dividend share, φ	30%	Dividends account for about 31% of S&P 500 total returns, 12/1926–02/2025 (S&P Dow Jones Indices, 2025, Exhibit 1).
Statutory tax rate, τ	25%	OECD realization-tax benchmark; Mengden (2025, Table 2).
Annual death probability, β	3%	Assumed constant mortality in the cohort model, Section VI.

Returns and inflation are annual. Dispersion applies to nominal total returns before dividend payouts; φ is a share of returns, not earnings. Aggregate risk is absent, reinvestment is frictionless ($\delta_I = 1$), and LUL carryforward is perfect ($\delta_F = 1$). These are maintained assumptions.

Figure 3 shows how this gap varies with tax rates and market conditions. In the top row, greater dispersion creates more losses and small gains that can fund consumption while appreciated positions remain invested. This lowers effective taxation through loss harvesting and gain deferral. The zero-dispersion curve gives the single-fund benchmark described above.

The next two rows confirm the effects discussed above: higher inflation and dividend shares raise effective rates over the displayed range. Refunds and carryforwards produce similar curves, although refunds are more valuable at high dispersion. Given the weak sensitivity to expected returns noted above, the figure omits that comparison.

Turning to pass-through, the curves are approximately linear. The legend's OLS fits through the origin give slopes of 0.60 under LUL and 0.59 under FUL at baseline market conditions. Thus a one-percentage-point statutory-rate increase raises the effective rate by about 0.60 percentage points in this approximation. The slight upward curvature nevertheless implies greater local pass-through at higher rates.[23]

Borrowing. Wealthy investors may also borrow against their portfolios rather than sell appreciated assets. Online Appendix Section V.A.3 accommodates borrowing as a fixed proportion of initial own wealth. With non-deductible interest, borrowing changes sustainable consumption but leaves the effective rate unchanged. Deductible interest instead offsets taxable gains and can substantially lower the effective rate.

[23] At baseline market conditions, the pointwise LUL derivatives are approximately 0.55, 0.60, and 0.72 at statutory rates of 10%, 25%, and 50%, respectively. These local derivatives differ from the figure's OLS slopes over the full tax-rate grid.

V.B Return alphas

The MGET holds the expected return fixed and adjusts the accrual tax rate. Here I instead fix the accrual tax rate at the statutory realization rate τ and increase the expected return until the same perpetuity is sustainable under either tax system. This required increase is the *return alpha*: the expected-return equivalent of the advantage from optimal realization trading. It provides a theoretical counterpart to empirical measures of the gains from tax-aware trading (Chaudhuri et al., 2020, Mainardi, 2026).

I compare A (accrual), S (single asset), and M (many assets). Under A, real total returns face the statutory accrual tax. Under S, the investor liquidates a single diversified fund proportionally, deferring gains but not selecting individual gains and losses within it. Under M, the investor also chooses which tax lots to sell. As discussed above, M reduces to S as cross-sectional dispersion vanishes.

Let $\bar{c}_j$ denote sustainable consumption under regime j at common market and tax parameters. Since $\bar{c}_A = \mu(1-\tau)$, the *total alpha* for many assets, α_M, and the *single-fund alpha*, α_S, are

$$\alpha_M := \frac{\bar{c}_M}{1-\tau} - \mu, \qquad \alpha_S := \frac{\bar{c}_S}{1-\tau} - \mu. \tag{21}$$

Subtracting the single-fund alpha from the total alpha isolates the additional contribution of tax-lot selection:

$$\Delta\alpha := \alpha_M - \alpha_S = \frac{\bar{c}_M - \bar{c}_S}{1-\tau}. \tag{22}$$

An extra unit of accrual-taxed return yields $1-\tau$ units of consumption. Dividing both consumption gains by this factor therefore gives common return units, so the components add exactly.

Substituting $\bar{c}_M = \mu(1-\tau^*)$ into (21) gives $\alpha_M = \mu(\tau - \tau^*)/(1-\tau)$. The total alpha and the MGET are therefore dual measures of the same consumption advantage: one expresses it in expected-return units, the other in tax-rate units.

Quantitative results. At the baseline in Table 1 and under LUL, the total alpha α_M is 73.8 bp per year. The single-fund alpha is 44.4 bp, and tax-lot selection adds 29.4 bp, about 40% of the total. These values have a similar order of magnitude to historical estimates of tax alpha, although the tax provisions and comparison criteria differ.[24]

Figure 4 reports total alpha, single-fund alpha, and their difference under LUL. In the top row, single-fund consumption is unchanged as dispersion rises, so the increase in total alpha comes entirely from tax-lot selection. At zero dispersion, the difference is zero.

The remaining rows show that inflation and dividends reduce total alpha mainly by reducing the benefit of single-asset deferral. Their effect on the contribution of tax-lot selection is smaller over the displayed range. Higher statutory rates raise both contributions.

[24] Chaudhuri et al. (2020) estimate a tax-loss-harvesting alpha of 108 bp before transaction costs, or 82 bp when wash-sale restrictions apply. Their historical strategy comparison is not a direct calibration target for the consumption equivalents here.

FUL gives similar results, with a baseline total alpha of 75.4 bp. Immediate refunds modestly increase total alpha relative to LUL, so separate FUL panels are omitted.

V.C Discussion

My results show that statutory rates alone are a poor basis for comparing realization-taxed capital gains with accrual-taxed labor income. At the baseline, a 25% realization tax is equivalent to a 13.9% accrual tax. Conversely, matching the 35.1% OECD labor tax wedge cited in the introduction requires a realization rate of about 56.6%. The gap is wider with more return dispersion and smaller with more inflation or dividends, while expected real returns have little effect on the MGET over the ranges considered. In return units, the baseline advantage is 73.8 bp per year, with about 40% attributable to tax-lot selection.

Finally, the measures developed here allow models with accrual taxation to reflect the advantage of realization without tracking tax lots. One can either retain τ and raise the expected return by α_M, or retain μ and lower the accrual tax rate to τ^*. These adjustments are alternatives; for single-fund deferral alone, use α_S or its corresponding rate. Unlike the mark-to-market heuristic of Fischer et al. (2013), this approach adjusts the tax rate or return in the accrual-tax model itself. In a life-cycle model, the adjustment need not precisely match consumption paths or welfare, but it offers a tractable way to reflect the different effective tax burdens on realization-taxed gains and accrual-taxed labor income.

VI Government-side effective tax rates

To measure the revenue cost of realization taxation, I build a cohort model from the perpetuity policies. Two issues arise. First, tax payments vary with portfolio age and must be aggregated into a constant revenue-equivalent rate. Second, tax collection may reduce invested wealth and future taxable returns, offsetting revenue gains. The comparison must therefore distinguish the timing of taxation from changes in the economy's tax base.

Model setup. The demographic structure follows Yaari (1965) and Blanchard (1985). Investors face annual mortality $\beta = 3\%$ (Table 1) and leave accidental bequests. While alive, investors retain the optimal perpetuity policy: mortality changes aggregation, not their consumption or trading choices.[25] The main results use LUL with perfect carryforward and frictionless reinvestment.

I consider three treatments of gains at death. Under *step-up*, inherited assets receive market-value basis, forgiving embedded gains. Under *carry-over*, by contrast, heirs retain the basis and unused losses, continuing the portfolio's tax history. Under *constructive realization*,

[25] Dammon et al. (2001a) study consumption and investment choices over the life cycle under capital gains taxation. Bennhoff and Scheuer (2026) use the Tax Portfolio Model to study such choices with many assets and aggregate risk.

remaining gains are taxed after loss offsets and inherited assets receive market-value basis. Unused losses left after either basis reset expire.

Recycling. To compare tax systems on the same real accrual base, I *recycle* consumption payments and tax receipts: both go to entering investors and are reinvested before the next return. Withdrawals therefore reduce individual portfolios but not aggregate wealth. Estates also remain invested, starting new tax histories after a basis reset or continuing with heirs under carry-over.[26]

This addresses the tax-base issue: aggregate accumulation is held fixed rather than determined by how public revenue is spent. Recycling thus isolates the tax comparison from spending effects and avoids taking a stance on the productivity of public spending.

Consequently, opening real wealth satisfies $W_{t+1} = (1+\mu)W_t$ under every tax system, with real accruals $A_t = \mu W_t$ including dividends. Given the same initial wealth, any realization-tax economy and any accrual-tax economy thus share the same wealth path and real accrual base.

Aggregation. To aggregate receipts across portfolio ages, I reuse the unit-investment path computed in Section V.A for each loss treatment; no new optimization is needed. At portfolio age a, this path gives consumption $c_a = \bar{c}_M$, living taxes tax_a, and newly accrued returns accr_a. Here accr_a is a one-period flow, including dividends. The portfolio and loss carryforward also determine dtax_a: the taxes the investor would pay if she died after trading at age a. All flows are in real units. Portfolio paths, tax payments, accrued returns, and entrant endowments depend on the statutory realization rate τ. I suppress this dependence on the right-hand sides below.

The perpetuity problem is homogeneous of degree one in initial wealth. Write N_{t-a} for the endowment of a cohort entering at date $t-a$. This cohort generates N_{t-a} times the unit-investment flows. Under step-up and constructive realization, a fraction $(1-\beta)^a$ of the cohort survives to age a. Total tax receipts $T_t(\tau)$ and newly accrued real returns A_t are therefore

$$\begin{aligned} T_t(\tau) &= \sum_{a\geq 0} (1-\beta)^a N_{t-a}\big(\text{tax}_a + \beta\,\text{dtax}_a\big), \\ A_t &= \sum_{a\geq 0} (1-\beta)^a N_{t-a}\,\text{accr}_a\,. \end{aligned} \tag{23}$$

Taxes paid at death satisfy $\text{dtax}_a = 0$ under both step-up and carry-over. Under constructive realization, I apply τ to the remaining nominal embedded gains after trading. Under LUL, unused losses are deducted and only the positive balance is taxed; the payment is then expressed in real units. Step-up retains $(1-\beta)^a$ in (23) because death resets basis and starts a new tax history. Carry-over instead replaces this factor by one because tax histories continue with heirs. Thus N_t includes inherited wealth under step-up but excludes it under carry-over.

[26] For intuition, imagine durable fashion items that give their owner utility for one period. Once the owner tires of them, an entering investor receives them and finds them useful. Government agents likewise wear tax-financed items for one period before passing them on. Items held in portfolios also produce new items, like trees bearing fruit. Consumption and taxes then record uses and transfers, not the destruction of resources.

Revenue equivalence. On the balanced-growth path, endowments at entry grow at the gross rate $1+\mu$. Hence

$$N_{t-a} = N_t(1+\mu)^{-a}, \qquad a \geq 0. \tag{24}$$

Substituting (24) into (23) makes both aggregates proportional to N_t, with weights q^a. Here $q = (1-\beta)/(1+\mu)$ under step-up or constructive realization; under carry-over, $q = 1/(1+\mu)$ and $\text{dtax}_a = 0$. At fixed τ, the weights q^a and unit-investment flows tax_a, dtax_a, and accr_a do not depend on calendar time t. Thus the cohort aggregation gives a constant ratio $T_t(\tau)/A_t$. The government-side effective rate is the tax rate on newly accrued real returns that raises the same revenue as realization taxation at rate τ. It is therefore

$$\tau_G^*(\tau) := \frac{T_t(\tau)}{A_t} = \frac{\sum_{a\geq 0} q^a\left(\text{tax}_a + \beta\,\text{dtax}_a\right)}{\sum_{a\geq 0} q^a\,\text{accr}_a}. \tag{25}$$

By construction, taxing current real accruals at $\tau_G^*(\tau)$ collects $\tau_G^*(\tau)A_t = T_t(\tau)$, the revenue from realization taxation at rate τ. This establishes revenue equivalence, whereas the investor-side MGET in Definition 5 establishes consumption equivalence.

The equivalence is stronger than a comparison of current receipts. Starting from the realization-tax economy's balanced-growth path, a permanent switch at date t to taxing newly accrued real returns at the fixed rate $\tau_G^*(\tau)$ leaves revenue unchanged at every subsequent date $t+s$. Recycling preserves the future accrual base, so $\tau_G^*(\tau)A_{t+s} = T_{t+s}(\tau)$ for every $s \geq 0$, where the right-hand side is revenue if realization taxation continues. This comparison excludes a separate tax on pre-switch gains. Proposition VI.A.1 in Online Appendix Section VI.A gives the statement and proof.

To compute (25), I use the unit-investment tax and accrual flows generated by the optimal-policy calculation in Section V.A. I sum these flows over 200 periods with weights q^a and take their ratio. The common factor N_t cancels, so neither its level nor the date t is needed.

Quantitative results. At the baseline statutory rate of 25%, the government-side effective rates are about 9.1% under step-up, 12.8% under carry-over, and 17.3% under constructive realization. Constructive realization thus raises about 1.9 times as much revenue as step-up, although its effective rate remains well below the statutory rate. Relative to accrual taxation at the same statutory rate, the LUL revenue shortfalls are 64%, 49%, and 31%, respectively.[27]

Figure 5 shows that this ranking persists across the displayed market conditions and statutory rates. The columns compare death provisions; the rows vary expected returns, dispersion, inflation, and dividends.

Higher expected returns have little effect on the carry-over rate, but raise the step-up rate and lower the constructive-realization rate over the displayed range. They change both tax payments and real accruals along each portfolio history. In addition, faster aggregate growth reduces the weight of older histories in (25). The balance of these effects differs across the three treatments at death. Revenue rates can therefore change even when the investor's effective rate changes little.

[27]The corresponding FUL revenue shortfalls are 66%, 50%, and 33% (Table VI.B.1).

The remaining rows largely follow the investor-side results. Greater dispersion lowers revenue because investors can realize losses and small gains while retaining appreciated positions. This effect is strongest under step-up, where gains still embedded at death escape taxation. Inflation and dividends work in the opposite direction: inflation raises nominal taxable gains relative to real accruals, while dividends bring more returns into current taxation. Both raise government effective rates under all three death provisions.

FUL gives similar baseline revenue, so Figure 5 concentrates on LUL. Online Appendix Section VI.B compares loss treatments, including the case without dividends or inflation where they differ more.

Discussion. The baseline government-side effective rate is close to the investor-side MGET. With LUL and carry-over basis at death, the revenue-equivalent rate is 12.8%, compared with a consumption-equivalent rate of 13.9%.

A further implication is that capital gains can remain perpetually deferred at the aggregate level, even without forgiveness at death. Aguiar et al. (2026) argue that borrowing against appreciated assets ultimately requires sales if the appreciation is to finance consumption or repay loans. At first sight, this seems inconsistent with perpetual deferral.

The resolution lies in the distinction between individual assets and the aggregate portfolio. Investors fund consumption by realizing losses and selling the least appreciated positions, while retaining more appreciated assets. Individual assets can eventually be sold, but their positions in the distribution of cumulative returns change over time. As gains are realized, new unrealized gains take their place. Thus, the aggregate stock of unrealized gains can remain a constant share of wealth without any particular gain having to remain deferred forever.[28]

At the baseline calibration, unrealized gains account for 55.6% of aggregate portfolio wealth under step-up, 66.9% under carry-over, and 55.6% under constructive realization, all under LUL.[29] For comparison, Msall and Næss (2026, Table 1, Panel (a)) estimate that unrealized gains comprise 63% of Norwegian wealth in listed and unlisted shares and taxable real estate, a share in the same ballpark as the model's despite differences in asset composition. Continual deferral sustains these large stocks of untaxed gains and keeps revenue below the accrual benchmark.

[28] For a simple derivation, set $\pi = \varphi = 0$, assume FUL and carry-over basis, and let H_t denote aggregate unrealized gains. Then $H_{t+1} - H_t = A_t - T_t/\tau$. On the balanced-growth path, write $H_t = hW_t$, with constant $h \in [0, 1)$. Because $W_{t+1} = (1+\mu)W_t$ and $A_t = \mu W_t$, we obtain $H_{t+1} - H_t = hA_t$ and hence $\tau_G^*/\tau = 1 - h$. Nothing requires $h = 0$.

[29] Under FUL, the corresponding shares are 55.6%, 66.8%, and 55.6%. Let u_a and v_a denote unrealized gains and market value at the same post-trading point on the real unit-investment path. The cohort-weighted share is $h = (\sum_{a=0}^{T-1} q^a u_a)/(\sum_{a=0}^{T-1} q^a v_a)$, with $q = (1-\beta)/(1+\mu)$ under step-up or constructive realization and $q = 1/(1+\mu)$ under carry-over. Both stocks use $T = 200$ periods, and new tax histories begin with a fresh basis after a basis reset.

VII Conclusion

Investors' discretion over realizations substantially changes the effective burden of capital gains taxation. At the baseline, a statutory rate of 25% translates into an investor-side effective rate of about 14%, while government revenue under carry-over basis is 49% below what accrual taxation at the same statutory rate would collect. These advantages can persist indefinitely as investors continually postpone realizing the most appreciated remaining gains. The return-alpha decomposition attributes about 40% of the investor advantage to selecting realizations across multiple assets; the remaining 60% comes from deferral within a single diversified fund. These results highlight why models used to compare capital and labor taxation and assess tax progressivity need to account for investors' choices across multiple assets.

The theoretical results clarify the conditions under which familiar tax-aware trading rules are optimal. Under suitable tax and return conditions, harvesting losses and selling the least appreciated remaining positions is justified independently of investor preferences. Outside these conditions, preferences can matter for the structure of trading, and aggregate downside risk can favor selling appreciated positions and retaining losses.

One direction for future work is to identify combinations of statutory rates, loss provisions, and treatments of inherited gains that raise revenue most effectively once investors adjust their trading. Another is to study who benefits from realization discretion. Differences in liquidity needs, portfolio composition, and access to tax management may generate different effective burdens under identical statutory rules, with consequences for wealth concentration and its transmission across generations. In ongoing work, Bennhoff and Scheuer (2026) build on the framework in a quantitative life-cycle model with many assets and aggregate risk. Their aim is to connect optimal trading, saving, and asset-price responses to the distributional and welfare effects of capital taxation.

Fig. 3. Investor-side effective tax rates

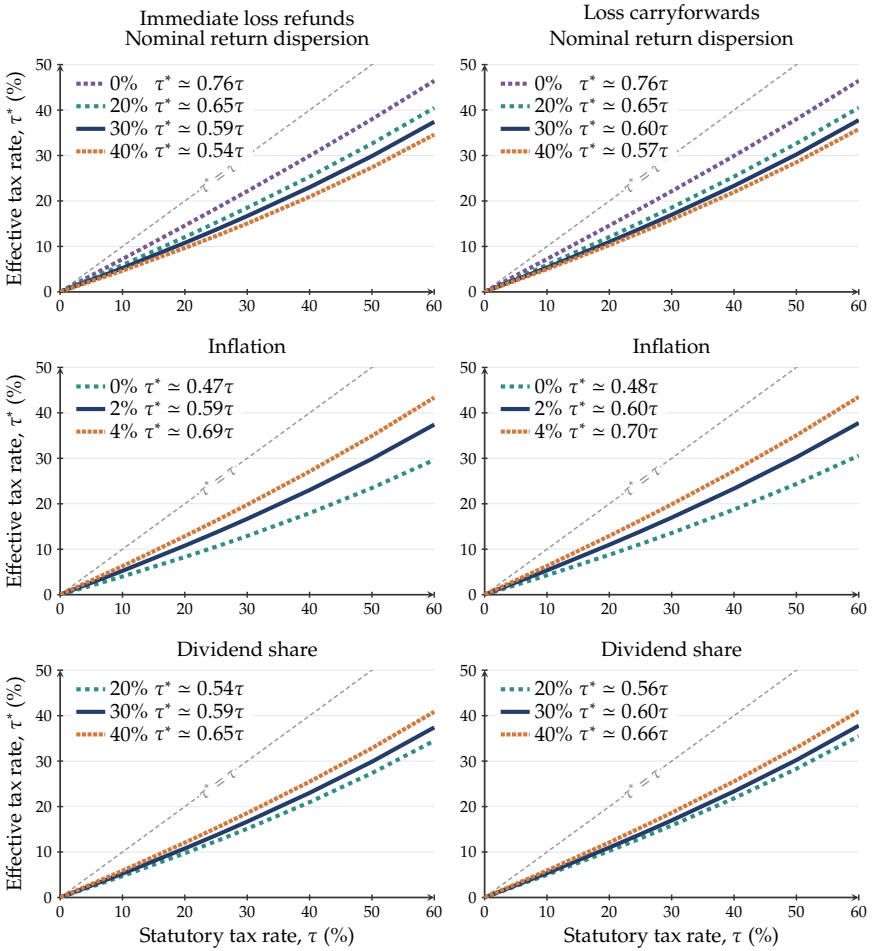
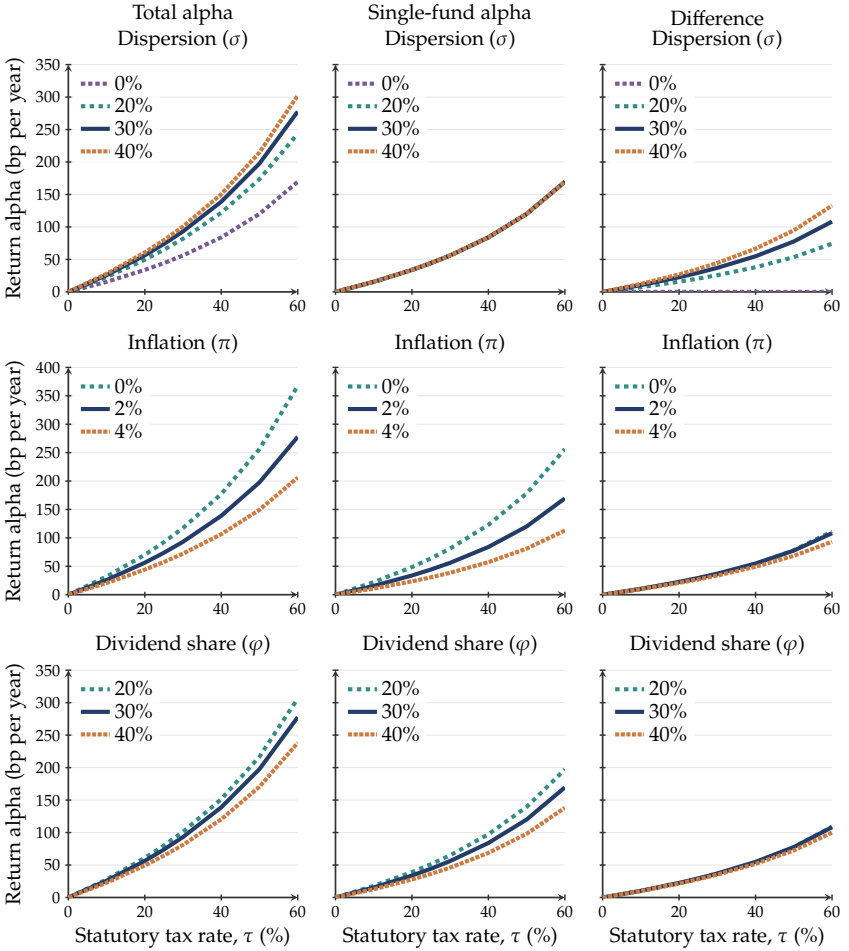
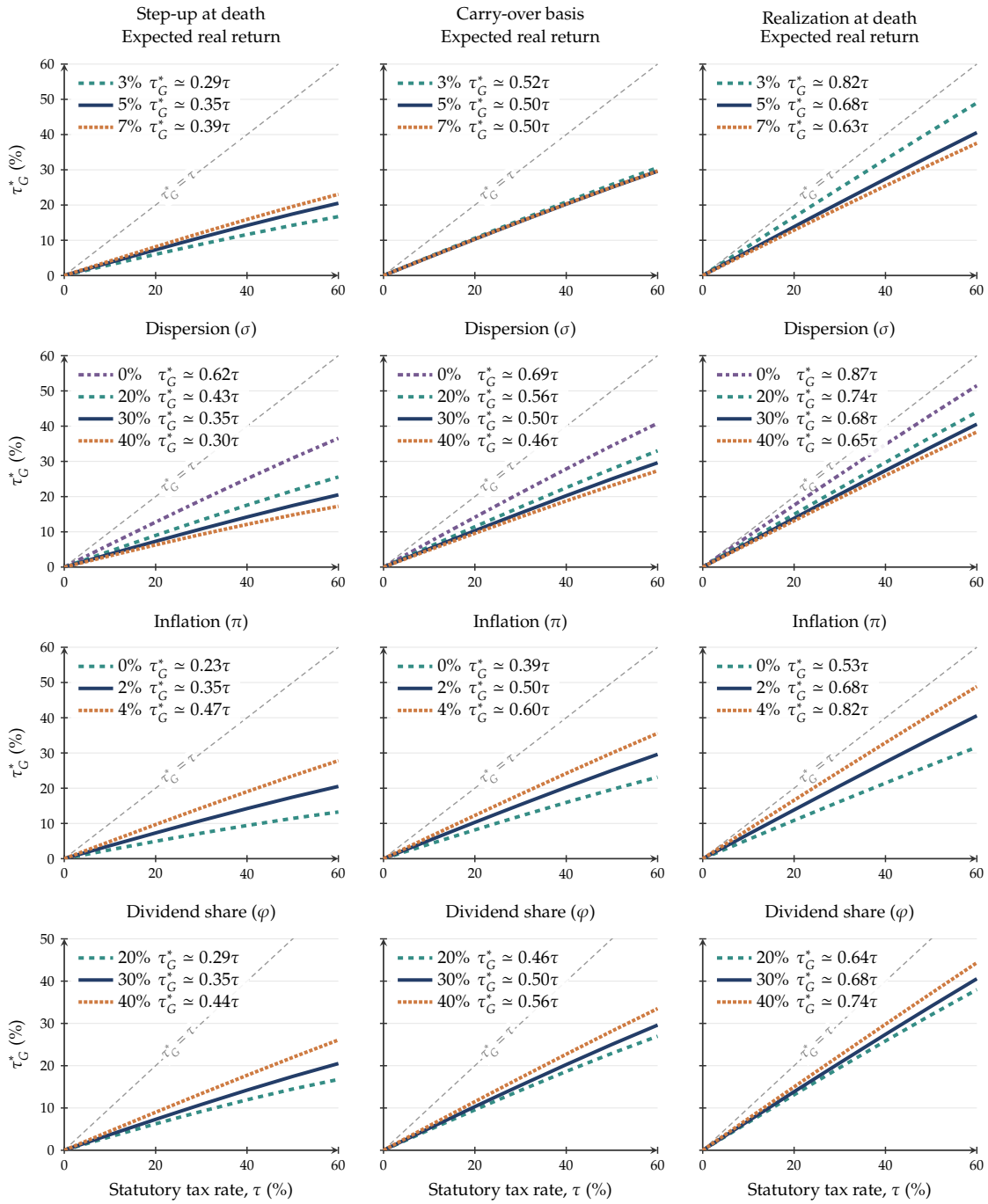


Fig. 4. Return alphas under limited use of losses



Notes: Both figures vary the statutory rate along the horizontal axis. Rows vary nominal return dispersion, inflation, and dividend share. Baseline parameters are $\mu = 5\%$, $\sigma = 30\%$, $\pi = 2\%$, and $\varphi = 30\%$; other parameters remain at baseline. Reinvestment is frictionless and LUL carryforward is perfect. Figure 3 compares FUL and LUL and reports $\tau^* = 1 - \bar{c}_M/\mu$; its legend equations are least-squares fits through the origin over the displayed tax-rate grid, not marginal derivatives. Figure 4 uses LUL. Its columns show total alpha $\alpha_M = \bar{c}_M/(1-\tau) - \mu$, single-fund alpha $\alpha_S = \bar{c}_S/(1-\tau) - \mu$, and their difference $\Delta\alpha = (\bar{c}_M - \bar{c}_S)/(1-\tau)$, in bp per year. Single-fund curves coincide across dispersion values. Scales are common across columns within each row.

Fig. 5. Government-side effective rates under limited use of losses



Notes: The tax system is LUL with perfect carryforward. Columns compare step-up, carry-over, and constructive realization at death. Rows vary the expected real return (3%, 5%, 7%), nominal return dispersion (0%, 20%, 30%, 40%), inflation (0%, 2%, 4%), and dividend share (20%, 30%, 40%). Other parameters remain at $\mu = 5\%$, $\sigma = 30\%$, $\pi = 2\%$, $\varphi = 30\%$, and annual mortality $\beta = 3\%$. The vertical axis is $\tau_G^*(\tau) = T_t(\tau)/A_t$, with A_t the real accrual base. Taxes and consumption are recycled. Legend equations are least-squares fits through the origin over the displayed tax-rate grid; they are not marginal derivatives. Scales are common across columns within each row.

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Online Appendix to the Paper

The Untaxables? Effective Capital Gains Taxes with Diversified Portfolios

The appendix groups proofs and supplementary material by the corresponding sections of the main paper, with Appendix IV also following the subsection structure of Section IV.

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II Tax Motives Dominate in Large Portfolios

II.A Proofs of the Large-Portfolio Results

II.A.1 Proof of Theorem 1: Asymptotic Sufficiency of the Tax-Lot Distribution

Proof of Theorem 1. For each tax lot r, set

$$w_\alpha^n(r) := \sum_{i \in I^n(r)} \alpha_i^n \omega_i^n, \qquad \bar{w}_{\bar{\alpha}}^n(r) := \sum_{i \in I^n(r)} \bar{\alpha}_i^n \bar{\omega}_i^n.$$

Let

$$B_{\text{tot}}^n := \sum_i \frac{\omega_i^n}{r_i}, \qquad \bar{B}_{\text{tot}}^n := \sum_i \frac{\bar{\omega}_i^n}{r_i}.$$

Because the set of tax lots is finite and both endowment sequences converge lot by lot to the same limit w, we have $B_{\text{tot}}^n - \bar{B}_{\text{tot}}^n \to 0$.

Assume statement 1. Then

$$\mathrm{M}_1^n - \bar{\mathrm{M}}_1^n \to 0, \qquad \mathrm{CG}_1^n - \bar{\mathrm{CG}}_1^n \to 0.$$

Continuity of T_1 gives $c_1^n - \bar{c}_1^n \to 0$.

For period 2, write $\mu_Z := Z(1+\mu)$ and define the residuals $\varepsilon_\alpha^n := \sum_i (1-\alpha_i^n)\omega_i^n (ZR_i^n - \mu_Z)$ and $\varepsilon_{\bar{\alpha}}^n := \sum_i (1-\bar{\alpha}_i^n)\bar{\omega}_i^n (Z\bar{R}_i^n - \mu_Z)$. The common-mean condition makes both residuals conditionally mean zero. Moreover, $\mathrm{M}_2^n = \mu_Z(M - \mathrm{M}_1^n) + \varepsilon_\alpha^n$ and $\bar{\mathrm{M}}_2^n = \mu_Z(M - \bar{\mathrm{M}}_1^n) + \varepsilon_{\bar{\alpha}}^n$, where M is the common total market value.

Conditional on Z, the covariance matrix of $Z\boldsymbol{R}^n$ is $Z^2\Sigma^n$, so

$$\begin{aligned}\mathbb{E}[(\varepsilon_\alpha^n)^2 \mid Z] &\le Z^2 \lambda_{\max}(\Sigma^n) \sum_i ((1-\alpha_i^n)\omega_i^n)^2 \\ &\le Z^2 \lambda_{\max}(\Sigma^n) M \max_{i \le n} \omega_i^n.\end{aligned}$$

The analogous bound uses $\bar{\Sigma}^n$ for $\varepsilon_{\bar{\alpha}}^n$. Balancedness, diversifiability, and $\mathbb{E}[Z^2] < \infty$ therefore imply that both residuals converge to zero in L^2. It follows that $\mathrm{M}_2^n - \bar{\mathrm{M}}_2^n = -\mu_Z(\mathrm{M}_1^n - \bar{\mathrm{M}}_1^n) + \varepsilon_\alpha^n - \varepsilon_{\bar{\alpha}}^n \to 0$ in L^2.

Next,

$$B_1^n := \mathrm{M}_1^n - \mathrm{CG}_1^n = \sum_i \frac{\alpha_i^n \omega_i^n}{r_i}, \qquad \bar{B}_1^n := \bar{\mathrm{M}}_1^n - \bar{\mathrm{CG}}_1^n,$$

so $B_1^n - \bar{B}_1^n \to 0$. Since

$$\mathrm{CG}_2^n = \mathrm{M}_2^n - (B_{\text{tot}}^n - B_1^n), \qquad \bar{\mathrm{CG}}_2^n = \bar{\mathrm{M}}_2^n - (\bar{B}_{\text{tot}}^n - \bar{B}_1^n),$$

we also have $\mathrm{CG}_2^n - \bar{\mathrm{CG}}_2^n \to 0$ in L^2. Hence

$$(\mathrm{M}_2^n, \mathrm{CG}_2^n, \mathrm{CG}_1^n) - (\bar{\mathrm{M}}_2^n, \bar{\mathrm{CG}}_2^n, \bar{\mathrm{CG}}_1^n) \to 0 \quad \text{in probability.}$$

By continuity of T_2, $c_2^n - \bar{c}_2^n \to 0$ in probability. Therefore

$$u(c_1^n, c_2^n) - u(\bar{c}_1^n, \bar{c}_2^n) \to 0 \quad \text{in probability.}$$

The maintained boundedness and integrability conditions provide uniform integrability of the utility differences. Their convergence to zero in probability therefore implies $|U^n - \bar{U}^n| \to 0$, proving statement (ii). ■

II.A.2 Proof of Corollary 1: $\sqrt{n}$-Decay of Security-Selection Effects

Proof of Corollary 1. Write $v_i := (\alpha_i^n - \bar{\alpha}_i^n)\omega_i^n / M$. Identical lot-level liquidations imply

$$\sum_i v_i = \sum_i \frac{v_i}{r_i} = 0,$$

so current consumption is identical and, under linear taxes,

$$g_2^n = (1-\tau) Z \sum_i v_i R_i^n.$$

The common-mean assumption gives $\mathbb{E}[g_2^n \mid Z] = 0$. Moreover,

$$\sum_i v_i^2 \le \sum_i \left(\frac{\omega_i^n}{M}\right)^2 \le \frac{\max_i \omega_i^n}{M} \le \frac{C}{n}.$$

The covariance bound therefore yields

$$\mathrm{sd}(g_2^n) \le (1-\tau)\sqrt{\frac{\bar{\lambda} C\, \mathbb{E}[Z^2]}{n}}.$$

Finally, bounded marginal utility in the second argument and identical current consumption imply

$$|U^n - \bar{U}^n| \le LM\, \mathbb{E}|g_2^n| \le LM\, \mathrm{sd}(g_2^n) = O(n^{-1/2}).$$

■

III The Tax Portfolio Model

III.A Portfolios, Trading, and Taxation

III.A.1 Proof of Lemma 1: Laws of Motion for Portfolio Statistics

Proof of Lemma 1. Set $x := w - \widetilde{w}$ and $u := Qx + \delta_I I\chi$. Tonelli's theorem and the substitutions $r = sR$ and $r = zs$ give $\mathrm{M}(Qx) = \mathbb{E}[R]\,\mathrm{M}(x)$, $\mathrm{B}(Qx) = \mathrm{B}(x)$, $\mathrm{M}(A_z u) = z\,\mathrm{M}(u)$, and $\mathrm{B}(A_z u) = \mathrm{B}(u)$. Because $\chi = Q\boldsymbol{\delta}_1$, also $\mathrm{M}(\chi) = \mathbb{E}[R]$ and $\mathrm{B}(\chi) = 1$. Substitution yields the first two identities; $\mathrm{CG} = \mathrm{M} - \mathrm{B}$ and $\mathrm{K}_{\mathrm{FUL}} = \mathrm{M} - \tau\,\mathrm{CG}$ yield the remaining two. ■

III.A.2 Heterogeneous Factor Exposures

The baseline gives assets a common marginal return distribution. This extension allows heterogeneous exposure to an aggregate factor while retaining a common idiosyncratic return law (cf. Al-Najjar, 1998, Khan and Sun, 2003, Lintner, 1965, Sharpe, 1964). Let R_i have density f, let z be the realization of the aggregate factor, and let asset i have loading β_i. Its gross return is $(1 + \beta_i z)R_i$, with $1 + \beta_i z > 0$.

At the current trading date, suppose the same market-value-weighted distribution of loadings, q, applies within every tax lot and to new investment. Trades within each tax lot preserve this distribution. Reinvestment efficiency is δ_I, equal to one in the baseline. Let Q apply the idiosyncratic return law to the portfolio, and let $\chi = Q\boldsymbol{\delta}_1$ be the resulting portfolio from a unit investment at basis. The operator A_s scales market values and cumulative gross returns by s. Under the stated independence condition, the one-period law of motion is

$$w' = \int A_{1+\beta z}\big(Q(w - \widetilde{w}) + I\delta_I\chi\big)\, q(\beta)\,\mathrm{d}\beta.$$

Conditional on z, total market value therefore satisfies

$$\mathrm{M}(w') = \big(1 + \mathbb{E}_q[\beta]z\big)\mathbb{E}_R[R]\big(\mathrm{M}(w - \widetilde{w}) + \delta_I I\big).$$

Thus the mean loading determines aggregate exposure, whereas the distribution of loadings determines the additional dispersion of returns across assets.

Replacing a common loading $\bar{\beta}$ with a nondegenerate distribution q satisfying $\mathbb{E}_q[\beta] = \bar{\beta}$ preserves the aggregate return conditional on z. For nonzero z, however, it increases cross-sectional return dispersion. This creates more opportunities to sell losses or small gains while leaving appreciated positions invested. The calculations in Section V.A illustrate this mechanism with exogenous return dispersion.

This simplification has a limitation: representing the portfolio by accrued returns alone requires the same loading distribution in every tax lot at every trading date. With fixed heterogeneous loadings, realized returns generally make loadings and accrued returns correlated. The condition at one date therefore need not hold later. When it fails, the state must record portfolio value jointly over (r, β).

III.A.3 Short Selling

Short selling offers another way to create tax losses. Consider a dollar long and a dollar short in disjoint diversified portfolios with the same distribution of factor exposures. Their aggregate exposures offset before liquidation, but both legs contain gains and losses. Conditional on z, let f_z be their common cross-sectional density of gross returns $\mathcal{R}$. The market-value densities of the long positions and the securities owed on the short leg are

$$w_{\text{long}}(r \mid z) = w_{\text{short}}(r \mid z) = r f_z(r).$$

Selling depreciated long positions and covering appreciated securities on the short leg realizes a total loss of

$$\begin{aligned} L(z) &= -\int_0^1 w_{\text{long}}(r \mid z)(1 - 1/r)\,\mathrm{d}r + \int_1^\infty w_{\text{short}}(r \mid z)(1 - 1/r)\,\mathrm{d}r \\ &= \mathbb{E}[|\mathcal{R} - 1| \mid z]\,. \end{aligned} \tag{III.A.1}$$

Without aggregate risk, this becomes $L = \mathbb{E}[|R - 1|]$ under the baseline return law.

I call this construction *shorting the aggregate box*. Scaling both legs by a generates $aL(z)$ of realizable losses while the winning positions remain open. In the baseline case with $L > 0$, a position of size G/L on each leg generates losses equal to a realized gain G. These losses can offset gains realized elsewhere in the portfolio.

The construction uses distinct securities on the two legs. By contrast, shorting against an individual asset offsets that same appreciated position. The calculation abstracts from financing costs, dividends, and margin requirements. Selectively closing the losing positions can also change the remaining aggregate exposure.

IV The Recursive Framework

IV.A Two Recursive Problems

This appendix accompanies Section IV.A. It collects the portfolio and tax-share assumptions, proves the perpetuity interpretation, and summarizes the methods used below.

IV.A.1 Portfolio Space and Primitive Conditions

All Tax Portfolio Model primitives are inherited from Section III: portfolios w, order intervals $[0, w]$, operators M, B, CG, K, Q, A_z, G_z, parameters f, χ, δ_I, and feasible set Γ. The recursive analysis restricts these objects to a weighted state space and uses the following regularity conditions. To define the state space, fix exponents $\alpha < 0 < 1 < \gamma$ and let $\phi(r) := r^{\alpha}\mathbf{1}_{(0,1]}(r) + r^{\gamma}\mathbf{1}_{(1,\infty)}(r)$ for $r > 0$.

The portfolio space on which we interpret the TPM is the positive cone of the weighted function space $\mathcal{L}^{\infty}_{\phi}$, equipped with the *portfolio norm* $\|\cdot\|_{\infty,\phi}$:

$$\mathcal{W} := \mathcal{L}^{\infty}_{\phi,+}, \qquad \|w\|_{\infty,\phi} := \operatorname*{ess\,sup}_{r>0} |w(r)|\phi(r),$$
$$\mathcal{L}^{\infty}_{\phi} := \Big\{ w : (0,\infty) \to \mathbb{R} \text{ measurable} : \|w\|_{\infty,\phi} < \infty \Big\}.$$

The portfolio law of motion maps $\mathcal{W}$ into itself if return distributions do not send too many assets close to $r = 0$ or $r = \infty$, which would imply unbounded liquidation values even under linear taxes. Assumption 1 precludes this and is easy to satisfy—for example, with log-normal returns. To stay in $\mathcal{W}$ at all times, the Dirac-measure investments are as before passed through Q and create the investment profile χ.

Assumption 1 (Law of motion, aggregate risk, and investment profile)**.** *The return density and investment profile satisfy $m_Q := \mathbb{E}[\phi(R)] < \infty$ and $\|\chi\|_{\infty,\phi} < \infty$. For the finite-support aggregate-risk distribution from Definition 3 and $p \in [0,1]$, set $m_{A,p} := \sum_z p_z \phi(z)^p$ and $m_{A,0} := 1$.*

For tax operators T in the tax-share family, liquid resources are generated by a lot-specific payoff share $k = 1 - t$, so $\mathrm{K}(x) = \int_0^{\infty} x(r)k(r)\,dr$. The fixed-point and attainment results in Section IV.B use only the regularity conditions collected in Assumption 2. Under these conditions, market value and liquidation value are positive and bounded on $\mathcal{L}^{\infty}_{\phi}$, with $\mathrm{M}(w) \le c_{\mathrm{M}}\|w\|_{\infty,\phi}$ and $\mathrm{K}(w) \le c_{\mathrm{K}}\|w\|_{\infty,\phi}$; see Proposition IV.B.1 in the Online Appendix.

Assumption 2 (Tax-share liquidation regularity)**.** *The tax operator T belongs to the tax-share family, and its tax-share payoff function $k = 1 - t$ is nonnegative and belongs to $\mathcal{L}^1_{1/\phi}$. Equivalently, the induced liquidation functional $\mathrm{K} : \mathcal{L}^{\infty}_{\phi,+} \to [0,\infty)$ is positive, linear and weak-$*$ continuous. (Hence satisfies $0 \le \mathrm{K}(x) \le c_{\mathrm{K}}\|x\|_{\infty,\phi}$ for some $c_{\mathrm{K}} < \infty$, with $c_{\mathrm{K}} = \int_0^{\infty} k(r)\phi(r)^{-1}\,dr$.)*

IV.A.2 Perpetuity Interpretation

The proof uses the finite-horizon continuity and attainment results established in Appendix IV.B.5, together with compactness of feasible controls. It does not use the stationary fixed-point result.

Proof of Corollary 2. With no aggregate risk, finite-horizon attainment and induction on n show that $V_n^{\mathbb{P}}(w)$ is the largest constant consumption sustainable for the current period and n further periods. These values are nonnegative and decrease in n, so write $L(w) := \lim_{n\to\infty} V_n^{\mathbb{P}}(w)$. Every feasible perpetuity is bounded above by each finite-horizon value, hence $\sup \mathcal{C}(w) \leq L(w)$.

Conversely, fix $0 \leq c < L(w)$ and define $S_n(c) = \{x : V_n^{\mathbb{P}}(x) \geq c\}$. At each $x \in \cap_n S_n(c)$, the controls that deliver at least c now and return the state to $S_n(c)$ form a nonempty nested family of weak-$*$ compact sets. Nonemptiness follows from $V_{n+1}^{\mathbb{P}}(x) \geq c$ and finite-horizon attainment. Each set is closed by finite-horizon continuity and compact because $\Gamma(x)$ is compact; these properties are established in Appendix IV.B.5 under Assumptions 1 and 2. The finite-intersection property gives a control that returns the state to $\cap_n S_n(c)$. Choosing such a control recursively sustains c forever. Letting $c \uparrow L(w)$ gives $\sup \mathcal{C}(w) \geq L(w)$. If $L(w) = 0$, the equality follows because zero consumption is feasible. ■

IV.A.3 Methods

The recursive analysis uses the methods of Stokey and Lucas (1989) and Bertsekas and Shreve (1978), the functional-analytic results of Aliprantis and Border (2006), and the weighted sup-norm contraction argument of Bertsekas (2012). These references provide the analytical foundations for convergence and policy attainment. I verify their applicability to the Tax Portfolio Model and establish the perpetuity interpretation. The optimal trading restrictions are derived from the model-specific dominance and replacement arguments developed here.

IV.B Convergence and Optimal Policies

This appendix accompanies Section IV.B. It gives the utility and contraction conditions, then proves convergence, shape preservation, and policy attainment.

IV.B.1 Utility, Weights, and Contraction Moduli

For any state space $\mathcal{S}$ and weight function $W : \mathcal{S} \to [1, \infty)$, define the weighted *value function norm* $\|\cdot\|_W$ and the corresponding function space by

$$\mathcal{B}_W(\mathcal{S}) := \{V : \mathcal{S} \to \mathbb{R} : \|V\|_W := \sup_{s \in \mathcal{S}} |V(s)|/W(s) < \infty\}.$$

Assumption 3 determines the admissible utility functions.

Assumption 3 (Period utility). *The period utility $u : [0, \infty) \to \mathbb{R}$ is continuous, increasing, and strictly concave. If $u(0) = -\infty$, we use the affine extension below a small threshold $\underline{c} > 0$. On the unaffected interval, it is one of the following:*

i) bounded: there is $\bar{u} < \infty$ such that $|u(c)| \leq \bar{u}$ for all $c \geq 0$;

ii) logarithmic: $u(c) = \log c$ on the economically relevant consumption region;

iii) sublinear: there are constants $a_u, b_u \geq 0$ and $p \in (0, 1]$ such that $|u(c)| \leq a_u + b_u c^p$ for $c \geq 0$.

Let $\lambda_{I,p}$ be the growth bound entering the contraction modulus $\beta\lambda_{I,p}$, and define the weight function of the value function space to be $W_p(w) := \xi + \|w\|^p_{\infty,\phi}$ with $\xi \geq 1$ for $p > 0$ and $W_p(w) := 1$, otherwise. All parameters α, γ, ξ, p are ingredients of $\lambda_{I,p}$, of which the first three can be chosen freely subject to their respective one-sided inequality. The possible choices of p are dictated by Assumption 3. Thus, $\lambda_{I,p}$ can be made close to its infimum, $\inf_{\alpha,\gamma,\xi,p} \lambda_{I,p} = \underline{\lambda}_I$, which determines whether a given Bellman equation can be guaranteed to be a contraction mapping on $\mathcal{B}_W(\mathcal{L}^\infty_{\phi,+})$.

The tax-share transition has portfolio-growth bound $m_Q + \delta_I c_K \|\chi\|_{\infty,\phi}$ and income bound $\delta_I y \|\chi\|_{\infty,\phi}$ (Lemma IV.B.2). The first coefficient is the part of the law of motion that scales the current portfolio; the second is an additive term generated by exogenous income invested in the new tax lot. The growth bound is $\lambda_{I,p} := \max\{m_{A,p}(m_Q + \delta_I c_K \|\chi\|_{\infty,\phi})^p,\ 1 + m_{A,p}(\delta_I y \|\chi\|_{\infty,\phi})^p/\xi\}$. The discounted Bellman operator in (6) has contraction modulus $\beta\lambda_{I,p}$ when this is below one. Set $\lambda_{I,0} = 1$ for bounded utility. The additive-income term $m_{A,p}(\delta_I y \|\chi\|_{\infty,\phi})^p/\xi$ vanishes as ξ grows. Additionally, if p can be chosen arbitrarily close to 0 (log-case), then all remaining terms collapse to 1 and $\underline{\lambda}_I = 1$.

IV.B.2 Weighted Portfolio Space and Operators

Write

$$\mathcal{L}^1_{1/\phi} := \left\{\eta : \int_0^\infty \frac{|\eta(r)|}{\phi(r)}\, dr < \infty\right\}, \qquad \langle w, \eta\rangle := \int_0^\infty w(r)\eta(r)\, dr.$$

Multiplication by ϕ identifies $\mathcal{L}^\infty_\phi$ isometrically with ordinary L^∞, while division by ϕ identifies $\mathcal{L}^1_{1/\phi}$ with ordinary L^1. Hence $\mathcal{L}^\infty_\phi$ is the dual of $\mathcal{L}^1_{1/\phi}$ and is a Banach lattice with positive cone $\mathcal{L}^\infty_{\phi,+}$ (Aliprantis and Border, 2006, Section 9.1 and Theorem 13.28). We use the weak-$*$ topology $\sigma(\mathcal{L}^\infty_\phi, \mathcal{L}^1_{1/\phi})$. Its closed norm balls are compact by Banach–Alaoglu, and its norm-bounded order intervals are weak-$*$ compact (Aliprantis and Border, 2006, Theorem 6.21 and Lemma 8.56).

Lemma IV.B.1 (Weight-ratio bound). *For all $x, s > 0$, $\phi(x)/\phi(x/s) \le \phi(s)$.*

Proof. If x and x/s lie on the same side of one, the ratio is s^α or s^γ. If $s \le x \le 1$, it is $s^\gamma x^{\alpha-\gamma} \le s^\alpha$; if $1 \le x \le s$, it is $x^{\gamma-\alpha} s^\alpha \le s^\gamma$. ∎

Proposition IV.B.1 (Market value and liquidation functionals). *The market-value functional is positive and bounded on $\mathcal{L}^\infty_\phi$, with*

$$|\mathrm{M}(w)| \le c_{\mathrm{M}} \|w\|_{\infty,\phi}, \qquad c_{\mathrm{M}} = \frac{1}{1-\alpha} + \frac{1}{\gamma - 1}.$$

If $Kx = \int_0^\infty x(r)k(r)\,dr$, where $k \ge 0$ and $k \in \mathcal{L}^1_{1/\phi}$, then K is positive, linear, bounded, and weak-$*$ continuous, with $|Kx| \le \|k\|_{\mathcal{L}^1_{1/\phi}} \|x\|_{\infty,\phi}$.

Proof. The market-value bound follows by integrating ϕ^{-1} separately on $(0,1]$ and $(1,\infty)$. The second claim is the defining predual pairing; continuity follows directly from the dual representation above. ∎

Proposition IV.B.2 (Portfolio operators). *Under Assumption 1, Q and every A_z are positive, linear, bounded, and weak-$*$ continuous on $\mathcal{L}^\infty_\phi$, with*

$$\|Q\| \le m_Q, \qquad \|A_z\| \le \phi(z), \qquad \|A_z Q w\|_{\infty,\phi} \le \phi(z) m_Q \|w\|_{\infty,\phi}.$$

Proof. Lemma IV.B.1 gives

$$|Qw(x)|\phi(x) \le \|w\|_{\infty,\phi} \int_0^\infty \frac{\phi(x)}{\phi(x/R)} f(R)\,dR \le m_Q \|w\|_{\infty,\phi},$$

and, after writing $x = zy$, $\|A_z w\|_{\infty,\phi} \le \phi(z)\|w\|_{\infty,\phi}$. For $\eta \in \mathcal{L}^1_{1/\phi}$, the preadjoints are

$$Q^*\eta(y) = \int_0^\infty R f(R) \eta(Ry)\,dR, \qquad A_z^*\eta(y) = z\eta(zy).$$

The same weight-ratio bound gives $\|Q^*\eta\|_{\mathcal{L}^1_{1/\phi}} \le m_Q \|\eta\|_{\mathcal{L}^1_{1/\phi}}$ and $\|A_z^*\eta\|_{\mathcal{L}^1_{1/\phi}} \le \phi(z)\|\eta\|_{\mathcal{L}^1_{1/\phi}}$. The adjoint criterion of Aliprantis and Border (2006, Theorem 6.43) therefore yields weak-$*$ continuity. ∎

IV.B.3 Bellman Bounds and Backward Iteration

Lemma IV.B.2 (Tax-share drift and current-payoff bounds). *For every $p \in (0,1]$ and $\varepsilon > 0$, the normalization in W_p can be chosen so that every feasible $(w, \widetilde{w}, I)$ satisfies*

$$\sum_z p_z W_p\big(G_z(w, \widetilde{w}, I)\big) \le (\lambda_{I,p} + \varepsilon) W_p(w).$$

Moreover, $0 \le c(\widetilde{w}, I) \le y + c_{\mathrm{K}} \|w\|_{\infty,\phi}$.

Proof. Feasibility gives $I \le y + \mathrm{K}(\widetilde{w}) \le y + c_{\mathrm{K}}\|w\|_{\infty,\phi}$. Proposition IV.B.2 therefore implies

$$\|G_z(w, \widetilde{w}, I)\|_{\infty,\phi} \le \phi(z)\left[(m_Q + \delta_I c_{\mathrm{K}}\|\chi\|_{\infty,\phi})\|w\|_{\infty,\phi} + \delta_I y\|\chi\|_{\infty,\phi}\right].$$

Raise this inequality to $p \le 1$, average over z, and use $(a+b)^p \le a^p + b^p$. The first term gives $\lambda_{I,p}\|w\|^p_{\infty,\phi}$; choosing ξ sufficiently large absorbs the additive income term into $\varepsilon W_p(w)$. The consumption bound is immediate from feasibility. ■

Proof of Proposition 1. The discounted part is an application of the weighted sup-norm theorem in Bertsekas (2012, Proposition 2.1). Under bounded utility take $W_0 \equiv 1$ and obtain modulus β. For log utility, the affine extension removes the singularity at zero and $|\log c| \le C_p(1 + c^p)$ for every sufficiently small $p > 0$; Lemma IV.B.2 then gives a modulus approaching β as $p \downarrow 0$. Under sublinear utility, the same lemma gives modulus $\beta(\lambda_{I,p} + \varepsilon)$, which is below one when $\beta\underline{\lambda}_I < 1$ and the free parameters are chosen sufficiently close to their infima. The contraction mapping theorem then gives the asserted fixed point and convergence.

For $\mathbb{P}$, monotonicity of the operator and $\mathbb{P}V_0^{\mathbb{P}} \le V_0^{\mathbb{P}}$ imply $V_{n+1}^{\mathbb{P}} \le V_n^{\mathbb{P}}$. Nonnegativity gives the pointwise limit. To verify the fixed-point claim, fix w and let b_n denote the period objective on the compact set $\Gamma(w)$. The attainment argument below makes every b_n continuous and gives $b_n \downarrow b$, where b is upper semicontinuous. If a_n maximizes b_n, compactness gives a convergent subnet. Comparing this subnet with every fixed b_N yields $\lim_n \max b_n \le \max b$; the reverse inequality is immediate from $b_n \ge b$. Thus maxima and limits commute at each state, so $V^{\mathbb{P}} = \mathbb{P}V^{\mathbb{P}}$. ■

IV.B.4 Shape Preservation and Common Utilities

Proposition IV.B.3 (Monotonicity and concavity)**.** *Suppose Assumptions 1, 2, and 3 hold.*

i) If V is increasing, then $\mathbb{T}V$ and $\mathbb{P}V$ are increasing.

ii) If V is concave, then $\mathbb{T}V$ and $\mathbb{P}V$ are concave.

Consequently, whenever Proposition 1 yields the discounted-utility fixed point $V^{\mathbb{T}}$, it is increasing and concave.

Proof of Proposition IV.B.3. If $w_1 \le w_2$, every control feasible at w_1 is feasible at w_2, and positivity of Q and A_z makes every continuation state weakly larger. This proves monotonicity. For concavity, take convex combinations of two states and their feasible controls. Linearity of the tax-share liquidation functional makes the feasible graph convex, current consumption is affine, and the transition is affine. Concavity is preserved by u, by continuation values, and by the minimum of the two concave terms in $\mathbb{P}$. Taking suprema proves the claim. ■

Corollary IV.B.1 (CARA utility)**.** *Let $u(c) = \bar{u} - ae^{-\eta c}$ with $a, \eta > 0$, or take any positive affine transformation. Then, under Assumptions 1 and 2, $\mathbb{T}$ is a β-contraction on bounded value functions. The sup-min conclusions of Proposition 1 also apply.*

Corollary IV.B.2 (CRRA and log utility). *For $u_\sigma(c) = c^{1-\sigma}/(1-\sigma)$, $0 < \sigma < 1$, take $p = 1 - \sigma$ and require $\beta\underline{\lambda}_I < 1$. For log utility bounded away from zero, choose any sufficiently small $p > 0$. For $\sigma > 1$ and consumption bounded away from zero, utility is bounded and the bounded case applies. The sup-min conclusions are unchanged.*

IV.B.5 Policy Attainment

Proof of Theorem 2. Fix a norm-bounded set of states. Feasible liquidation portfolios then lie in a common weak-$*$ compact ball by Banach–Alaoglu, and each interval $[0, w]$ is weak-$*$ compact (Aliprantis and Border, 2006, Theorem 6.21 and Lemma 8.56). The investment coordinate is bounded by $y + c_{\mathrm{K}}\|w\|_{\infty,\phi}$. The budget condition is closed because K is a predual pairing. Thus Γ has nonempty compact values and a closed graph on the bounded set.

For lower hemicontinuity, if $w_\nu \to w$ and $\widetilde{w} \le w$, write $\widetilde{w} = \theta w$ for a measurable $0 \le \theta \le 1$, set $\widetilde{w}_\nu = \theta w_\nu$, and truncate the investment coordinate at $y + \mathrm{K}(\widetilde{w}_\nu)$. Multiplication by bounded θ is weak-$*$ continuous. The standard closed-graph and lower-hemicontinuity criteria therefore make Γ continuous on bounded state sets (Aliprantis and Border, 2006, Theorems 17.16–17.19 and 17.28).

Proposition IV.B.2 makes every transition weak-$*$ continuous. The current payoff is continuous, as are the finite-horizon Bellman objectives whenever the continuation value is. Berge's maximum theorem (Aliprantis and Border, 2006, Theorem 17.31) now gives, by induction, bounded-set weak-$*$ continuity of every finite-horizon value, attainment, compact argmax sets, and upper hemicontinuity on bounded state sets.

For $\mathbb{T}$, weighted-norm convergence is uniform on every norm-bounded state set, so the fixed point is weak-$*$ continuous there. At any fixed state its continuation set is norm bounded; Berge's theorem therefore gives a stationary maximizer and a compact argmax set. For $\mathbb{P}$, the decreasing limit of continuous functions is upper semicontinuous on every bounded state set. The compact maximizer argument in the proof of Proposition 1 gives the fixed point and a stationary maximizer. Its argmax is a closed subset of $\Gamma(w)$ and hence weak-$*$ compact. ■

IV.C Recursive Optimal Trading under Linear Taxation

This appendix accompanies Section IV.C. It follows the main argument from right dominance to dominance-inducing trades and recursive optimality. The broader tax-share extension closes the appendix.

IV.C.1 Right Dominance under Returns and Backward Iteration

The arguments below use the law-of-motion identities established in Lemma 1 in Section III.

Lemma IV.C.1 (Lower-interval residual dominance). *Under* $\mathrm{T}_{\mathrm{FUL}}$, *fix* $w \in \mathcal{W}$ *and* $x \in [0, \mathrm{K}(w)]$, *and choose* $a \in [0,\infty]$ *such that* $\mathrm{K}(w\mathbf{1}_{(0,a]}) = x$, *with* $a = \infty$ *meaning sale of the whole portfolio. If* $0 \le \widetilde{w} \le w$ *and* $\mathrm{K}(\widetilde{w}) = x$, *then* $w\mathbf{1}_{(a,\infty)} \succeq_R w - \widetilde{w}$. *Moreover, if* $w \succeq_R \bar{w}$, $x \le \mathrm{K}(\bar{w})$, *and* $a, \bar{a}$ *satisfy* $\mathrm{K}(w\mathbf{1}_{(0,a]}) = \mathrm{K}(\bar{w}\mathbf{1}_{(0,\bar{a}]}) = x$, *then* $w\mathbf{1}_{(a,\infty)} \succeq_R \bar{w}\mathbf{1}_{(\bar{a},\infty)}$.

Proof of Lemma IV.C.1. The cutoff a exists because the map $b \mapsto \mathrm{K}(w\mathbf{1}_{(0,b]})$ is continuous, starts at zero, and converges to $\mathrm{K}(w)$. For the single-portfolio claim, the K component holds with equality because $\mathrm{K}(w\mathbf{1}_{(a,\infty)}) = \mathrm{K}(w) - x = \mathrm{K}(w - \widetilde{w})$. It remains to compare upper tails. First approximate w and $\widetilde{w}$ by portfolios that are constant on finitely many ordered cells $0 < r_1 < \cdots < r_N$, and write l_j for the market value liquidated from cell j. If there are indices $i < j$ such that $l_i < w_i$ and $l_j > 0$, move an after-tax cash amount ε of liquidation from cell j to cell i. Since $k(r_i) \ge k(r_j)$, this changes sold market value by $+\varepsilon/k(r_i)$ in the lower cell and $-\varepsilon/k(r_j)$ in the higher cell. The liquidation target is unchanged, and every upper-tail residual weakly increases: tails above r_j are unchanged, tails between r_i and r_j gain $\varepsilon/k(r_j)$, and tails below r_i gain $\varepsilon/k(r_j) - \varepsilon/k(r_i) \ge 0$. Iterating until no such inversion remains gives the lower-interval liquidation. Passing from finite partitions to measurable portfolios follows by monotone approximation of the cumulative liquidation and tail integrals.

Now suppose $w \succeq_R \bar{w}$ and $x \le \mathrm{K}(\bar{w})$. The K component is $\mathrm{K}(w\mathbf{1}_{(a,\infty)}) = \mathrm{K}(w) - x \ge \mathrm{K}(\bar{w}) - x = \mathrm{K}(\bar{w}\mathbf{1}_{(\bar{a},\infty)})$. For upper tails, removing the first x units of after-tax $k(r)w(r)\,dr$-mass from the left tail preserves first-order right-tail ordering: a portfolio with weakly larger upper tails cannot have smaller residual upper tails after the same left-tail cash extraction. Equivalently, use the exchange argument above on common finite partitions and pass to the limit. Hence $w\mathbf{1}_{(a,\infty)} \succeq_R \bar{w}\mathbf{1}_{(\bar{a},\infty)}$. ∎

Proof of Proposition 3. Fix $z \in \mathcal{Z}$. For the market-tail component, a change of variables gives

$$\int_a^\infty A_z Q w(r)\,dr = z \int_0^\infty Rf(R) \left(\int_{a/(zR)}^\infty w(s)\,ds \right) dR,$$

which is weakly larger than the same expression with $\bar{w}$ whenever w has weakly larger upper tails. Taking $a \downarrow 0$ also gives $\mathrm{M}(w) \ge \mathrm{M}(\bar{w})$. For linear taxes, $\mathrm{K}_{\mathrm{FUL}}(A_z Q x) = \mathrm{K}_{\mathrm{FUL}}(x) + (1 - \tau)(z\mathbb{E}[R] - 1)\,\mathrm{M}(x)$. Hence $\mathrm{K}_{\mathrm{FUL}}(A_z Q w) \ge \mathrm{K}_{\mathrm{FUL}}(A_z Q \bar{w})$ whenever $z\mathbb{E}[R] \ge 1$ and $w \succeq_R \bar{w}$. Thus $A_z Q w \succeq_R A_z Q \bar{w}$. Adding the common nonnegative portfolio $A_z(I\chi)$ to both sides preserves upper-tail inequalities and K-dominance.

To prove that the Bellman operators preserve right-increasingness, let V be right-increasing and suppose $w \succeq_R \bar{w}$. Take any feasible trade $(\widetilde{w}, I)$ from $\bar{w}$, and set $x := \mathrm{K}(\widetilde{w})$. Since $x \leq \mathrm{K}(\bar{w}) \leq \mathrm{K}(w)$, choose lower-cutoff sales from both portfolios that raise x. Lemma IV.C.1 gives

$$w\mathbf{1}_{(a,\infty)} \succeq_R \bar{w}\mathbf{1}_{(\bar{a},\infty)} \succeq_R \bar{w} - \widetilde{w}.$$

Keep the original investment I. The replacement trade from w is feasible and produces the same current consumption, $y + x - I$. Stability under returns, followed by addition of the common investment portfolio, gives

$$G_z(w, w\mathbf{1}_{(0,a]}, I) \succeq_R G_z(\bar{w}, \widetilde{w}, I) \qquad \text{for every } z.$$

Right-increasingness of V therefore makes both Bellman objectives weakly larger at the replacement trade. Taking suprema proves that $\mathbb{T}V$ and $\mathbb{P}V$ are right-increasing. Starting from $V_0^{\mathbb{T}} = 0$ and $V_0^{\mathbb{P}} = y + \mathrm{K}$, induction establishes the claim for every iterate. The convergence results of Proposition 1 then pass the inequalities to the limiting value functions; convergence in the weighted norm implies pointwise convergence at each state. ■

IV.C.2 Harvesting and Cutoff Replacement

Proof of Proposition 4. Let $H := (0, r_H]$ and let $h := (w - \widetilde{w})\mathbf{1}_H$ be the part of the core harvest region that the original control leaves unsold. Set $\widetilde{w}^H := \widetilde{w} + h$ and $I^H := I + \mathrm{K}(h)$. Then $(\widetilde{w}^H, I^H) \in \Gamma^H(w)$, feasibility is preserved because $I^H \leq y + \mathrm{K}(\widetilde{w}) + \mathrm{K}(h) = y + \mathrm{K}(\widetilde{w}^H)$, and current consumption is unchanged.

Fix $z \in \mathcal{Z}$. Since $H \subseteq \mathcal{H}_z^L$, each unit tax lot $\boldsymbol{\delta}_r$ with $r \in H$ satisfies $A_z(\delta_I k_{\mathrm{FUL}}(r)\chi) \succeq_R A_z Q\boldsymbol{\delta}_r$. Integrating this comparison against the nonnegative measure $h(r)\,dr$ and adding the unchanged part of the continuation portfolio gives

$$G_z(w, \widetilde{w}^H, I^H) \succeq_R G_z(w, \widetilde{w}, I).$$

Thus the constructed core-harvesting control weakly improves every aggregate-state continuation while preserving current consumption. If V is right-increasing, both the discounted objective and the sup-min objective weakly increase under this replacement. Since $\Gamma^H(w) \subseteq \Gamma(w)$, the suprema over $\Gamma(w)$ and $\Gamma^H(w)$ are equal.

Let $(\widetilde{w}, I) \in \Gamma^H(w)$ and set $x := \mathrm{K}(\widetilde{w})$. Since the control harvests the core, $x \geq \mathrm{K}(w\mathbf{1}_{(0,r_H]})$. Choose $a \in [r_H, \infty]$ such that $\mathrm{K}(w\mathbf{1}_{(0,a]}) = x$. Then $(w\mathbf{1}_{(0,a]}, I) \in \Gamma_{r_H}^L(w)$ and current consumption is unchanged. By Lemma IV.C.1, $w\mathbf{1}_{(a,\infty)} \succeq_R w - \widetilde{w}$. Proposition 3 implies

$$A_z Q(w\mathbf{1}_{(a,\infty)}) \succeq_R A_z Q(w - \widetilde{w}) \qquad \text{for every } z \in \mathcal{Z}.$$

Adding the common investment portfolio $A_z(\delta_I I\chi)$ preserves right dominance, so

$$G_z(w, w\mathbf{1}_{(0,a]}, I) \succeq_R G_z(w, \widetilde{w}, I) \qquad \text{for every } z \in \mathcal{Z}.$$

If V is right-increasing, this proves equality of the suprema over $\Gamma^H(w)$ and $\Gamma_{r_H}^L(w)$.

Applying the first replacement and then the second turns any feasible full control into a $\Gamma^L_{r_H}(w)$ control with the same current consumption and weakly better continuation states. Hence any attained full Bellman maximum admits a $\Gamma^L_{r_H}$ maximizer; uniqueness of the full maximizer forces that maximizer to belong to $\Gamma^L_{r_H}(w)$. ■

IV.C.3 Recursive LHCL Optimality

Lemma IV.C.2 (Bellman restriction for right-increasing continuation values). *Suppose Assumptions 1 and 3 hold,* $\mathrm{K} = \mathrm{K}_{\mathrm{FUL}}$, *and* $z\mathbb{E}[R] \geq 1$ *for every* $z \in \mathcal{Z}$. *If* V *is right-increasing, then:*

i) $\mathbb{T}V = \mathbb{T}^L V$ *and* $\mathbb{P}V = \mathbb{P}^L V$.

ii) Whenever the supremum defining either full Bellman objective at any state w *is attained, it is attained by a* $\Gamma^L_{r_H}(w)$ *control. If the full maximizer is unique, it belongs to* $\Gamma^L_{r_H}(w)$.

iii) $\mathbb{T}V$, $\mathbb{T}^L V$, $\mathbb{P}V$, *and* $\mathbb{P}^L V$ *are right-increasing.*

Proof of Lemma IV.C.2. If V is right-increasing and $(\widetilde{w}, I) \in \Gamma(w)$, apply the two replacements in Proposition 4: first harvest the core, then complete the control as a lower cutoff. This gives a control $(\widetilde{w}^L, I^L) \in \Gamma^L_{r_H}(w)$ with the same current consumption and

$$G_z(w, \widetilde{w}^L, I^L) \succeq_R G_z(w, \widetilde{w}, I) \qquad \text{for every } z \in \mathcal{Z}.$$

Therefore $V(G_z(w, \widetilde{w}^L, I^L)) \geq V(G_z(w, \widetilde{w}, I))$ state by state. The discounted and sup-min objectives under the $\Gamma^L_{r_H}$ replacement are weakly larger than under the original control, so $\mathbb{T}V \leq \mathbb{T}^L V$ and $\mathbb{P}V \leq \mathbb{P}^L V$. The reverse inequalities are immediate from $\Gamma^L_{r_H}(w) \subseteq \Gamma(w)$.

If a full Bellman supremum is attained at $(\widetilde{w}, I)$, the same replacement gives a $\Gamma^L_{r_H}(w)$ control whose objective value is weakly higher. Since the full and restricted suprema are equal, this replacement is also a full maximizer. If the full maximizer is unique, it must already belong to $\Gamma^L_{r_H}(w)$.

The proof of Proposition 3 establishes that the full Bellman operators preserve right-increasingness for any right-increasing continuation value V. The equalities just proved therefore show that $\mathbb{T}^L V$ and $\mathbb{P}^L V$ are also right-increasing. ■

Proof of Theorem 3. Proposition 3 establishes that the full iterates and their limits are right-increasing. The full and restricted initial values coincide: $V_0^{\mathbb{T}} = V_0^{\mathbb{T},L} = 0$ and $V_0^{\mathbb{P}} = V_0^{\mathbb{P},L} = y + \mathrm{K}_{\mathrm{FUL}}$. For the latter equality, full liquidation with zero investment attains the maximal current payout in both feasible sets.

For either $\mathbb{H} \in \{\mathbb{T}, \mathbb{P}\}$, suppose $V_n^{\mathbb{H}} = V_n^{\mathbb{H},L}$. Lemma IV.C.2 then gives

$$V_{n+1}^{\mathbb{H}} = \mathbb{H}V_n^{\mathbb{H}} = \mathbb{H}^L V_n^{\mathbb{H}} = \mathbb{H}^L V_n^{\mathbb{H},L} = V_{n+1}^{\mathbb{H},L}.$$

Induction proves equality at every finite horizon.

Under the discounted convergence conditions, Proposition 1 gives $V_n^{\mathbb{T}} \to V^{\mathbb{T}}$ in the relevant norm. It also gives $V_n^{\mathbb{P}} \downarrow V^{\mathbb{P}}$ pointwise for the canonical sup-min sequence. Finite-horizon equality gives the same limits for the restricted sequences. Each full limit is a fixed point and is right-increasing, so Lemma IV.C.2 yields

$$\mathbb{H}^L V^{\mathbb{H}} = \mathbb{H} V^{\mathbb{H}} = V^{\mathbb{H}}, \qquad \mathbb{H} \in \{\mathbb{T}, \mathbb{P}\}.$$

The restricted discounted operator satisfies the same contraction estimate as the full operator, so its fixed point is unique under the same convergence conditions.

Finally, Theorem 2 supplies full Bellman maximizers at every finite horizon and at the stationary limits. Lemma IV.C.2 replaces each such maximizer by a control in $\Gamma^L_{r_H}(w)$ with the same current consumption and weakly better continuation states. The replacement is therefore also a full maximizer. If the full maximizer is unique, it must itself belong to $\Gamma^L_{r_H}(w)$. ■

IV.C.4 Frictionless Investment

Proof of Corollary 4. If $\delta_I = 1$, the formula for (12) gives $\mathcal{H}^L_z = (0, 1]$ for every $z \in \mathcal{Z}$ under $z\mathbb{E}[R] \geq 1$. Hence $r_H = 1$, and all loss lots are in the common harvest region.

It remains to rule out round trips above 1. Fix $r > 1$ and $z \in \mathcal{Z}$. Since $\delta_I = 1$, $k_{\mathrm{FUL}}(r) \leq 1$. For any upper-tail threshold q,

$$z \int_{q/(zr)}^{\infty} Rf(R)\, dR \geq k_{\mathrm{FUL}}(r) z \int_{q/z}^{\infty} Rf(R)\, dR,$$

so the market-tail part of $A_z Q \boldsymbol{\delta}_r \succeq_R A_z(k_{\mathrm{FUL}}(r)\chi)$ holds. For the liquidation component,

$$\mathrm{K}_{\mathrm{FUL}}(A_z Q \boldsymbol{\delta}_r) - \mathrm{K}_{\mathrm{FUL}}(A_z(k_{\mathrm{FUL}}(r)\chi)) = \tau(1 - r^{-1})\{z(1-\tau)\mathbb{E}[R] + \tau - 1\} \geq 0,$$

where the inequality uses $r > 1$ and $z\mathbb{E}[R] \geq 1$. Thus keeping any gain lot $r > 1$ right-dominates selling it and reinvesting its after-tax proceeds.

Now let $(w\mathbf{1}_{(0,a]}, I) \in \Gamma^L_1(w)$ and put $x := \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,a]}) - I$, the liquidation-and-investment contribution to current consumption apart from y. If $x \leq \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,1]})$, set $b = 1$ and $I' := \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,1]}) - x$. Original feasibility gives $x \geq -y$, so $I' \leq y + \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,1]})$ and $(w\mathbf{1}_{(0,1]}, I') \in \Gamma^{sL}_1(w)$. If $x > \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,1]})$, choose $b \in [1, a]$ with $\mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,b]}) = x$ and set $I' = 0$. In both cases, $\mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,b]}) - I' = \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,a]}) - I$, so current consumption is unchanged, and $I - I' = \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(b,a]})$.

For each $z \in \mathcal{Z}$, integrate the Dirac comparison over $r \in (b, a]$. Since $\mathrm{K}_{\mathrm{FUL}} = \mathrm{K}$ is the linear functional with kernel k_{FUL}, additivity of right dominance gives

$$A_z Q(w\mathbf{1}_{(b,a]}) \succeq_R A_z\big(\mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(b,a]})\chi\big) = A_z\big((I - I')\chi\big).$$

Adding the common portfolio $A_z\{Q(w\mathbf{1}_{(a,\infty)}) + I'\chi\}$ to both sides yields

$$G_z(w, w\mathbf{1}_{(0,b]}, I') \succeq_R G_z(w, w\mathbf{1}_{(0,a]}, I).$$

If V is right-increasing, the replacement weakly raises the continuation term and leaves current consumption fixed. Hence the suprema over $\Gamma_1^L(w)$ and $\Gamma_1^{sL}(w)$ coincide in both Bellman objectives. Combining this equality with the full-to-Γ_1^L restriction in Lemma IV.C.2 and the recursive conclusions of Theorem 3 gives the corresponding full-to-Γ_1^{sL} restriction and the stated value-iteration consequences. ■

IV.C.5 A Broader Tax-Share Sufficient Class

IV.C.5.1 Aggregate Compatibility

For a general tax-share kernel, the future liquidation value of a formal unit market-value tax lot is

$$H_{k,z}(r) := \mathrm{K}(A_z Q \boldsymbol{\delta}_r) = z \int_0^\infty R f(R) k(zRr)\, dR. \tag{IV.C.1}$$

Throughout this extension, write $\succeq_R^k$ for right dominance under the liquidation functional induced by the displayed kernel k. The following compatibility condition is sufficient for right dominance to be stable under idiosyncratic returns and aggregate dilation.

Assumption 4 (Tax-share kernel regularity and aggregate compatibility)**.** *The tax-share kernel k satisfies the following conditions.*

i) $k : (0,\infty) \to (0,\infty)$ *is measurable, belongs to* $\mathcal{L}^1_{1/\phi}$*, and is nonincreasing.*

ii) For every aggregate shock z, there are a scalar $\rho_z \geq 0$ and a nonnegative nondecreasing function h_z such that

$$H_{k,z}(r) = \rho_z k(r) + h_z(r), \qquad r > 0. \tag{IV.C.2}$$

Lemma IV.C.3 (Tail integral order)**.** *If w has weakly larger upper tails than $\bar{w}$, then for every nonnegative nondecreasing function h for which the integrals are finite, $\int_0^\infty h(r)w(r)\,dr \geq \int_0^\infty h(r)\bar{w}(r)\,dr$.*

Proof. This is the upper-tail formulation of first-order stochastic dominance; see Aliprantis and Border (2006, Section 19.8). For completeness, it holds first for nonnegative step functions by taking linear combinations of the tail inequalities. Approximate h monotonically by such functions and apply monotone convergence. ■

Proposition IV.C.1 (General aggregate propagation)**.** *Suppose Assumption 4(ii) holds. If $w \succeq_R^k \bar{w}$, then $A_z Q w \succeq_R^k A_z Q \bar{w}$ for every aggregate shock z. Consequently, for every $J \geq 0$, $A_z(Qw + J\chi) \succeq_R^k A_z(Q\bar{w} + J\chi)$ for every z.*

Proof. Fix an aggregate realization z. For the market-tail component, a change of variables gives

$$\int_a^\infty A_z Q w(r)\, dr = z \int_0^\infty R f(R) \left(\int_{a/(zR)}^\infty w(s)\, ds \right) dR,$$

which is weakly larger than the same expression with $\bar{w}$ whenever w has weakly larger upper tails. For the after-tax component, equation (IV.C.1) gives

$$\mathrm{K}(A_z Q w) - \mathrm{K}(A_z Q \bar{w}) = \int_0^\infty H_{k,z}(r)(w(r) - \bar{w}(r))\, dr.$$

By Assumption 4(ii), this equals

$$\rho_z\{\mathrm{K}(w) - \mathrm{K}(\bar{w})\} + \int_0^\infty h_z(r)(w(r) - \bar{w}(r))\,dr.$$

The first term is nonnegative by (11), and the second is nonnegative by Lemma IV.C.3. Thus $A_zQw \succeq_R^k A_zQ\bar{w}$. Adding the common nonnegative portfolio $A_z(J\chi)$ to both sides preserves upper-tail inequalities and K-dominance. ■

IV.C.5.2 A Primitive Harvest Test

Lemma IV.C.4 (Primitive statewise harvest condition). *Fix $r \le 1$ and an aggregate realization z. If $\delta_I k(r) \ge 1$ and*

$$\delta_I k(r) H_{k,z}(1) \ge H_{k,z}(r), \tag{IV.C.3}$$

then

$$A_z(\delta_I k(r)\chi) \succeq_R^k A_zQ\boldsymbol{\delta}_r. \tag{IV.C.4}$$

Proof of Lemma IV.C.4. Fix the aggregate realization z. For upper tails and $r \le 1$,

$$\int_a^\infty A_zQ\boldsymbol{\delta}_r(s)\,ds = z\int_{a/(zr)}^\infty Rf(R)\,dR \le z\int_{a/z}^\infty Rf(R)\,dR = \int_a^\infty A_z\chi(s)\,ds.$$

Multiplication by $\delta_I k(r) \ge 1$ gives the market-tail inequality for $A_z(\delta_I k(r)\chi)$ against $A_zQ\boldsymbol{\delta}_r$. For the after-tax component, $\mathrm{K}(A_z(\delta_I k(r)\chi)) = \delta_I k(r) H_{k,z}(1)$ and $\mathrm{K}(A_zQ\boldsymbol{\delta}_r) = H_{k,z}(r)$, so (IV.C.3) gives K-dominance. Hence r satisfies the state-z harvest condition (IV.C.4). ■

IV.C.5.3 Power Kernels

Proposition IV.C.2 (Power kernels). *A tractable sufficient class is obtained by taxing a power transform of the gain. For $a, \theta \in [0,1]$, set $k_{a,\theta}(r) := a + (1-a)r^{-\theta}$. Equivalently, for $r \ne 1$ this kernel is generated by the effective tax-rate function $\tau_{a,\theta}(r) := (1-a)(1-r^{-\theta})/(1-r^{-1})$, with removable value $\tau_{a,\theta}(1) = (1-a)\theta$. The usual linear tax with rate τ is the case $a = 1-\tau$ and $\theta = 1$.*

Let $\mu_q := \mathbb{E}[R^q]$ and suppose the moments μ_1 and $\mu_{1-\theta}$ are finite. For $k = k_{a,\theta}$, the following hold.

i) The tax-rate function satisfies $0 \le \tau_{a,\theta}(r) \le 1 - a \le 1$, and k is positive and nonincreasing.

ii) For every aggregate shock z,

$$H_{k,z}(r) = az\mu_1 + (1-a)z^{1-\theta}\mu_{1-\theta}r^{-\theta}. \tag{IV.C.5}$$

Hence Assumption 4(ii) holds if $a = 0$ or $z^\theta\mu_1 \ge \mu_{1-\theta}$ for every z. In particular, without aggregate risk this condition follows from $\mu_1 \ge 1$ by Jensen's inequality.

iii) For each z, the harvest test (IV.C.3) is equivalent to

$$\delta_I\left[a + (1-a)r^{-\theta}\right]\left[az\mu_1 + (1-a)z^{1-\theta}\mu_{1-\theta}\right] \ge az\mu_1 + (1-a)z^{1-\theta}\mu_{1-\theta}r^{-\theta}. \tag{IV.C.6}$$

The tax lots satisfying this K-part of the harvest test form a lower interval on $(0,1]$. The additional tail condition $\delta_I k(r) \ge 1$ is also lower-interval, and intersecting over the finite aggregate support preserves the lower-interval property.

Proof of Proposition IV.C.2. For $r \neq 1$, the identity $1 - \tau_{a,\theta}(r)(1 - r^{-1}) = a + (1 - a)r^{-\theta}$ is immediate, and the value at $r = 1$ follows from l'Hopital's rule. Since $0 \leq \theta \leq 1$, the ratio $(1 - r^{-\theta})/(1 - r^{-1})$ lies in $[0, 1]$ on both sides of $r = 1$, so $0 \leq \tau_{a,\theta} \leq 1 - a$. Positivity of k is clear, and $k'(r) = -(1 - a)\theta r^{-\theta-1} \leq 0$ wherever the derivative is taken.

Using (IV.C.1),

$$H_{k,z}(r) = z \int_0^\infty Rf(R)\{a + (1 - a)(zRr)^{-\theta}\}\, dR = azμ_1 + (1 - a)z^{1-\theta}μ_{1-\theta}r^{-\theta},$$

which proves (IV.C.5). If $a = 0$, this is $z^{1-\theta}μ_{1-\theta}k(r)$, so compatibility holds with $h_z = 0$. If $a > 0$, write

$$H_{k,z}(r) = z^{1-\theta}μ_{1-\theta}k(r) + a\{zμ_1 - z^{1-\theta}μ_{1-\theta}\}.$$

The residual term is a nonnegative constant whenever $z^\theta μ_1 \geq μ_{1-\theta}$, giving Assumption 4(ii). With no aggregate risk this condition is $μ_1 \geq μ_{1-\theta}$, which follows from Jensen's inequality and $μ_1 \geq 1$.

Substituting (IV.C.5) into (IV.C.3) gives (IV.C.6). For fixed z, write $q = r^{-\theta} \geq 1$. The difference between the left- and right-hand sides of (IV.C.6) is affine in q and is nonpositive at $q = 1$ unless $\delta_I = 1$. If its slope is nonpositive, the test is empty on $(0, 1]$ except for possible boundary cases; if its slope is positive, the test is of the form $q \geq q_i^*$. Thus the tested tax lots form a lower interval in r. A finite intersection of lower intervals is a lower interval. ■

IV.D Limited Use of Losses and Carryforward

This appendix accompanies Section IV.D. It proves the Bellman foundations for the augmented state and then the sLHCL restriction under perfect carryforward.

IV.D.1 Bellman Foundations

The carryforward Bellman operators $\mathbb{T}_{\mathrm{LUL}}$ and $\mathbb{P}_{\mathrm{LUL}}$ augment the portfolio state by the stock of unused losses, $(w, F) \in \mathcal{S} := \mathcal{W} \times [0, \infty)$, and use the corresponding LUL primitives in eqs. (6) and (8).

Proposition IV.D.1 (Carryforward Bellman foundations). *On the product state (w, F), the discounted and sup-min carryforward problems have the same contraction, backward-iteration, and statewise-attainment properties used in Proposition 5. In the absence of aggregate risk, the sup-min limit has the same perpetuity interpretation as Corollary 2.*

Proof. The only new coordinate is scalar. Since

$$c_{CG} := \int_0^\infty |1 - r^{-1}| \phi(r)^{-1}\, dr = \frac{1}{-\alpha(1-\alpha)} + \frac{1}{\gamma(\gamma-1)} < \infty,$$

both M and CG are predual pairings. The maps $\mathrm{T}_{\mathrm{LUL}}$, $\mathrm{K}_{\mathrm{LUL}}$, and F_{LUL} are therefore weak-$*$ continuous in liquidation and continuous in F. Order intervals remain compact in the product of the weak-$*$ and Euclidean topologies, so the Banach–Alaoglu and Berge arguments above apply without change.

For the contraction step use $W_p^F(w, F) = \xi + \|w\|_{\infty,\phi}^p + \eta F^p$. Taxes cannot increase proceeds, hence $I \le y + c_{\mathrm{M}} \|w\|_{\infty,\phi}$, while

$$F_{\mathrm{LUL}}(\widetilde{w}, F) \le F + c_{CG} \|w\|_{\infty,\phi}.$$

For $p \in (0, 1]$, the carryforward term in the next-period weight is therefore bounded by $\eta F^p + \eta c_{CG}^p \|w\|_{\infty,\phi}^p$. The portfolio drift estimate in Lemma IV.B.2 applies with c_{K} replaced by c_{M} and has coefficient at least one. Hence the term ηF^p is already controlled; choosing η small makes the additional portfolio term arbitrarily small, while choosing ξ large absorbs additive income. Thus the same strict weighted-drift condition suffices. Proposition 2.1 of Bertsekas (2012) then gives discounted contraction under the same three utility cases. Monotonicity and the compact-maximizer argument give the decreasing sup-min recursion and its fixed point. The nested-compact-set proof of Corollary 2 applies verbatim when aggregate risk is absent. ∎

IV.D.2 Withdrawals and Cutoff Controls

Throughout the trading argument, $\delta_F = \delta_I = 1$. For a feasible control $(\widetilde{w}, I)$ at (w, F), write

$$q := \mathrm{K}_{\mathrm{LUL}}(\widetilde{w}, F) - I, \qquad x := w - \widetilde{w} + I\boldsymbol{\delta}_1, \qquad F' := \left[F - \mathrm{CG}(\widetilde{w})\right]_+.$$

Here q is the net withdrawal, current consumption is $y+q$, and (x,F') is the state after trading but before returns. The unit-basis investment is written as a Dirac measure only at this intermediate step; applying Q gives the investment profile $I\chi$ in the next-period portfolio.

The resource and withdrawal identities are

$$\mathrm{K}_{\mathrm{LUL}}(w,F)=\min\{\mathrm{M}(w),\Psi(w,F)\}, \tag{IV.D.1}$$

$$\Psi(w-\widetilde{w}+I\boldsymbol{\delta}_1,F')=\Psi(w,F)-\big(\mathrm{K}_{\mathrm{LUL}}(\widetilde{w},F)-I\big). \tag{IV.D.2}$$

For the first identity, use $\Psi(w,F)=\mathrm{M}(w)-\tau\,\mathrm{CG}(w)+\tau F$ and $\mathrm{K}_{\mathrm{LUL}}(w,F)=\mathrm{M}(w)-\tau\big[\mathrm{CG}(w)-F\big]_+$. For the second, linearity of $\mathrm{K}_{\mathrm{FUL}}$ and $\mathrm{K}_{\mathrm{FUL}}(I\boldsymbol{\delta}_1)=I$ give

$$\Psi(x,F')=\mathrm{K}_{\mathrm{FUL}}(w)-\mathrm{K}_{\mathrm{FUL}}(\widetilde{w})+I+\tau\big[F-\mathrm{CG}(\widetilde{w})\big]_+=\Psi(w,F)-q.$$

Thus two controls delivering the same current net withdrawal leave the same pre-return Ψ value at a given starting state. A lower-cutoff replacement can improve the upper tails without sacrificing the second component of the order.

The perfect-carryforward sLHCL correspondence is

$$\Gamma^{sL}_{\mathrm{CF}}(w,F):=\big\{(w\mathbf{1}_{(0,a]},I)\in\Gamma(w,F):a\in[1,\infty],\ a>1\Rightarrow I=0\big\}. \tag{IV.D.3}$$

It harvests every loss; if liquidation extends into gains, it rules out simultaneous reinvestment.

IV.D.3 The Combined Return Condition

The support condition in Proposition 5 follows by applying idiosyncratic and aggregate returns together. Neither stage needs to preserve the order separately. For any portfolio x,

$$\mathrm{M}(A_zQx)=z\mathbb{E}[R]\,\mathrm{M}(x),\qquad \mathrm{B}(A_zQx)=\mathrm{B}(x).$$

The carryforward is unchanged during this return step. Since $\mathrm{K}_{\mathrm{FUL}}(x)=(1-\tau)\,\mathrm{M}(x)+\tau\,\mathrm{B}(x)$,

$$\Psi(A_zQx,F')=\Psi(x,F')+(1-\tau)\big(z\mathbb{E}[R]-1\big)\,\mathrm{M}(x). \tag{IV.D.4}$$

For two carryforward-ordered states, subtraction gives equation (14). Its right-hand side is nonnegative whenever $z\mathbb{E}[R]\geq 1$, because both the initial Ψ difference and the market-value difference are nonnegative.

Upper tails are preserved for every positive z, independently of this mean-return restriction. For density portfolios,

$$\int_a^\infty A_zQ(\bar{x}-x)(r)\,dr=z\int_0^\infty Rf(R)\int_{a/(zR)}^\infty(\bar{x}-x)(s)\,ds\,dR\geq 0.$$

The same identity holds for the finite measures used at the intermediate trading step. Hence the complete transition preserves carryforward right dominance state by state under $z\mathbb{E}[R]\geq 1$ for every z.

For example, $\mathbb{E}[R]=1.05$ and $z=0.98$ give $z\mathbb{E}[R]=1.029>1$. Such an aggregate decline is allowed by the combined condition. With no aggregate risk, the condition reduces to $\mathbb{E}[R]\geq 1$.

The condition is sharp for preservation of this order. At $\tau = 1/2$, the formal portfolios $\bar{x} = 4\boldsymbol{\delta}_2$ and $x = 3\boldsymbol{\delta}_1$, each with zero carryforward, satisfy the upper-tail order and $\Psi(\bar{x}, 0) = \Psi(x, 0) = 3$. Their market values differ by one, so after returns the Ψ difference is $\frac{1}{2}(z\mathbb{E}[R] - 1)$. It is negative if $z\mathbb{E}[R] < 1$. This establishes failure of order preservation below the bound; it does not assert that sLHCL fails in every such portfolio problem. Narrow density approximations to these lots give the same strict reversal.

IV.D.4 Cutoff Replacement and Bellman Induction

Proof of Proposition 5. The return condition enters through the complete transition $A_z Q$. The trading comparison below uses the two components of the carryforward order stated in Section IV.D.

Fix $(\bar{w}, \bar{F}) \succeq_{\mathrm{CFRD}} (w, F)$ and any feasible control at (w, F) delivering net withdrawal q. The identity $\mathrm{K}_{\mathrm{LUL}} = \min\{\mathrm{M}, \Psi\}$ implies that $(\bar{w}, \bar{F})$ can raise at least as much cash. Define $C(a) := \mathrm{K}_{\mathrm{LUL}}(\bar{w}\mathbf{1}_{(0,a]}, \bar{F})$. This function is continuous and nondecreasing in a, and $C(\infty) = \mathrm{K}_{\mathrm{LUL}}(\bar{w}, \bar{F}) \geq q$. If $q \leq C(1)$, sell all losses and set $\bar{I} = C(1) - q$. If $q > C(1)$, choose $a > 1$ with $C(a) = q$ and set $\bar{I} = 0$. This constructs an sLHCL control with the same current consumption. The boundary $a = \infty$ is included.

Let $(\bar{x}, \bar{F}')$ and (x, F') be the resulting pre-return states. Equation (IV.D.2) gives $\Psi(\bar{x}, \bar{F}') - \Psi(x, F') = \Psi(\bar{w}, \bar{F}) - \Psi(w, F) \geq 0$.

Write T and $\bar{T}$ for the current nonnegative LUL tax payments under the two controls. Portfolio market value after trading equals starting market value minus the withdrawal and the tax payment: $\mathrm{M}(x) = \mathrm{M}(w) - q - T$ and $\mathrm{M}(\bar{x}) = \mathrm{M}(\bar{w}) - q - \bar{T}$. If $\bar{T} = 0$, initial tail dominance and $T \geq 0$ imply $\mathrm{M}(\bar{x}) \geq \mathrm{M}(x)$ immediately.

If $\bar{T} > 0$, the sLHCL control must have $a > 1$, $\bar{I} = 0$, and $\bar{F}' = 0$. Above a, it has retained all of $\bar{w}$, so $D(t) := \int_t^\infty (\bar{x} - x)(r)\, dr \geq 0$ for $t \geq a$. Indeed, the initial tail of $\bar{w}$ dominates that of w, sales can only reduce the latter, and investment at $r = 1$ lies below a. Below a, $\bar{x}$ has no holdings. Since k_{FUL} is positive and decreasing, integration by parts above a, and the sign of $\bar{x} - x$ below a, imply

$$\mathrm{K}_{\mathrm{FUL}}(\bar{x}) - \mathrm{K}_{\mathrm{FUL}}(x) \leq k_{\mathrm{FUL}}(a)\big[\mathrm{M}(\bar{x}) - \mathrm{M}(x)\big]. \tag{IV.D.5}$$

Specifically, the integral above a is $k_{\mathrm{FUL}}(a)D(a) + \int_a^\infty k'_{\mathrm{FUL}}(t)D(t)\, dt \leq k_{\mathrm{FUL}}(a)D(a)$, while multiplying the nonpositive difference below a by $k_{\mathrm{FUL}}(r) \geq k_{\mathrm{FUL}}(a)$ gives the remaining bound. The case $a = \infty$ follows by the same inequality with $k_{\mathrm{FUL}}(\infty) = 1 - \tau > 0$.

The Ψ comparison and $\bar{F}' = 0$ imply $\mathrm{K}_{\mathrm{FUL}}(\bar{x}) \geq \mathrm{K}_{\mathrm{FUL}}(x) + \tau F' \geq \mathrm{K}_{\mathrm{FUL}}(x)$. Equation (IV.D.5) therefore yields $\mathrm{M}(\bar{x}) \geq \mathrm{M}(x)$ also when $\bar{T} > 0$.

For $a > 1$, every upper tail of $\bar{x}$ below a equals its total market value, which is at least $\mathrm{M}(x)$; tails above a have already been compared. For $a = 1$, the same reasoning applies below one, while above one the original tail order is preserved because only losses were sold and investment is at basis. Thus $(\bar{x}, \bar{F}') \succeq_{\mathrm{CFRD}} (x, F')$ in all cases.

This construction covers both comparisons needed for dynamic programming. Taking the same starting state on both sides replaces any feasible trade by an sLHCL trade. Taking two ordered starting states shows that a trade from the dominated state can be matched from the dominant state with the same consumption and an ordered continuation state.

The zero terminal value is increasing in carryforward right dominance. Suppose the continuation value at a finite horizon has this property. The cutoff comparison, followed by the combined return step in equation (IV.D.4), preserves current consumption and weakly raises continuation value in every aggregate state. It follows both that the Bellman update is increasing in the order and that restricting controls to sLHCL leaves its value unchanged. Induction gives the finite-horizon statements of Proposition 5.

Under the proposition's convergence assumptions, equality of full and restricted values and their order monotonicity pass to the limit. Given an attained stationary full maximum, the same replacement yields an sLHCL maximizer. Uniqueness, when it holds, forces the full maximizer to be that restricted control. This proves the discounted-utility claims in Proposition 5. ■

V Investor-Side Tax Measures

V.A Effective Tax Rates

V.A.1 Proof of Proposition 6: Perpetuities with Dividends and Inflation

Proof of Proposition 6. For part (i), consider a nominal portfolio p entering a return period. After dividend payouts, each asset earns the retained return in (16). Write

$$a := \mathbb{E}[\widetilde{R}] = 1 + (1-\varphi)[(1+\pi)(1+\mu) - 1], \qquad b := \varphi[(1+\pi)(1+\mu) - 1].$$

Without aggregate risk, retained value is $\mathrm{M}(Qp) = a\,\mathrm{M}(p)$ and dividends equal $b\,\mathrm{M}(p)$. Thus every reachable state $w = Qp$ pays $d(w) = (b/a)\,\mathrm{M}(w)$. Since $a + b = (1+\pi)(1+\mu)$, the ratio also equals $b/a = (1+\pi)(1+\mu)/a - 1$, which gives (17). The same calculation applies to retained holdings and new investment.

The unit investment at date zero gives $w_1 = Q\boldsymbol{\delta}_1 = \chi$, with no carried losses. Since the post-trade portfolio is $w_t - \widetilde{w}_t + I_t\boldsymbol{\delta}_1$, linearity of Q gives the transition in (19). Writing $d_t = d(w_t)$ also yields

$$d_1 = b, \qquad F_1 = 0, \qquad d_{t+1} = b\{\mathrm{M}(w_t - \widetilde{w}_t) + I_t\}. \tag{V.A.1}$$

Dividends therefore require no additional state.

Next, let $G_t = \mathrm{CG}(\widetilde{w}_t)$ denote realized nominal gains. The FUL tax is $\tau(G_t + d_t)$. Under LUL, netting these gains and dividends against unused losses gives

$$\mathrm{tax}_t^n = \tau\big[G_t + d_t - F_t\big]_+, \qquad F_{t+1} = \big[F_t - G_t - d_t\big]_+. \tag{V.A.2}$$

Subtracting taxes from sale proceeds and dividends gives (18). Hence every feasible trading sequence satisfies the cash and transition constraints in (19). Conversely, any sequence satisfying those constraints consists of feasible sales and reinvestments and finances at least $c(1+\pi)^t$ at every date. These constraints therefore characterize the feasible real perpetuities, proving (i).

For part (ii), define $\widehat{w}_t = (1+\pi)^{-t} w_t$, $\widehat{\widetilde{w}}_t = (1+\pi)^{-t}\widetilde{w}_t$, $\widehat{I}_t = (1+\pi)^{-t} I_t$, and $\widehat{F}_t = (1+\pi)^{-t} F_t$. This rescales values, not the cumulative nominal return r indexing each tax lot; the tax basis is not indexed. Homogeneity of M, CG, d, and the positive-part map gives

$$K(\widehat{\widetilde{w}}_t, \widehat{F}_t; \widehat{w}_t) \geq c + \widehat{I}_t, \tag{V.A.3}$$

$$\widehat{w}_{t+1} = (1+\pi)^{-1}\{Q(\widehat{w}_t - \widehat{\widetilde{w}}_t) + \widehat{I}_t\chi\}. \tag{V.A.4}$$

Under LUL, the loss balance becomes

$$\widehat{F}_{t+1} = (1+\pi)^{-1}\big[\widehat{F}_t - \mathrm{CG}(\widehat{\widetilde{w}}_t) - d(\widehat{w}_t)\big]_+. \tag{V.A.5}$$

Under FUL it remains zero. Because $1+\pi > 0$, this change of units is invertible and preserves feasible real withdrawals. The stationary problem starts from $(\chi/(1+\pi), 0)$. Its canonical sup-min value $V^{\mathbb{P}}$ therefore characterizes the same set of perpetuities. Theorem 2 in Section IV.B gives attainment; Corollary 2 in Section IV.A identifies the perpetuity value, with the carryforward extension in Section IV.D. Their regularity conditions are imposed on the primitives expressed in real units. Hence

$$\bar{c}_M = V^{\mathbb{P}}(\chi/(1+\pi), 0) = \max \mathcal{C}(\mathrm{T}; f, \tau, \pi, \varphi). \tag{V.A.6}$$

Definition 5 then gives $\tau^* = 1 - \bar{c}_M/\mu$, proving (ii). ■

V.A.2 Numerical Computation

The numerical return law is log-normal before discretization. Its nominal arithmetic mean is $(1+\pi)(1+\mu)$ and its nominal arithmetic standard deviation is σ. Inflation comparisons hold μ and nominal σ fixed. They consequently change real dispersion as well as the nominal tax base.

The numerical procedure implements the trading rules established in Sections IV.C and IV.D of the manuscript. It approximates the perpetuity problem as follows.

i) Calibrate the log-normal nominal total-return distribution to mean $(1+\pi)(1+\mu)$ and standard deviation σ. Approximate it by a binomial grid with N substeps per period, and apply the dividend transformation in (16).

ii) Start from a unit investment before the first return. Set $w_1 = \chi$, $F_1 = 0$, and d_1 as in (V.A.1).

iii) For a candidate real withdrawal c, apply the relevant optimal liquidation rule at each trading date. Update holdings using (19), dividends using (V.A.1), and carried losses using (V.A.2) under LUL. Track residual liquidation value over 200 periods.

iv) Apply the growth screen to the residual liquidation-value sequence: over the simulated horizon, the value must be non-decreasing or remain bounded away from zero. If the candidate passes, raise c; if it fails, lower c. Repeating this step by bisection approximates the largest accepted withdrawal.

v) Compute $\tau^* = 1 - \bar{c}_M/\mu$ from the accepted withdrawal.

The canonical perpetuity is obtained by backward iteration of the perpetuity operator defined in Section IV.A, with convergence established in Section IV.B. The $T = 200$ forward simulation is a numerical approximation to this infinite-horizon value. Its growth screen alone does not prove infinite-horizon feasibility. The single-fund comparison uses the deterministic aggregate-fund recursion and proportional liquidation, so it does not depend on cross-sectional dispersion.

V.A.3 Borrowing and Interest Deductions

Borrowing provides a further way to manage taxes by financing a larger portfolio. If interest is deductible, it also creates deductions that can offset taxable returns. As in the short-selling example in Appendix III.A.3, I take the size of the position as given.

Let $x \geq 0$ be debt per unit of initial own wealth, and let r_b be the real borrowing rate. The investor invests $1 + x$, maintains debt at a constant real level, and pays interest from the portfolio. Let $\zeta = 1$ if interest is deductible and $\zeta = 0$ otherwise. The effective rate is the accrual tax rate that sustains the same maximal real perpetuity $\bar{c}(x)$:

$$\bar{c}(x) = (1 + x)\mu\big(1 - \tau^*(x)\big) - xr_b + \zeta x r_b \tau^*(x). \tag{V.A.7}$$

The normalization requires $(1 + x)\mu - \zeta x r_b > 0$, as in the reported calculations.

Without deductibility, homogeneity gives

$$\bar{c}(x) = (1 + x)\bar{c}_0 - xr_b, \qquad \tau^*(x) = \tau_0^*. \tag{V.A.8}$$

Here $\bar{c}_0$ and τ_0^* are the values without debt. Borrowing changes sustainable consumption but leaves the effective rate unchanged: the portfolio scales in the same way under realization and accrual taxation. Borrowing raises consumption if $r_b < \bar{c}_0$. At the baseline, this threshold is about 4.30%, compared with 3.75% under statutory real-accrual taxation.

With deductibility under LUL, however, interest adds to the carried loss balance. The investor must therefore account for interest deductions and harvested losses together. Table V.A.1 shows that more debt lowers the effective rate. At $r_b = 5\%$, the rate falls from 0.557τ without debt to 0.123τ at $x = 3$. At $r_b = 2.5\%$, the interest deduction is smaller and the effective rate falls to 0.372τ. Consumption is nevertheless higher at the lower borrowing cost.

To determine when borrowing raises consumption, define the break-even rate $r_b^*(x)$ by $\bar{c}(x, r_b^*(x)) = \bar{c}_0$. With deductions, it is about 5.64% near zero debt and declines with debt over the reported range. Larger deductions compete with harvested losses and can remain unused for longer. Without deductions, the break-even rate is exactly $\bar{c}_0$ at every x.

Table V.A.1: Deductible borrowing under LUL: baseline market environment

	$r_b = 5\%$		$r_b = 2.5\%$		Break-even
x	$\bar{c}(x)$	$\tau^*(x)/\tau$	$\bar{c}(x)$	$\tau^*(x)/\tau$	$r_b^*(x)$
0.00	.0430	.557	.0430	.557	—
0.25	.0442	.467	.0490	.515	5.57%
0.50	.0451	.396	.0549	.484	5.50%
1.00	.0463	.295	.0667	.441	5.38%
2.00	.0477	.182	.0901	.396	5.26%
3.00	.0485	.123	.1134	.372	5.19%

Baseline: $\mu = 5\%$, $\sigma = 30\%$, $\tau = 25\%$, $\varphi = 30\%$, $\pi = 2\%$, and perfect carryforward. The near-zero-debt calculation uses $x = .0025$ and gives $r_b^* = 5.64\%$. Calculations use 200 periods, 20 grid substeps, and consumption tolerance 10^{-8}; the near-zero-debt calculation uses 10^{-10} and the break-even root tolerance is 10^{-6}.

Without dividends or inflation, the same pattern holds. At $r_b = 5\%$, the effective-rate ratio falls from 0.411 without debt to 0.025 at $x = 3$. The deductible break-even rate declines from about 5.59% near zero debt to 5.17% at $x = 3$.

Under LUL, taxes cannot be negative, so

$$\bar{c}(x) \leq (1 + x)\mu - xr_b. \tag{V.A.9}$$

Borrowing and deductions can therefore eliminate the tax burden but cannot generate net tax refunds. FUL, by contrast, removes this restriction. With immediately refundable deductions,

$$\bar{c}(x) = (1 + x)\bar{c}_0 - xr_b(1 - \tau), \qquad r_b^* = \frac{\bar{c}_0}{1 - \tau}. \tag{V.A.10}$$

If $r_b = \mu$ and $\tau_0^* < \tau$, consumption increases without bound with debt. The corresponding effective rate can become negative. Stiglitz (1983) discusses this use of borrowing and interest deductions. The calculations here take debt as given and abstract from aggregate risk.

For example, at $\tau = 25\%$ and $\varphi = \pi = 0$, FUL gives a break-even borrowing cost of 4.59% without deductions and 6.13% with deductions. At $r_b = \mu = 5\%$, the effective rate is already slightly negative at $x = 0.5$.

VI Government-Side Effective Tax Rates

VI.A Recycling and Permanent Revenue Equivalence

Let $g = 1 + \mu$ and let W_t be aggregate real wealth at the start of period t. Total returns give gW_t before consumption and taxes. Write C_t and T_t for these payments under the applicable tax system. Surviving investors and heirs retain $gW_t - C_t - T_t$, while recycling reinvests $C_t + T_t$. Therefore

$$W_{t+1} = (gW_t - C_t - T_t) + (C_t + T_t) = gW_t. \tag{VI.A.1}$$

Inheritance changes ownership and possibly tax basis, but removes no wealth. Thus this identity holds under all three death provisions and under accrual taxation.

For the unit-investment paths, let v_a denote real wealth at the start of age a. Newly accrued real returns are $\text{accr}_a = \mu v_a$, including dividends. Summing across portfolios gives $A_t = \mu W_t$. If paths are recorded in nominal units relative to entry prices, opening stocks are divided by $(1+\pi)^a$ and period flows by $(1+\pi)^{a+1}$. Under this timing, the balanced-growth weights in (25) imply a constant ratio of revenue to the real accrual base. The following result shows why this is more than a date-specific accounting comparison.

Proposition VI.A.1 (Revenue equivalence after a permanent switch)**.** *Fix the statutory realization rate τ, market conditions, loss treatment, and treatment at death. Suppose the realization-tax economy is on a balanced-growth path with constant $T_u^{\mathrm{R}} / A_u^{\mathrm{R}} = \tau_G^*(\tau)$. At the start of any date t, replace realization taxation permanently by a tax at the fixed rate $\tau_G^*(\tau)$ on newly accrued real returns only. Impose no separate settlement or later realization tax on gains accrued before t. Maintain recycling and the common gross real return g in both economies. Relative to continuing realization taxation, the switch preserves revenue at every subsequent date:*

$$T_{t+s}^{\mathrm{A}} = \tau_G^*(\tau)\mu g^s W_t = T_{t+s}^{\mathrm{R}}, \qquad s \geq 0. \tag{VI.A.2}$$

Proof. Both economies have wealth W_t at the switch. Equation (VI.A.1) applies under either tax system, irrespective of the division between consumption and tax payments. Induction therefore gives

$$W_{t+s}^{\mathrm{A}} = W_{t+s}^{\mathrm{R}} = g^s W_t, \qquad A_{t+s}^{\mathrm{A}} = A_{t+s}^{\mathrm{R}} = \mu g^s W_t. \tag{VI.A.3}$$

The accrual tax collects $\tau_G^*(\tau)\mu g^s W_t$. The continuation of the realization-tax economy remains on its balanced-growth path, where its revenue-to-accrual ratio equals $\tau_G^*(\tau)$ at every date. Its revenue is therefore the same expression. This proves (VI.A.2). ∎

VI.B Comparison of Loss Treatments

Table VI.B.1: Revenue relative to statutory real-accrual taxation, τ_G^*/τ

	LUL			FUL		
Market environment	Step-up	Carry-over	Realization	Step-up	Carry-over	Realization
Baseline	.36	.51	.69	.34	.50	.67
$\varphi = \pi = 0$	.20	.37	.51	$< .01$	.28	.40

$\mu = 5\%$, $\sigma = 30\%$, $\tau = 25\%$, and $\beta = 3\%$. The baseline has $\varphi = 30\%$ and $\pi = 2\%$. LUL permits perfect carryforward and both systems permit frictionless reinvestment. The joint $\varphi = \pi = 0$ case is not included in Figure 5.

At the baseline, immediate refunds and perfect carryforward give similar collections. Without dividends or inflation, however, the difference is larger. Under FUL and step-up, living loss refunds almost offset living tax receipts, while embedded gains are forgiven at death. The near-zero revenue in that case is specific to the combination of loss refunds and forgiveness.